

STUCK PIPE PREVENTION

NEOTEK INOVASI GLOBAL



NEOTEK INOVASI GLOBAL - YOUR INTEGRATED ENERGY PARTNER



Alam Alsahara Group founded (Al Khobar, Saudi Arabia) heritage in petroleum training, consulting & supply chain.



Rebranded and incorporated as PT Neotek Inovasi Global (NIG), Indonesia.

2016

2018

2024
September

October
2024



PT Alam Alsahara Indonesia established.



HQ in South Tangerang, part of an international group spanning Indonesia, Saudi Arabia, Egypt, Türkiye & Algeria.

Core Services: Training, Consulting, Supply Chain, and Inspection & Asset Integrity

NIG AND THE CHALLENGES WE SOLVED

Training & Certifications	Inspection Services	Supply Chain Services	Consulting Services
Problem: "Field certifications are scattered, and it's unclear which ones actually count for drilling or geothermal work, at home or abroad."	Problem: "Aging equipment and missed inspection cycles put safety and uptime at risk on rigs and geothermal sites."	Problem: "Operators managing K3 (HSE) and geothermal compliance struggle to source the right PPE, spare parts, and equipment fast enough."	Problem: "Drilling projects need objective, experienced advisory but not every team has that expertise in-house."
Solution: Accredited Training , internationally recognized programs (UK & US bodies) for drilling and geothermal field roles, from entry-level to Train-the-Trainer.	Solution: Inspection & Asset Integrity - NDT, pressure vessel verification, and DROPS inspection built for drilling and geothermal assets.	Solution: Supply Chain Solutions - procurement, PPE, spare parts, and rental equipment tailored for K3 and geothermal operators, meeting API/ISO/ANSI standards.	Solution: Consulting - strategic advisory for drilling operations, HSE compliance, and technical performance optimization.

NIG CORE SERVICES

WHAT WE DO



TRUSTED BY



GIVING BACK TO COMMUNITY



**Universitas Islam
Negeri Syarif
Hidayatullah Jakarta**



**Universitas Islam
Negeri Syarif
Hidayatullah Jakarta**



**Universitas
Indonesia**



**Universitas
Indonesia**



**Universitas
Padjadjaran**



**Institut Teknologi
Bandung**



**Universitas
Trisakti**



**Universitas Bina
Nusantara**

ANAB ACCREDITATION SERVICES



Neotek Inovasi Global is in the process of being approved by ANAB (ANSI National Accreditation Board) the leading US accreditation body for 21 training programs. This milestone places NiG among an elite group of globally accredited training providers.

1 Advanced Rig Inspection

4 H2S Intensive

7 Hazard Communication (HAZCOM)

10 HSE Inspection

13 Permit to Work

16 Rig Orientation

19 Scaffolding Awareness Program

2 H2S Awareness

5 H2S Master Trainer

8 Hazard Identification (HAZID)

11 Job Safety Analysis

14 Rig Electrician

17 Rigging and Lifting

20 Scaffolding Erection and Dismantling

3 H2S Breathing Apparatus (BA)

6 Hazard and Operability Study (HAZOP)

9 Hazardous Materials (HAZMAT)

12 Lock Out Tag Out

15 Rig Mechanic

18 Safety Leadership

21 Working and Rescue at Height

NIG GALLERY



COURSE AGENDA

SESSION 1

Mohammad Antar Moursi

Client Relationship Manager

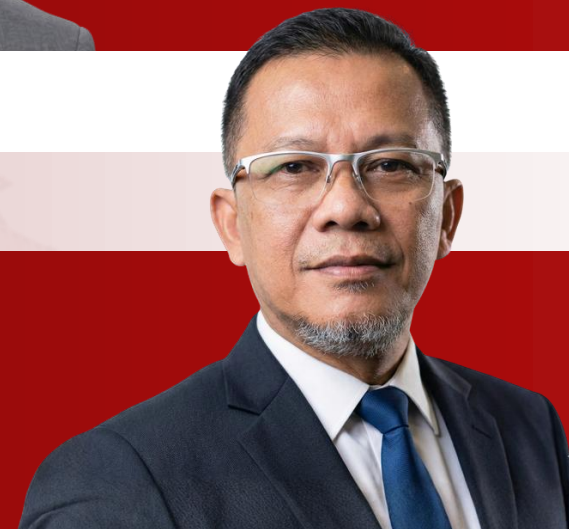


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SESSION 2

Irwin Wardiman

Technical Advisor



2

SESSION 3

Abdullah Adel Abdel Aziz

Country Manager



3

OBJECTIVES

- 1. Identify the cost and impact of stuck pipe**
- 2. Explore the role communication and teamwork play in preventing stuck pipe issues**
- 3. Recognize the mechanisms and warning signs that lead to stuck pipe incidents**
- 4. Demonstrate skills needed to prevent, recognize and resolve stuck pipe due to the following:**
 - Differential Sticking
 - Wellbore Geometry
 - Wellbore Instability



Mohammad Antar Moursi

Client Relationship Manager

Definition

Incidents and Impact

Indication, Causes, and Factors

Formation Characteristics

Troublesome formation

Stuck Pipe Case and Prevention

Fundamentals of Teamwork and Team

DEFINITION

What is Stuck Pipe?

When normal operations are interrupted for 1 hour or more because of the inability to retrieve the drill string from the hole.

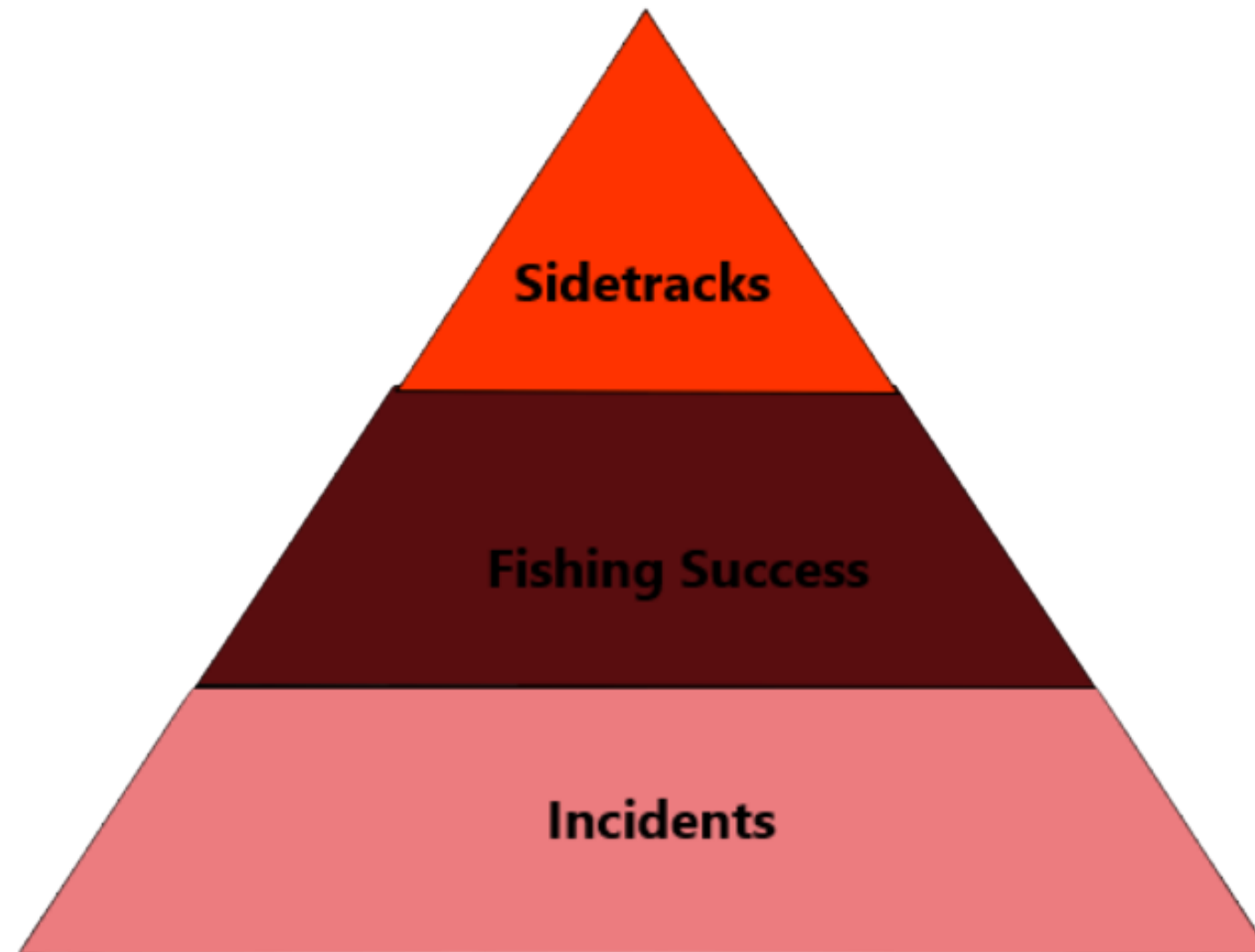
Stuck Pipe is referred to a situation when the drill string get stuck and cannot be pulled out of the hole. If not detected and treated in a short time, this problem can impose significant unforeseen operational costs, from loss of bottom hole assembly (BHA) and tools to unplanned side-tracking operations. (Zhong et al., 2022)

What is the difference between “stuck pipe” and pipe sticking?



IMPACT OF STUCK PIPE

WHAT IS THE COST OF STUCK PIPE?



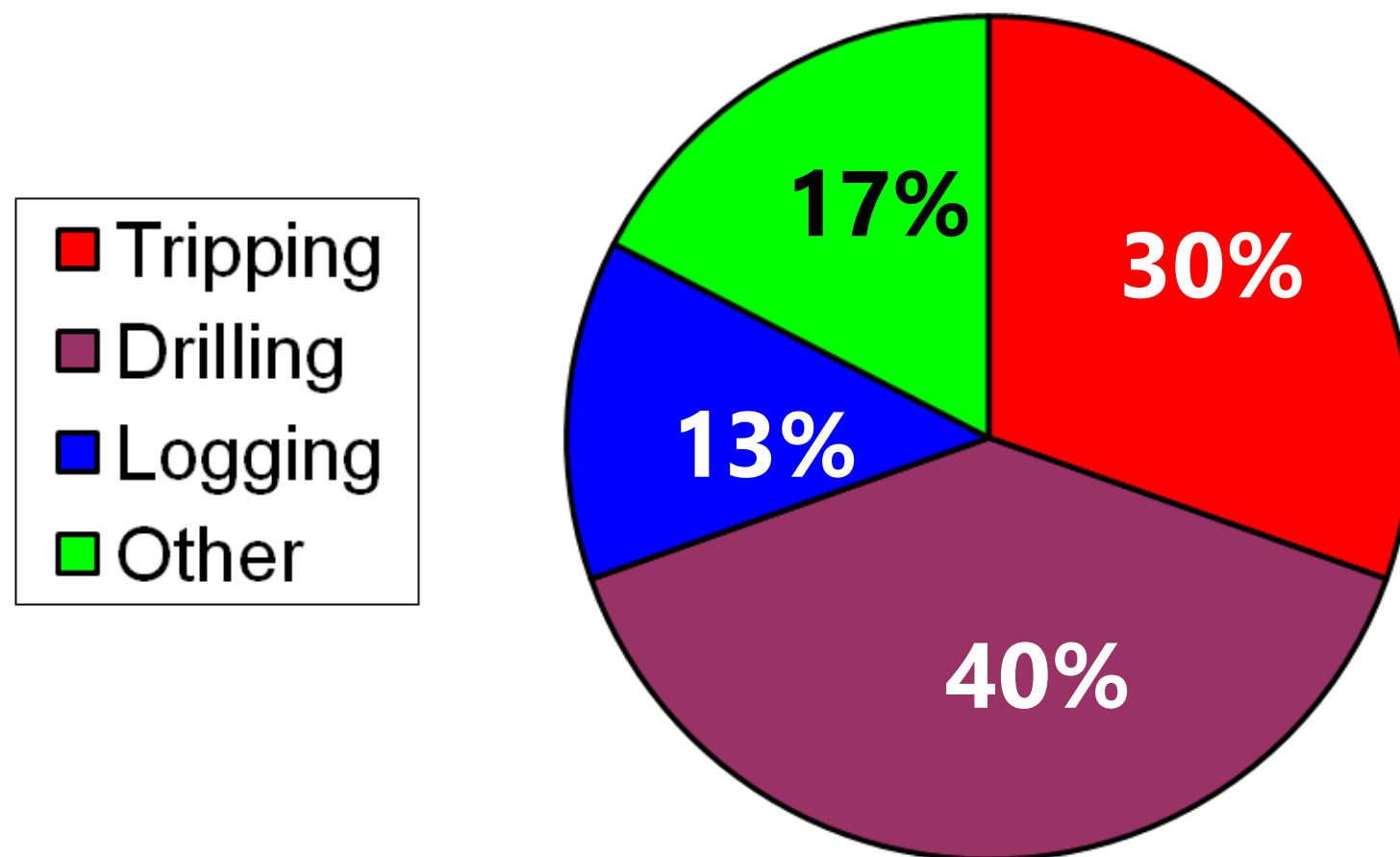
If we can reduce incidents
we may save a sidetrack!!



Stuck Pipe Cost
~\$200 - \$500 million/year

Stuck Pipe Incidents

What operations?



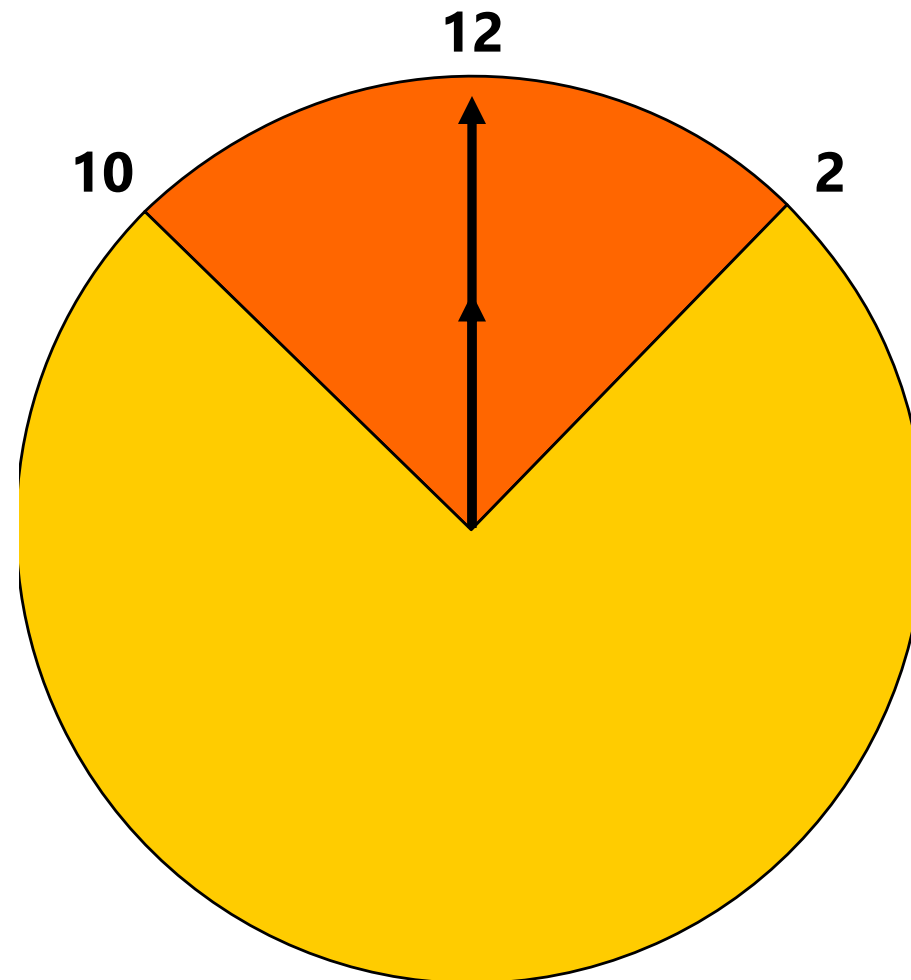
“Other” includes:

- **Setting cement plug**
- **Milling anchor seal**
- **Back-reaming**
- **Running liner**

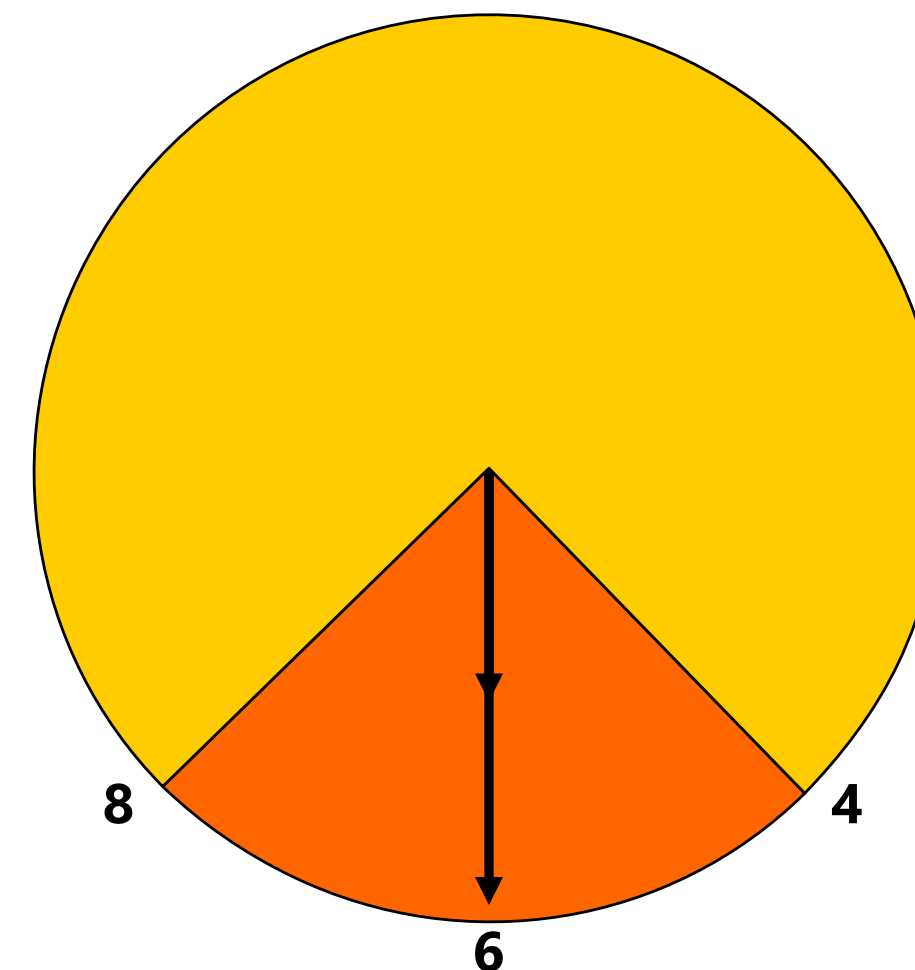
Impact of Tour Change

68% +/- 2 Hours of TOUR CHANGE

12:00 TOUR CHANGE



6:00 TOUR CHANGE



Beware of crew change hour!!!

Indication & Causes

Indication:

- Cannot pick up pipe

Causes:

- Cave – ins
- Keyseat – crooked hole
- Differential pressure sticking
- Filter cake
 - Deposited after circulation stops – while still on bottom

Results:

- Fishing operations (Back off, POH, RIG with fishing string)
- Loss of Hole

Factors

1. Formation Characteristics

- Porosity
- Permeability
- Pore Pressure
- Rock Stress
- Balance (Mud Weight)



Cannot controlled

But, can be prevent

2. Mechanical Factors

- Differential Pressure
- Hole Pack-off/Bridging
- Wellbore Geometry

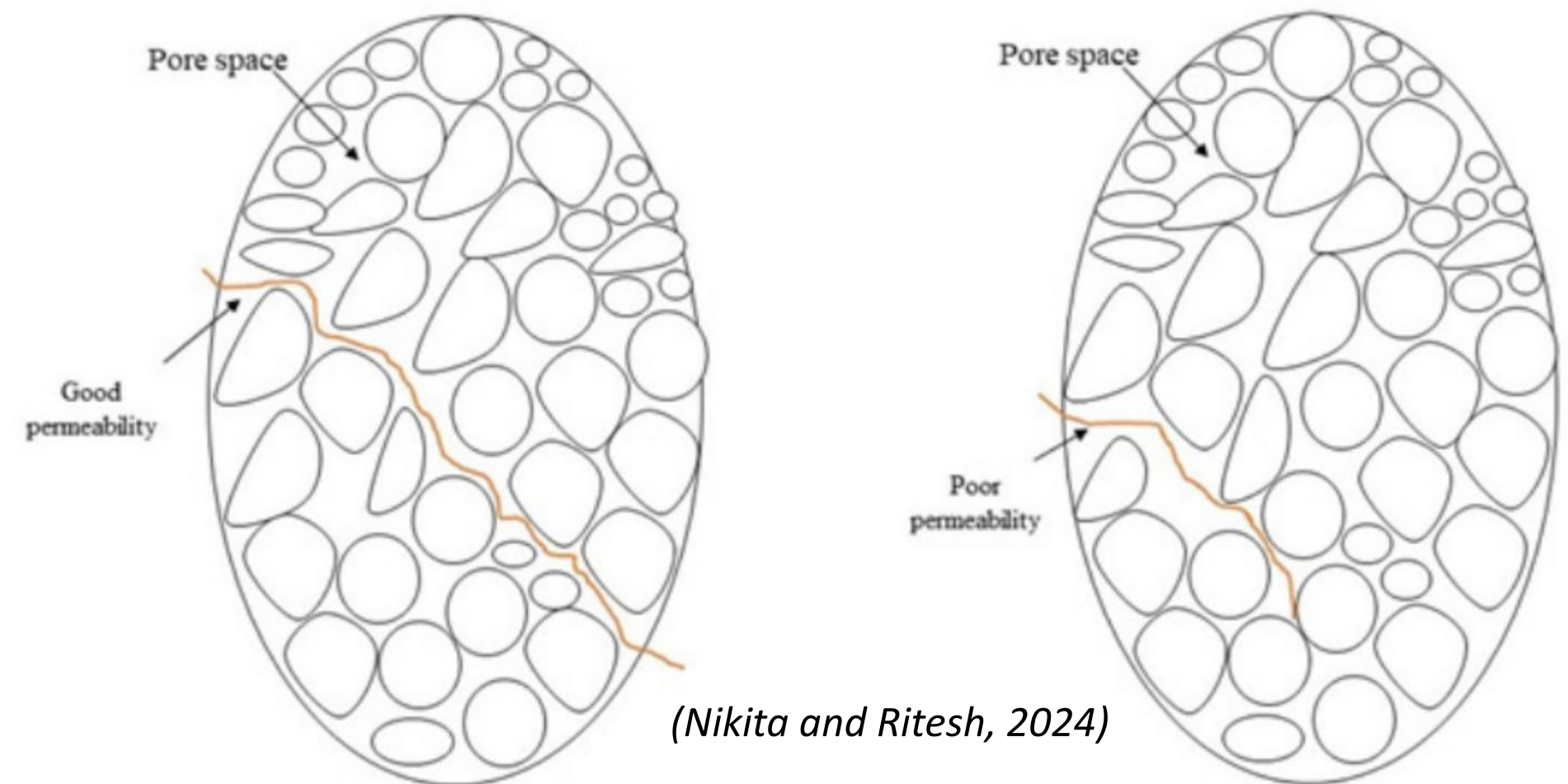
Porosity

- The “pore” spaces within the rock matrix
- Measured in % by volume
- Ratio of the fluid to the total rock volume

Permeability

- Capacity of a porous medium to transmit fluids.
- Pore spaces are “connected” allowing pore fluids to flow through the rock. Measured in “darcies” (milli-darcies typical).

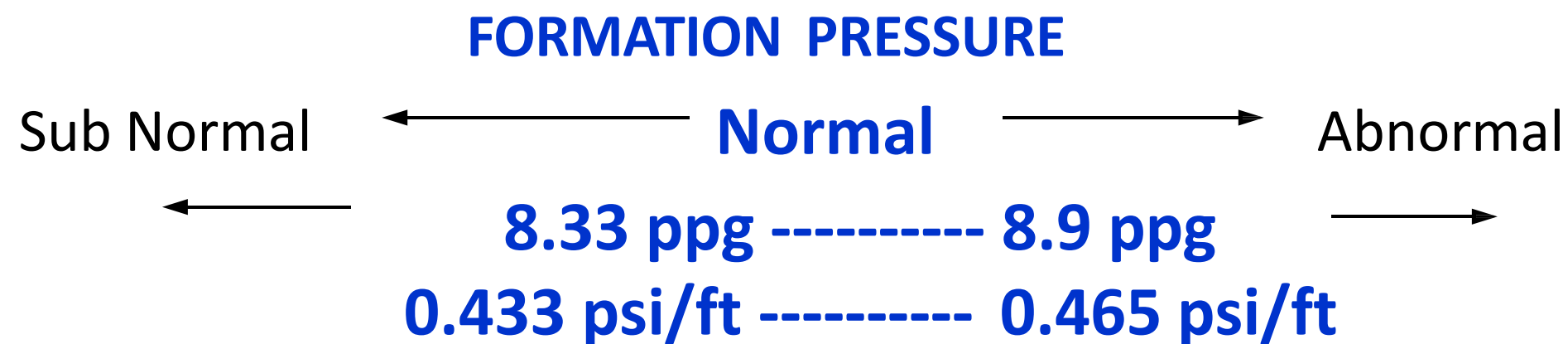
**Pores must be connected in order to have measurable permeability
– not unlike a sponge or foam rubber.**



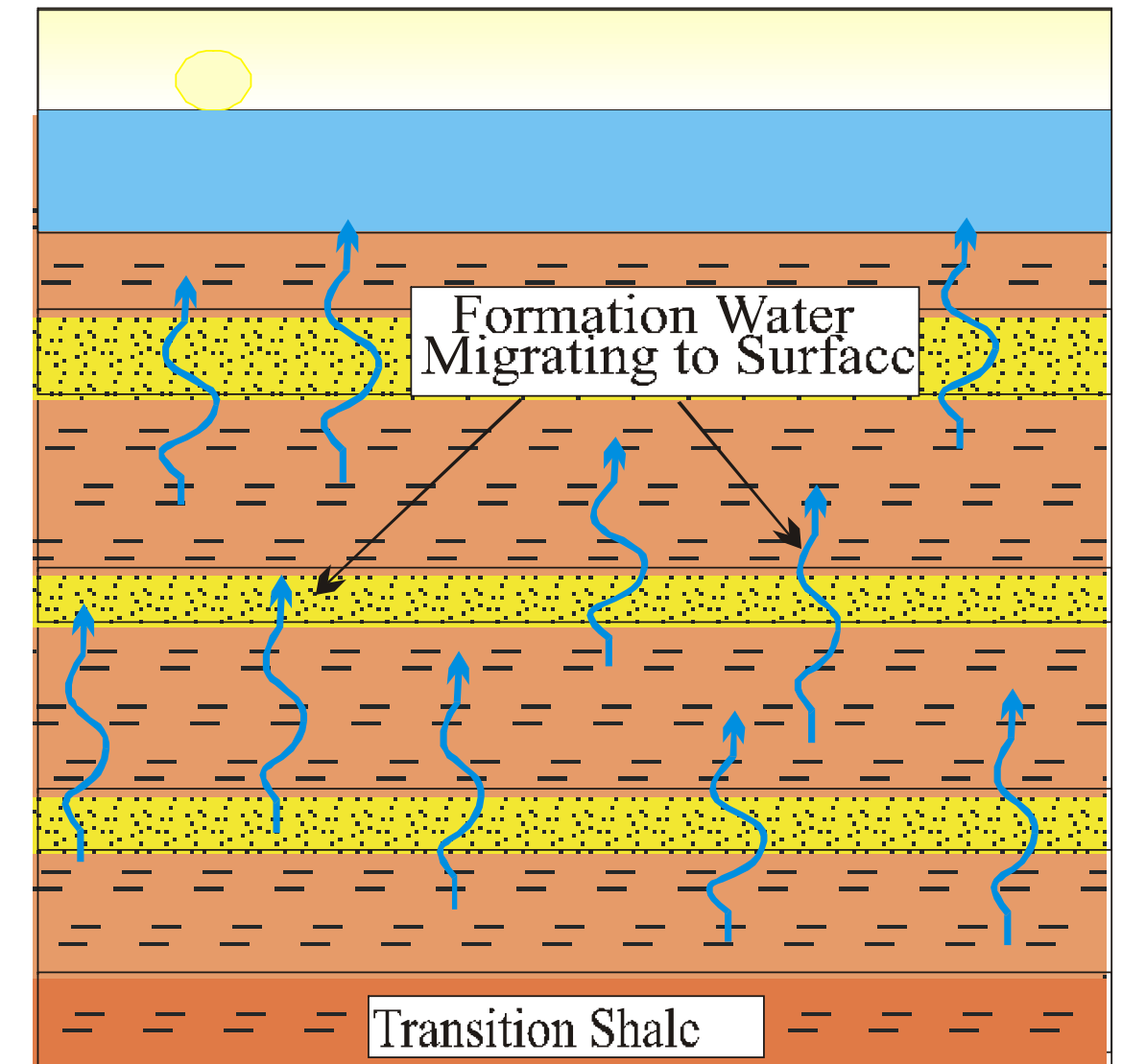
Pore Pressure

Pressure imposed on fluids contained in permeable formations. The main contributors to formation pressure are: overburden, rapid deposition, and tectonics.

Subnormal Pressure	(Gradient < 0.433 psi/ft)
Normal Pressure	(Gradient 0.433 – 0.465 psi/ft)
Abnormal Pressure	(Gradient > 0.465 – 1.0 psi/ft)

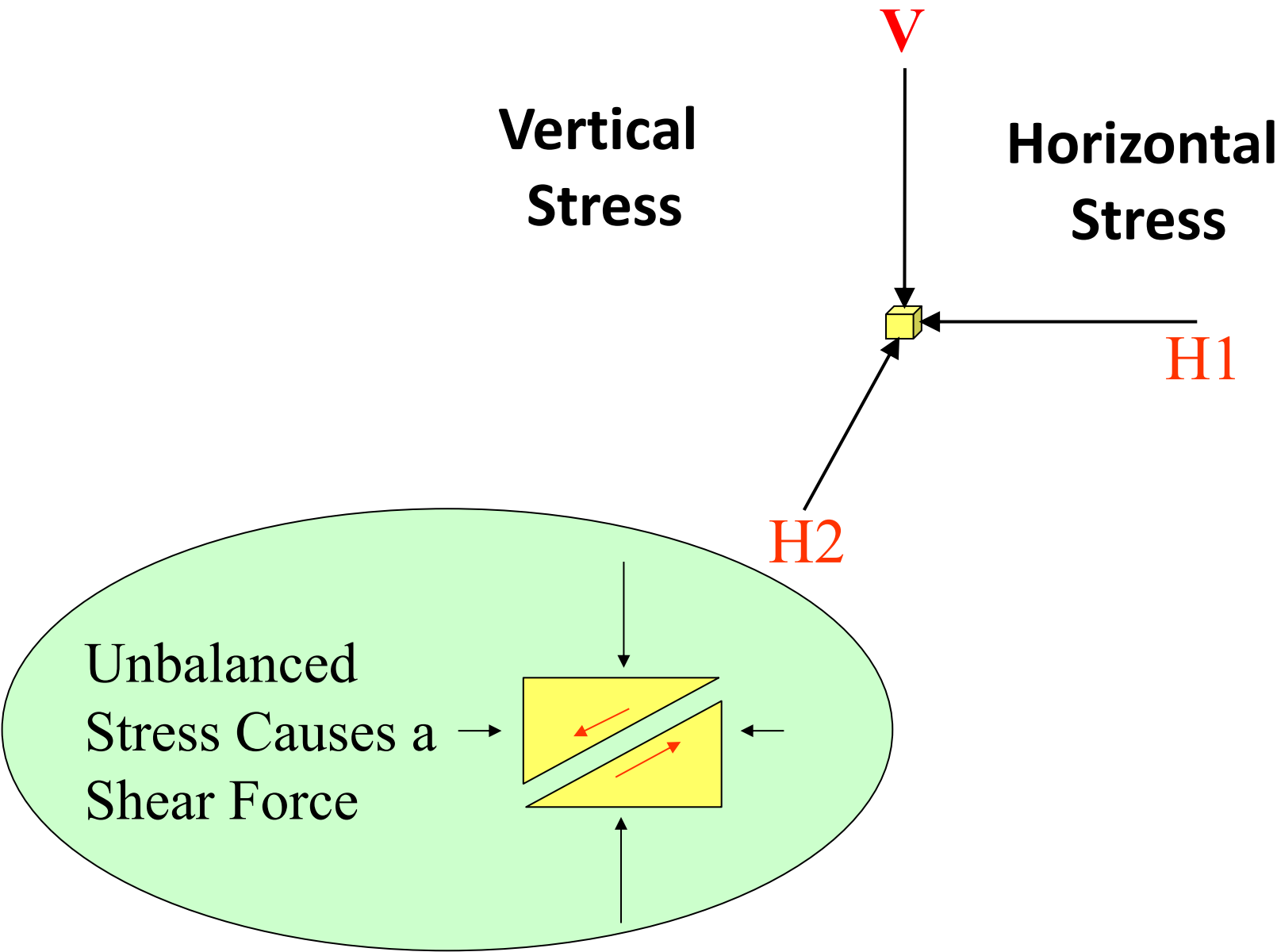


Fresh Water Formation ----- Salt Water Formation

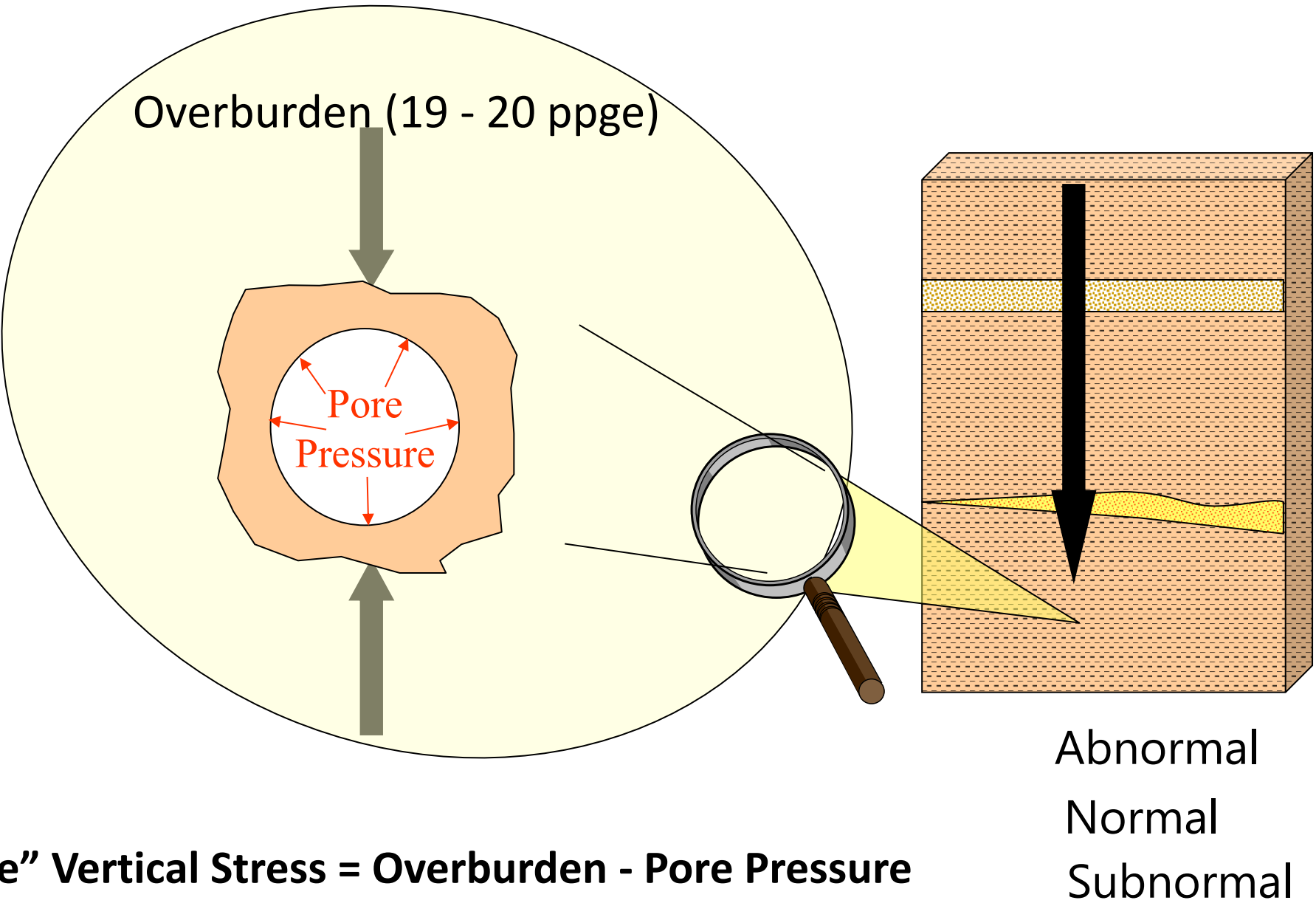


Rock Stress

Rock Stress in the earth Vertical and Horizontal components

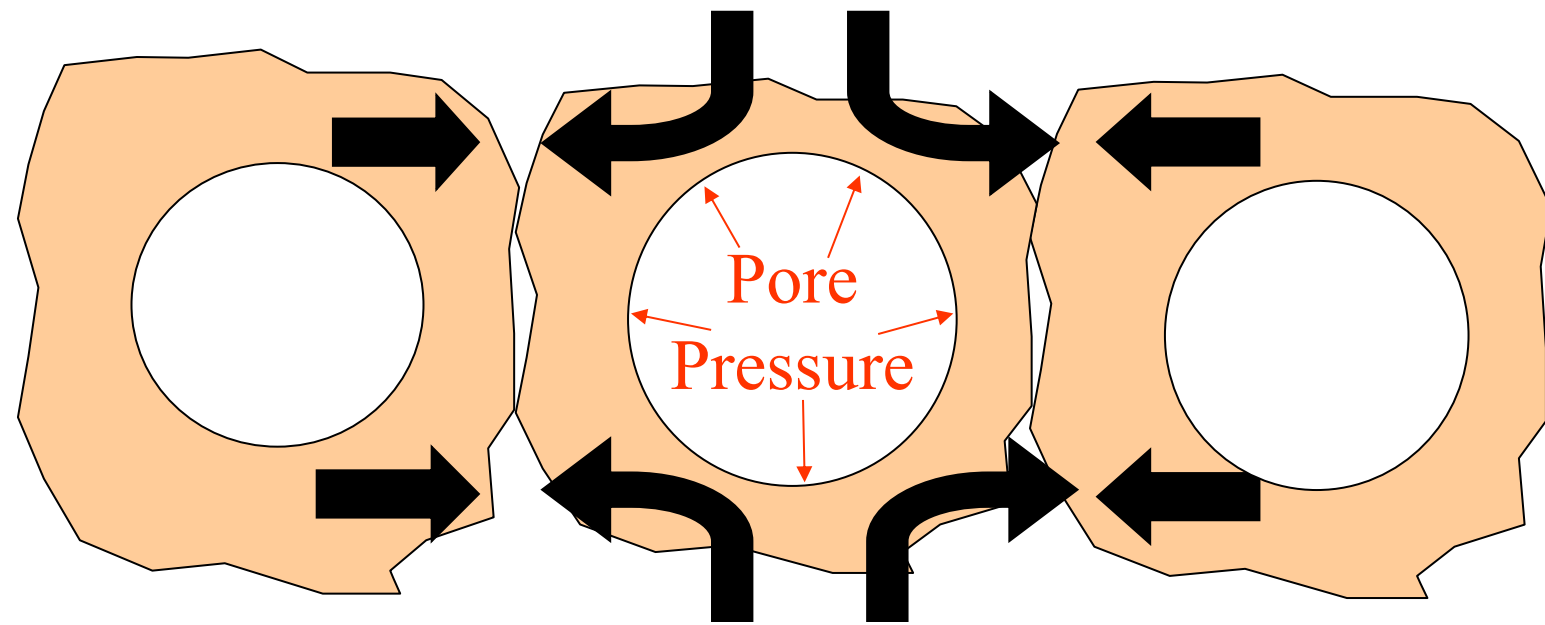


Rock Stress *In-Situ* Vertical Stress Due to Overburden



Rock Stress

Rock Stress *In-Situ* Horizontal Stress Due to Overburden

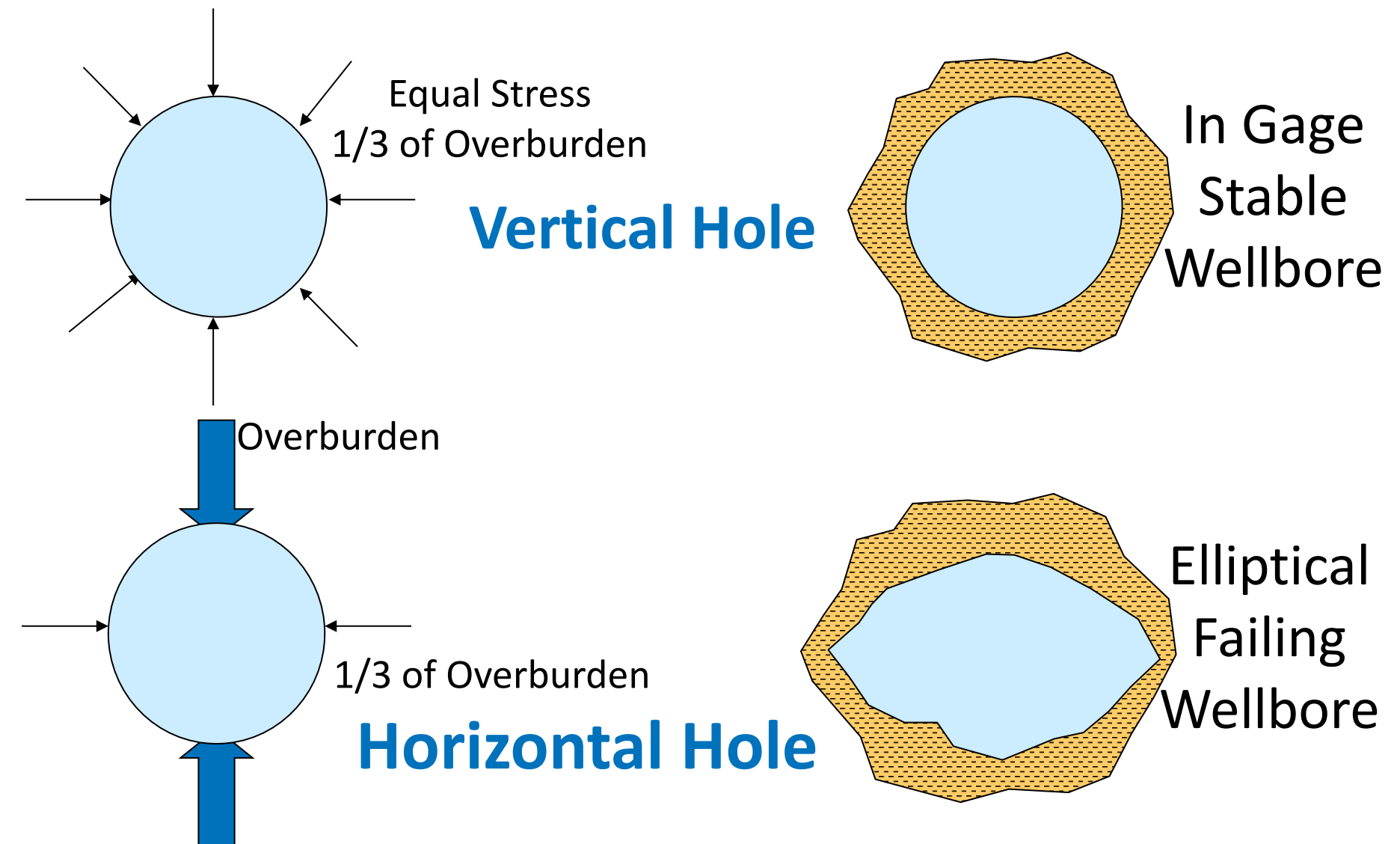


Horizontal Stress = Effective Vertical Stress x Lithology Factor

Shale - 1/3

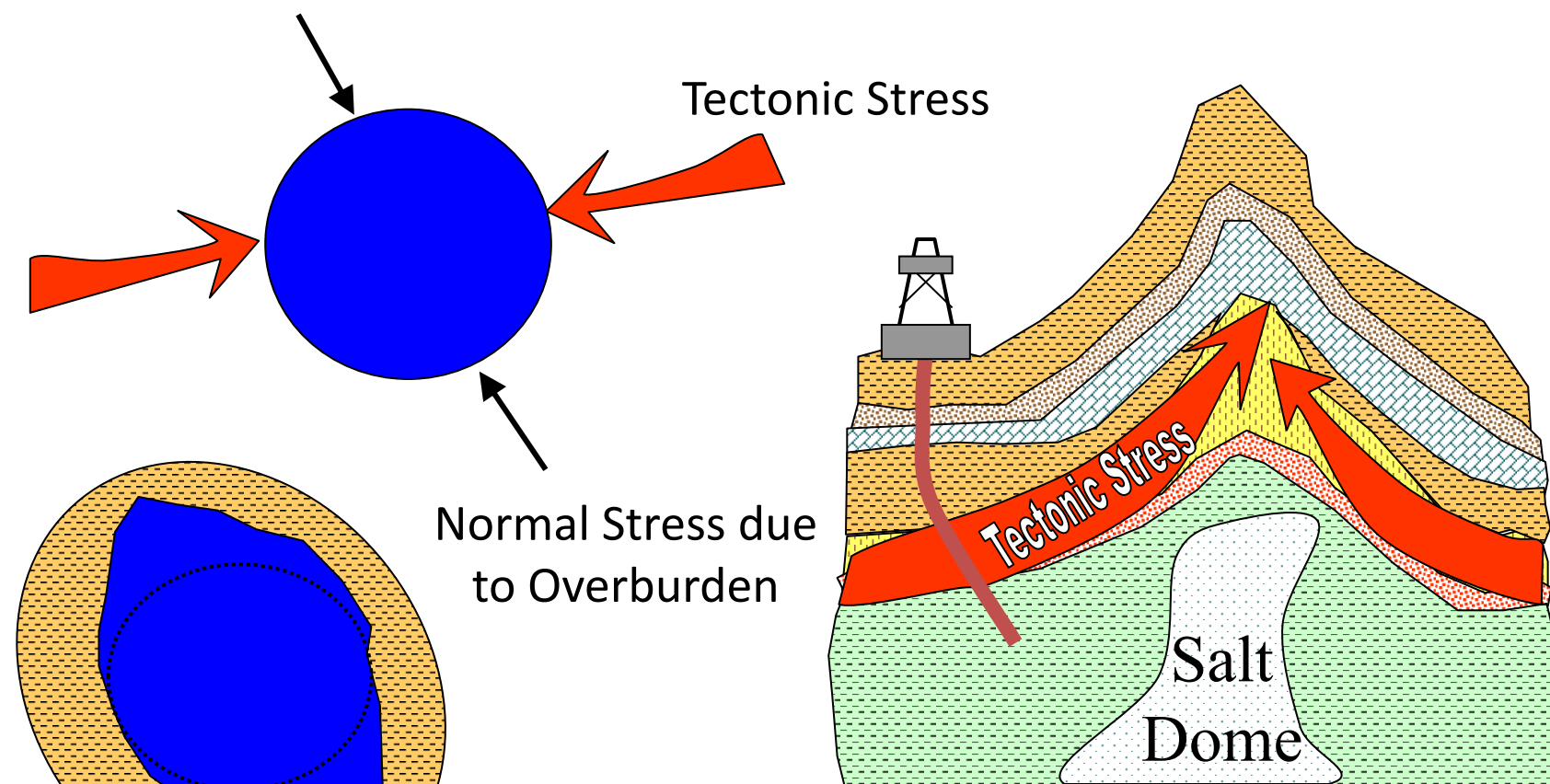
Horizontal Stress = 1/3 of Vertical Stress

Rock Stress Near Wellbore Stress Due to Overburden



Rock Stress

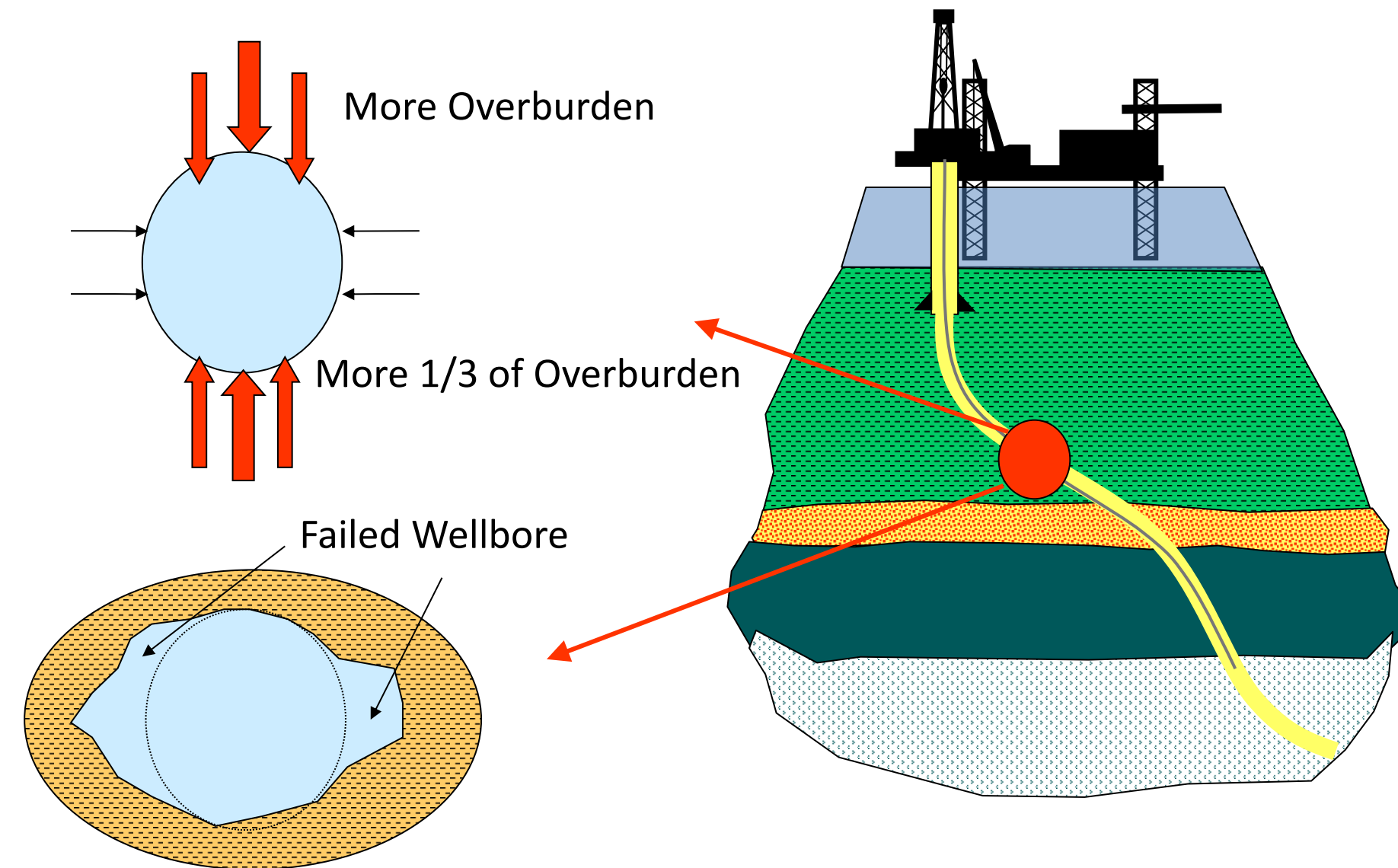
Rock Stress Unequal *In-Situ* Horizontal Stress



A hole deformed in this way can lead to:

- Stuck Tubulars
- Ineffective Hole Cleaning
- Ledges & Breakouts
- High Torque & Slip-Stick
- Drill string Failures

Rock Stress Shale Failure in High Angle Hole at Shallow Depth



Balance - Mud Weight

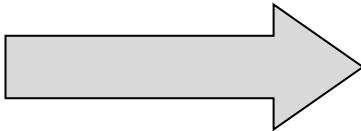
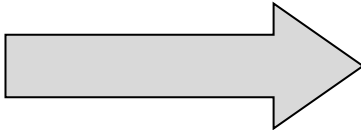
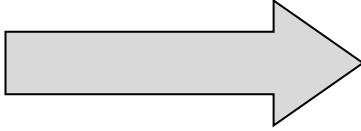
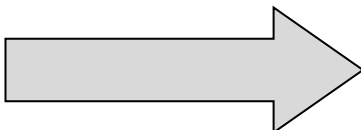
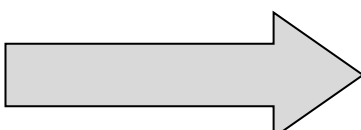
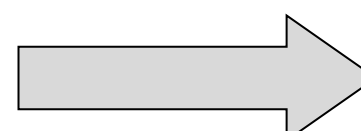
Mud weight is critical in controlling hydrostatic pressure.

- Designed to balance the stresses in the rock
- Designed to balance the pore pressure

There is usually a Mud Weight “window” that will optimise the stability of the wellbore.

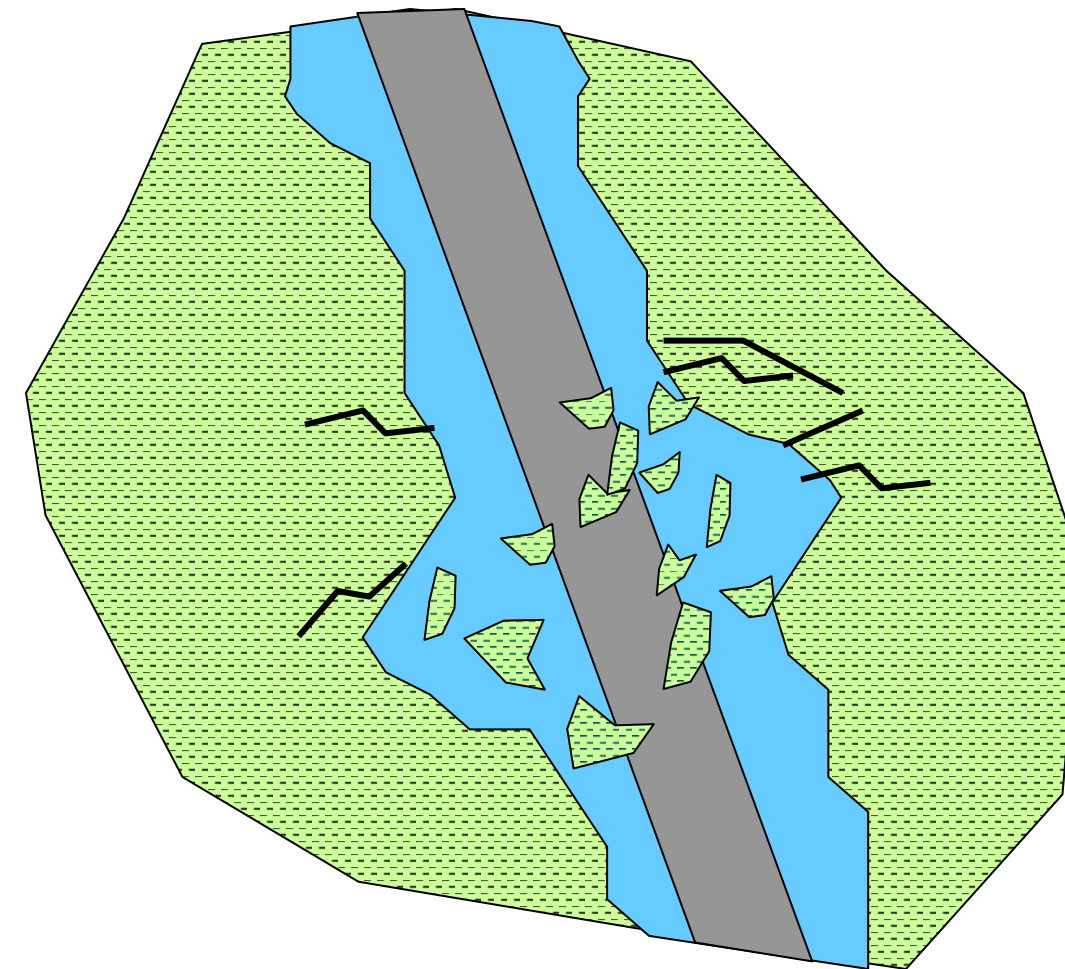
- Too little can result in bore hole collapse
- Too much can result in loss circulation and may allow shale failure and possible squeezing salt problems

Troublesome Formations

Mobile		Like toothpaste, can flow into the wellbore. This can cause geometry related stuck pipe. Soft clays and squeezing salts.
Reactive		Reacts with water and swells. Gumbo. Soft clay and shale.
Pressured		Shale which contains geopressured fluids. Commonly appear as splintery shale and sharp caving over shakers.
Depleted		Sands which have been produced to the point that the pressure is depleted. Overbalance hydrostatic can cause differential sticking
Abrasive		Sharp sands can wear bit and stabilizers down and cause the hole to become undergauged. Leads to tight hole conditions.
Fractured		Limestone, dolomites, and other brittle formations may be naturally fractured. Large blocks may fall into the hole.

Causes of Settled Cavings

- Reactive Shale (Gumbo)
- Geopressured Shale
- Overburden Stress
- Tectonic Stress
- Unconsolidated Formation
- Fractured Formations
- Soft Cement



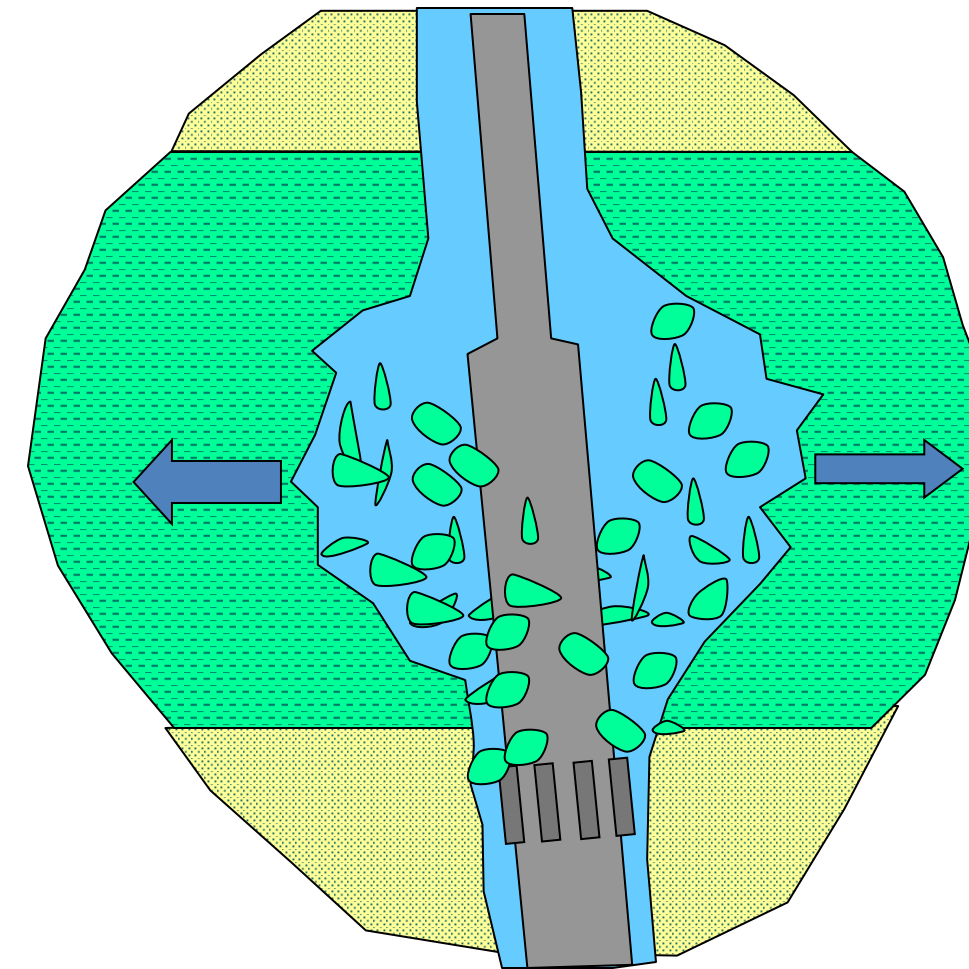
Reactive Shale Gumbo

Cause

- ▶ Water sensitive shale drilled with inadequate mud inhibition
- ▶ Shale absorbs water and swells into the wellbore
- ▶ Reaction is worsened with time

Warning signs

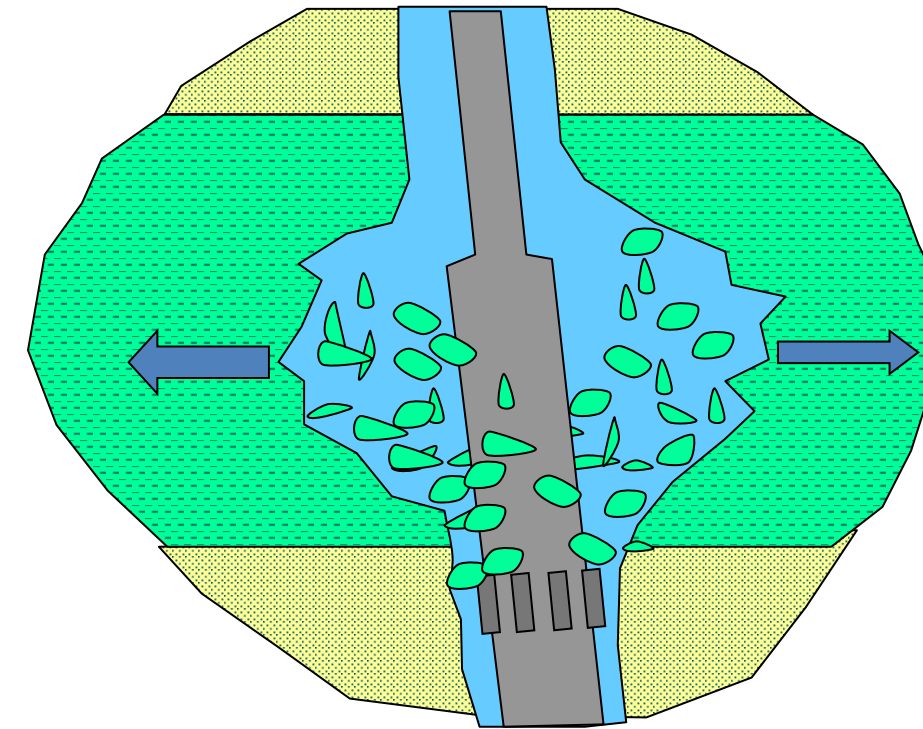
- ▶ Funnel viscosity, PV, YP, CEC increase
- ▶ Torque, drag, and pressure increases
- ▶ Clay balls, soft mushy cuttings at shaker
- ▶ Overpulls and swabbing
- ▶ BHA balling (mud rings)



Reactive Shale

Preventive Actions

- ✓ Use inhibited mud (high salt, polymers, OBM)
- ✓ Maintain mud properties
- ✓ Plan wiper trips
- ✓ Minimize open hole exposure time
- ✓ Keep BHA to minimum



Pressured Shale

Cause

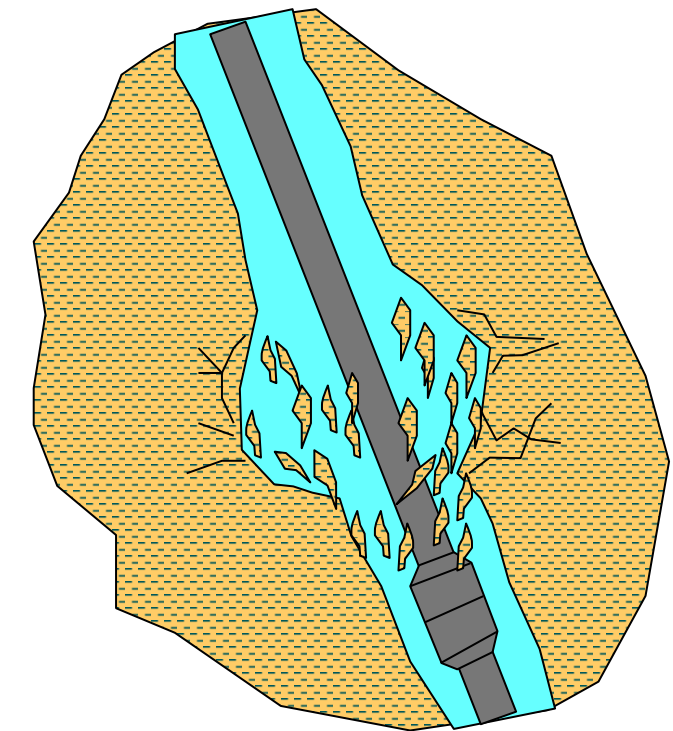
- ▶ Drilling pressured shale with insufficient MW
- ▶ Shale fractures and caves into the wellbore

Warning Signs

- ▶ Mud logger trends indicate increasing pore pressure
- ▶ ROP increasing when drilled
- ▶ Torque increase and drag on connections
- ▶ Hole fill - splintery shale over shakers
- ▶ Restricted circulation

Preventive Actions

- ✓ Increase MW as pore pressure increases
- ✓ Minimize surge/swab pressure
- ✓ Minimize exposure time
- ✓ Increase pump rate to lift cavings and cuttings out of well
- ✓ Regular wiper trips can help
- ✓ Ream on connections
- ✓ Monitor shakers and pump pressure trends



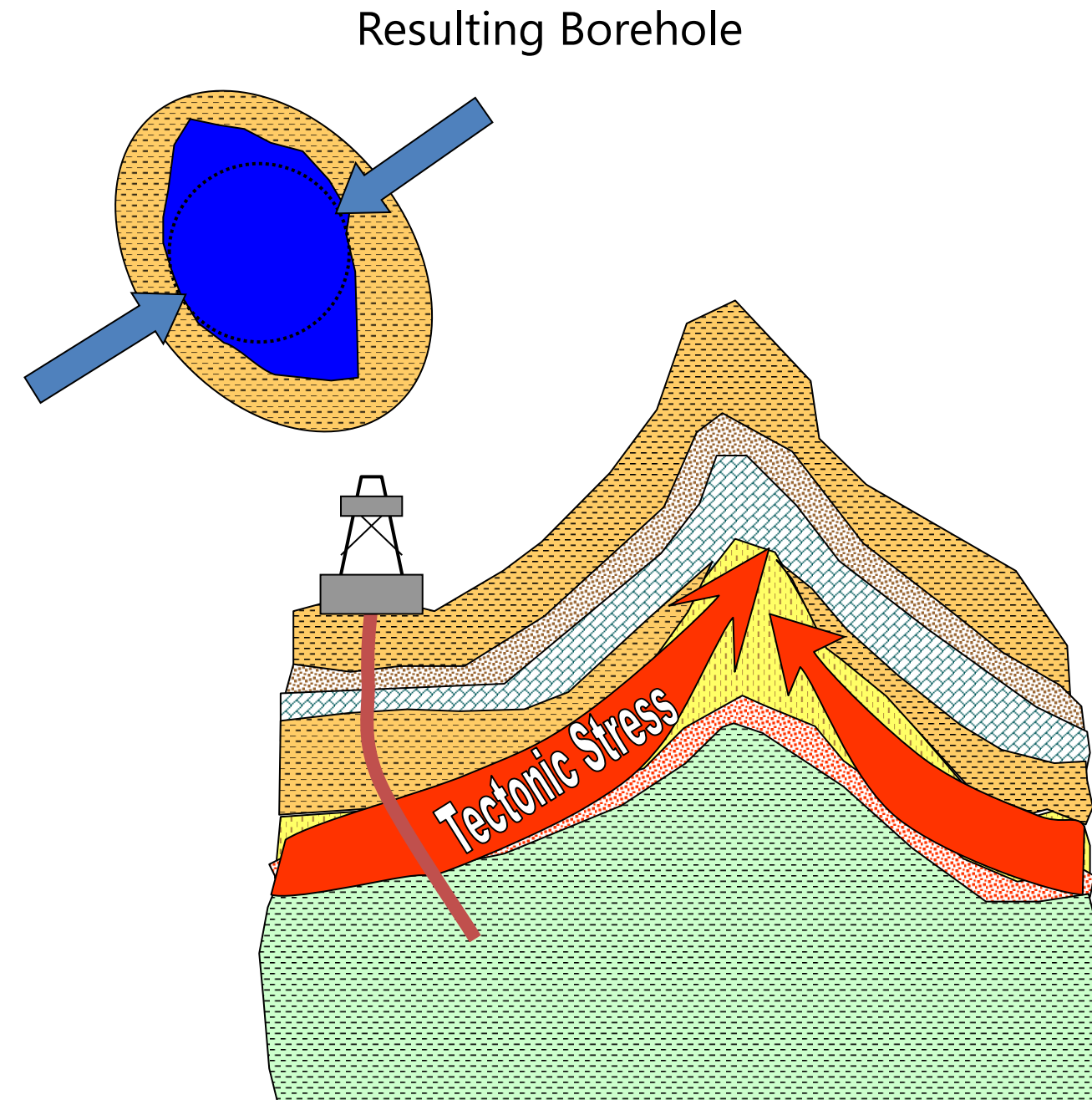
Tectonic Stress

Cause

- ▶ Naturally occurring lateral forces in the formations
- ▶ Stressed shale fractures and falls into the wellbore causing an elliptical shaped bore

Warning Signs

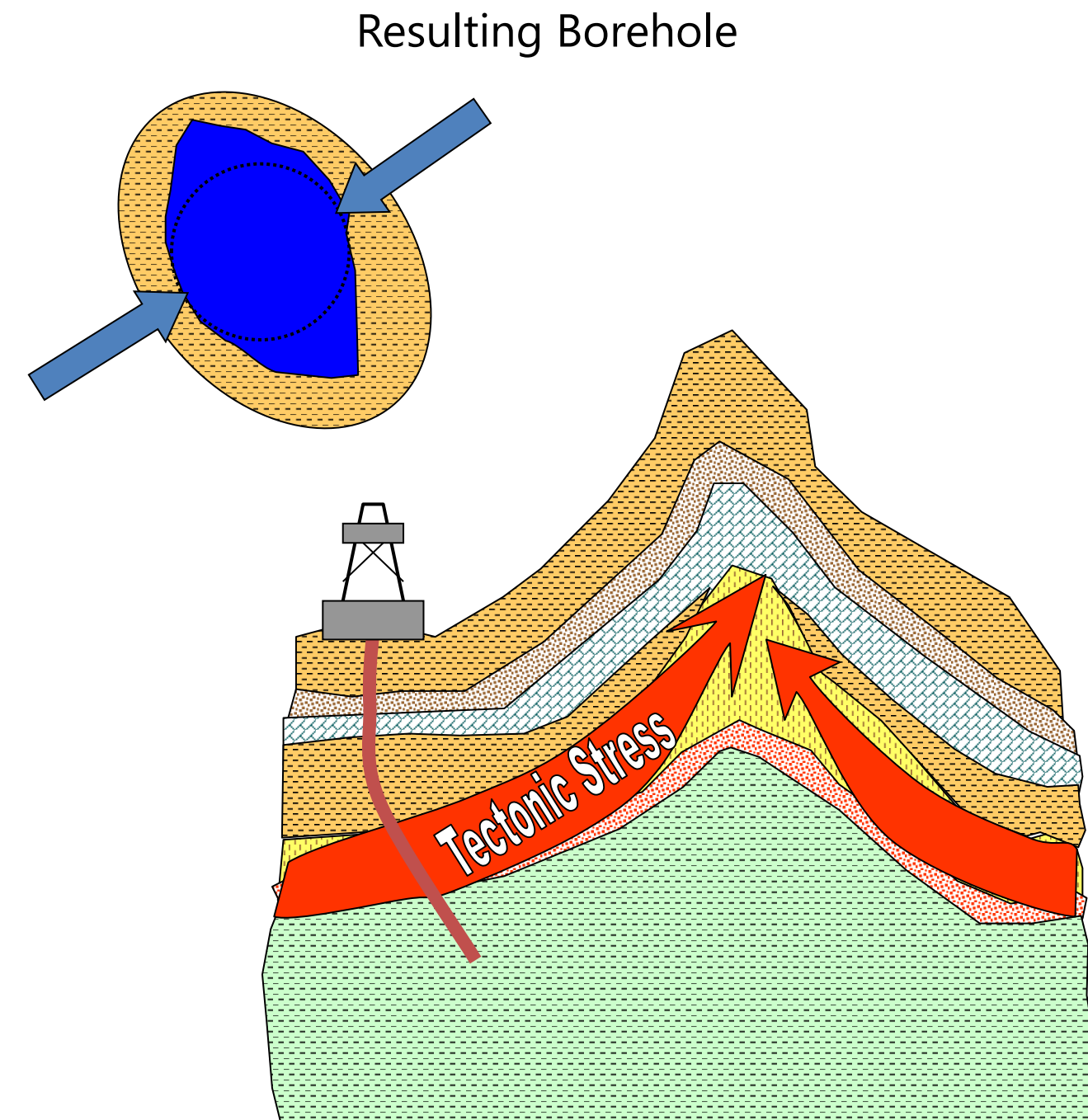
- ▶ Mountainous locations
- ▶ Active tectonic plate movement
- ▶ Salt domes
- ▶ High and erratic torque/drag
- ▶ Blocky shale caving



Tectonic Stress

Preventive Actions

- ✓ Increase MW if possible
- ✓ Circulate high density sweeps
- ✓ Minimize ECD/surge/swab
- ✓ Minimize hole exposure time



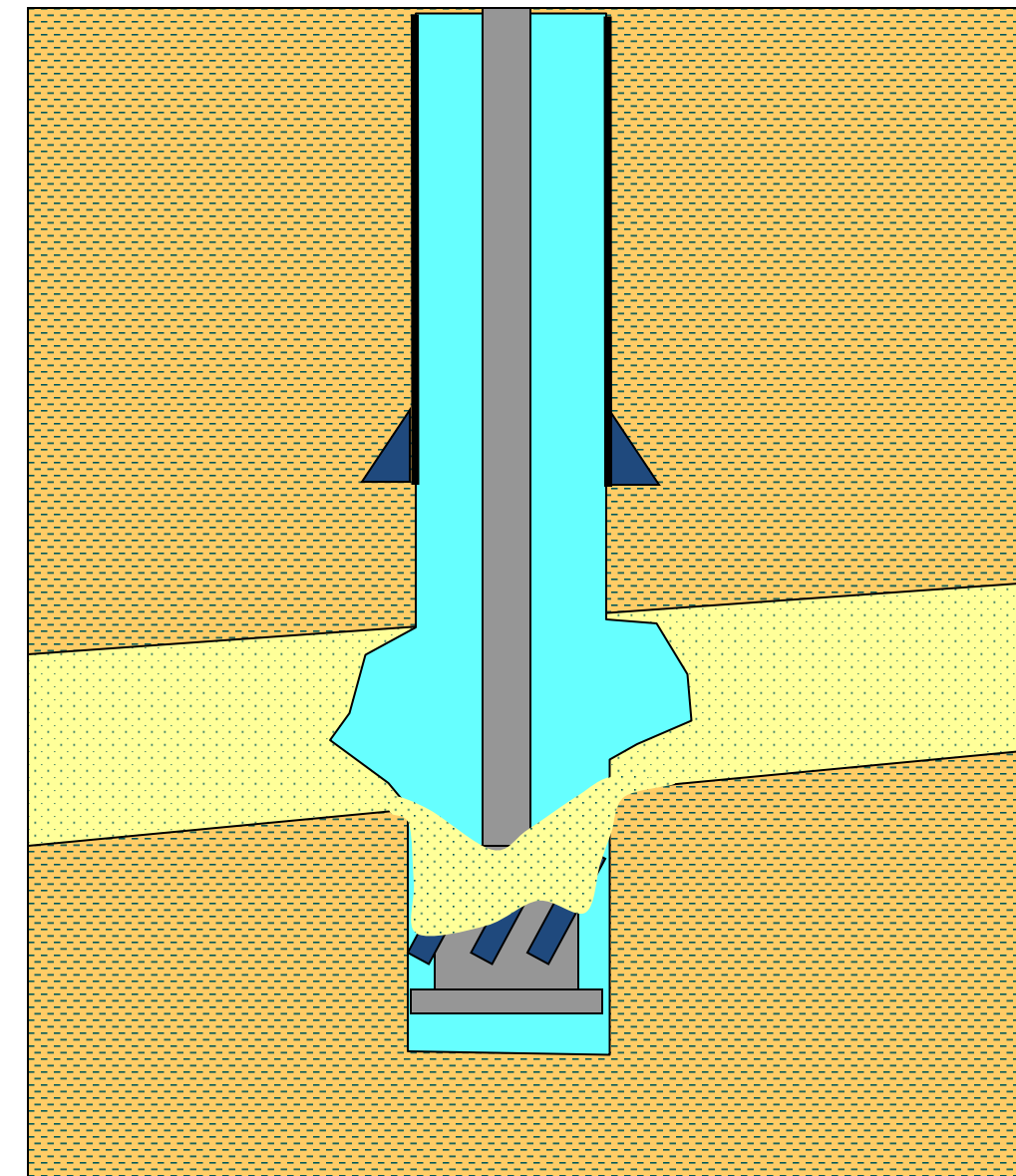
Unconsolidated Formations

Cause

- ▶ Poorly cemented sands or pea gravel falls into the wellbore and packs off the drill string
- ▶ Usually occurs at shallow depths
- ▶ Little or no filter cake and insufficient hydrostatic support

Warning Signs

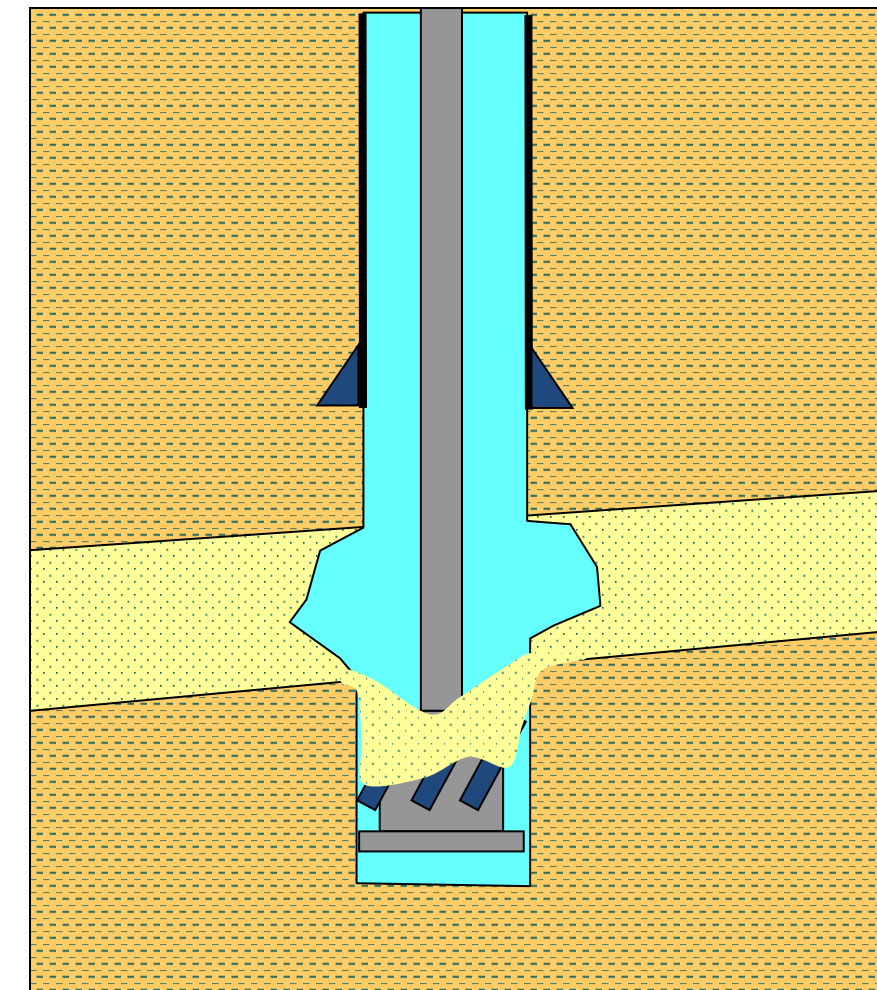
- ▶ Seepage losses
- ▶ Increase in torque/drag
- ▶ Pump pressure fluctuations
- ▶ Hole fill
- ▶ Shaker and de-sander overload
- ▶ Large overpulls at connections



Unconsolidated Formations

Preventive Actions

- ✓ Identify sand or porous formations
- ✓ Maintain a high gel mud in the slug tank
- ✓ Monitor pump pressure and drill cuttings
- ✓ Use high viscosity sweeps (often)
- ✓ Spot gel pill prior to POOH
- ✓ Control fluid loss if applicable
- ✓ Case off known problem zones
- ✓ Control ROP
- ✓ Keep flow rate to minimum
- ✓ Ensure low permeability filter cake
- ✓ Pick off bottom and circulate



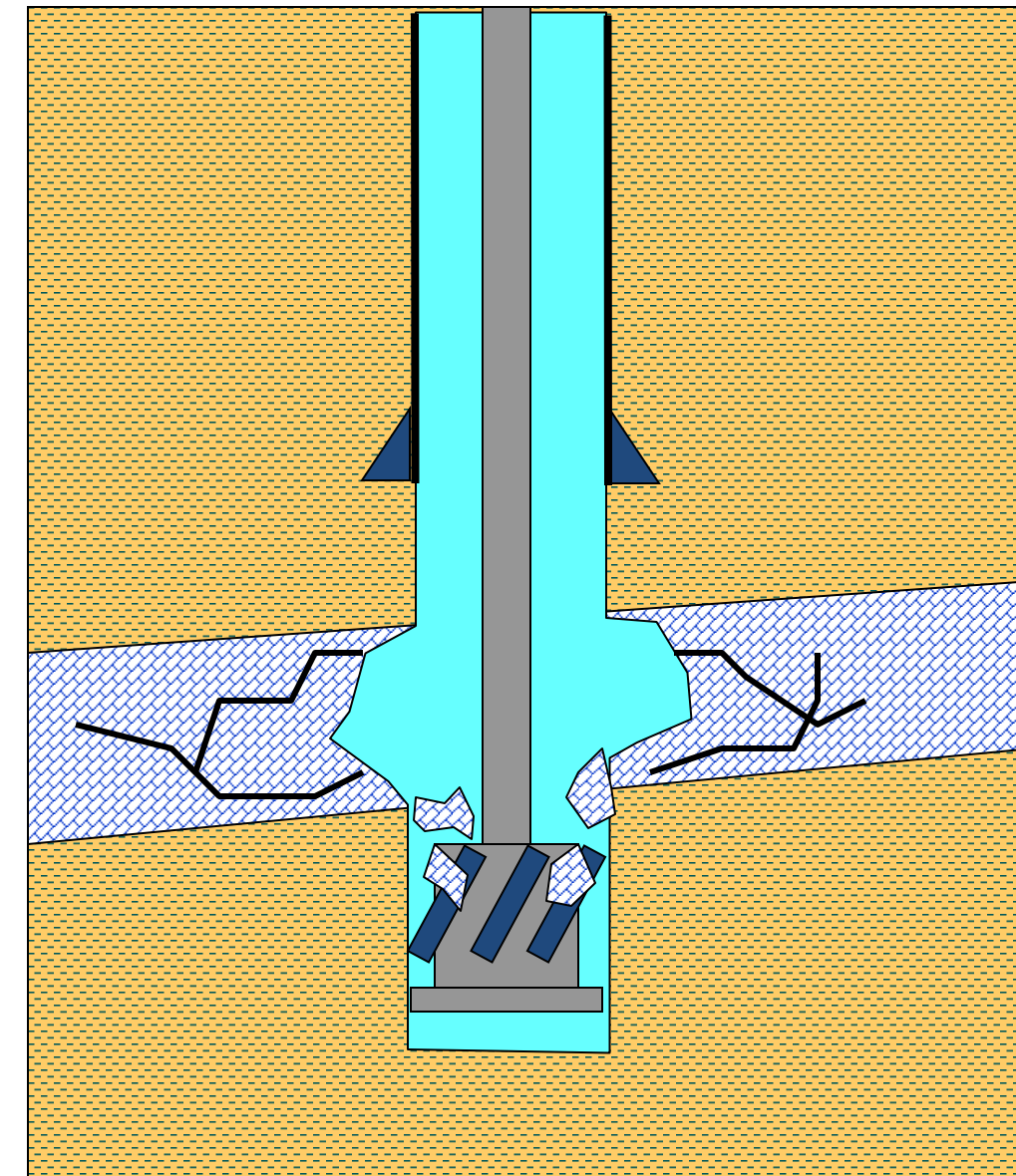
Fractured/Faulted Formations

Causes

- ▶ Naturally occurring fractures
- ▶ Pieces of formation fall into the wellbore and jam the drill string

Warning Signs

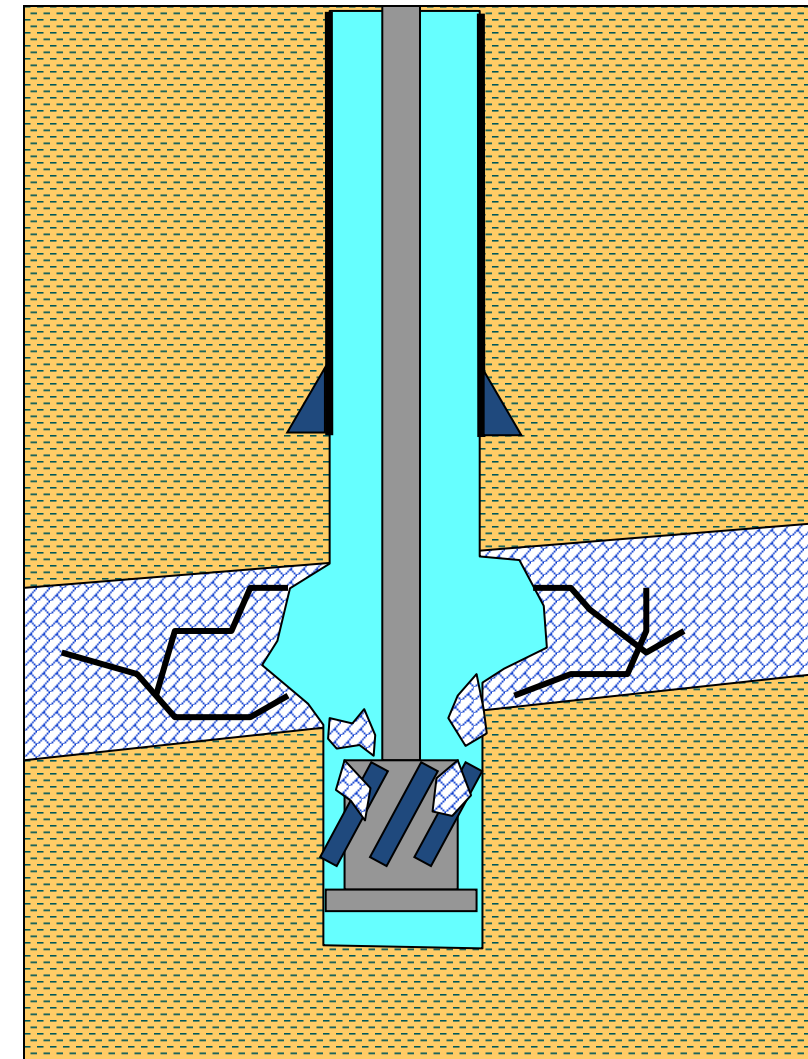
- ▶ Prognosed limestone or fractured shale and/or faults
- ▶ Mud logger sample interpretation
- ▶ Blocky caving at shaker
- ▶ Hole fill on connections and trips
- ▶ Erratic and high torque and drag
- ▶ Pump pressure fluctuations



Fractured/Faulted Formations

Preventive Actions

- ✓ Check hole condition constantly
- ✓ Control drill
- ✓ Circulate hole clean before drilling ahead
- ✓ Minimize mud seepage loss
- ✓ Minimize surge/swab pressures
- ✓ Design BHA to minimize risk of ledge formation
- ✓ Trip slowly when BHA in suspected zone
- ✓ Start/stop drill string slowly to avoid pressure surges to well bore
- ✓ Minimize drill string vibration
- ✓ Ream fractured zones cautiously



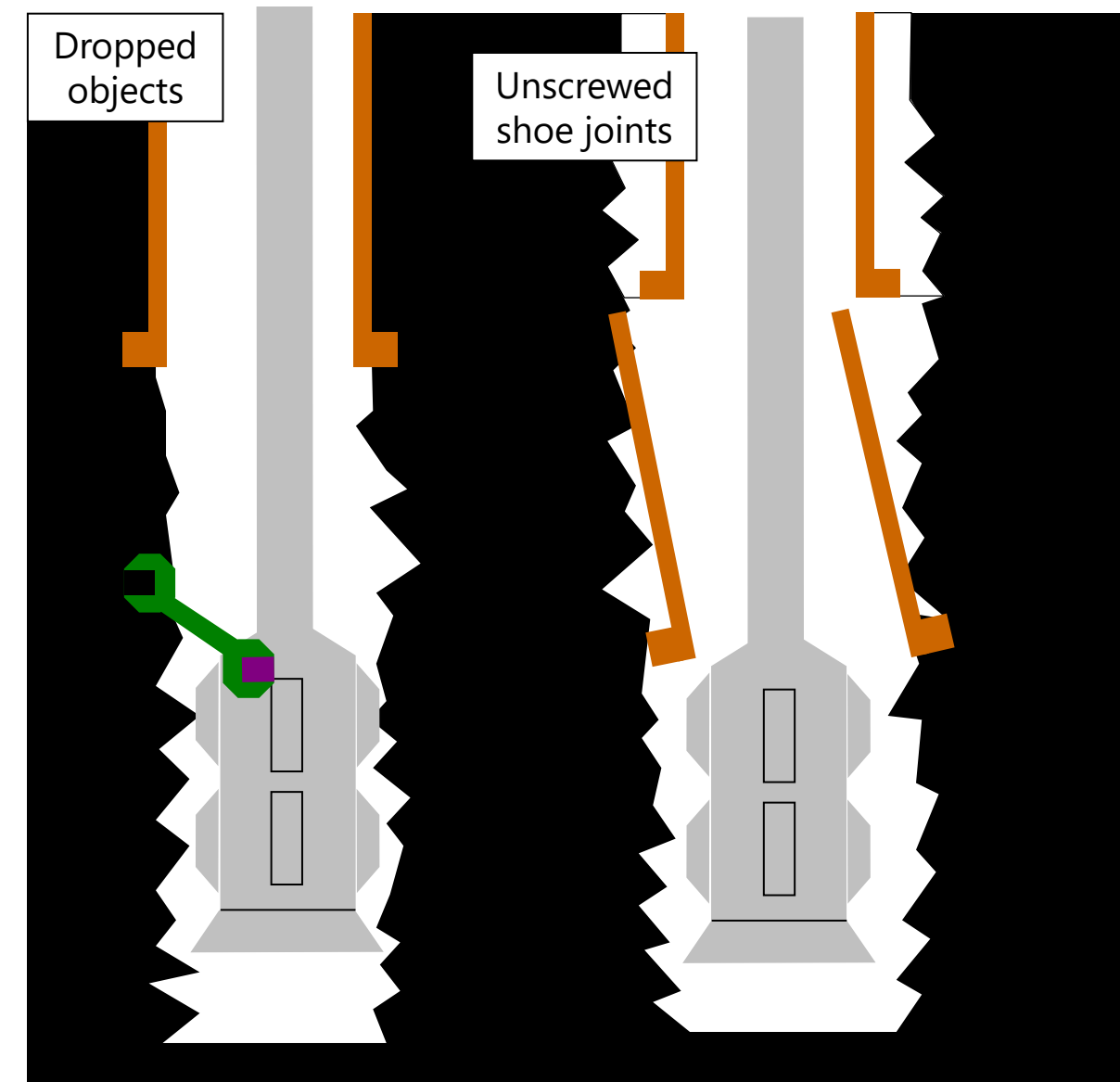
Junk in the Hole

Cause

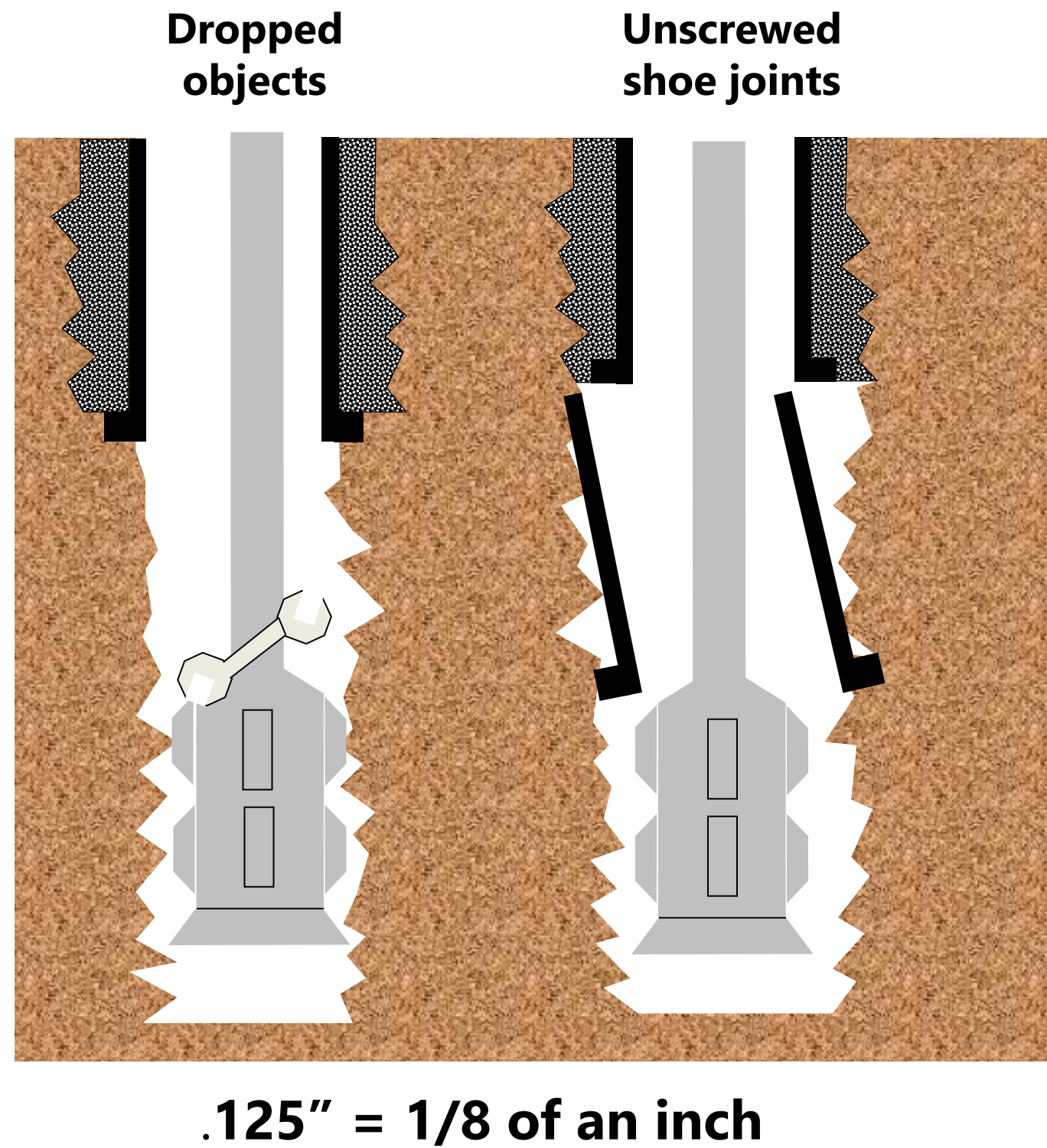
- ▶ Debris that has fallen into the hole and jams the drill string
- ▶ Poor housekeeping on the rig floor
- ▶ Hole cover not being installed
- ▶ Downhole equipment failure

Warning Signs:

- ▶ Missing hand tools/Equipment
- ▶ Metal shavings at shaker
- ▶ Sudden erratic torque
- ▶ Inability to make hole
- ▶ Circulation unrestricted



Junk in the Hole



Always keep hole covered when out of hole and practice good housekeeping with tong dies and other parts likely to fall into the wellbore.

Even a PDC cutter is large enough to cause a stuck drill string.

Consider the clearance between 9 5/8", 47 lb/ft casing and an 8 1/2" stabilizer. The drift Diameter = 8.525".

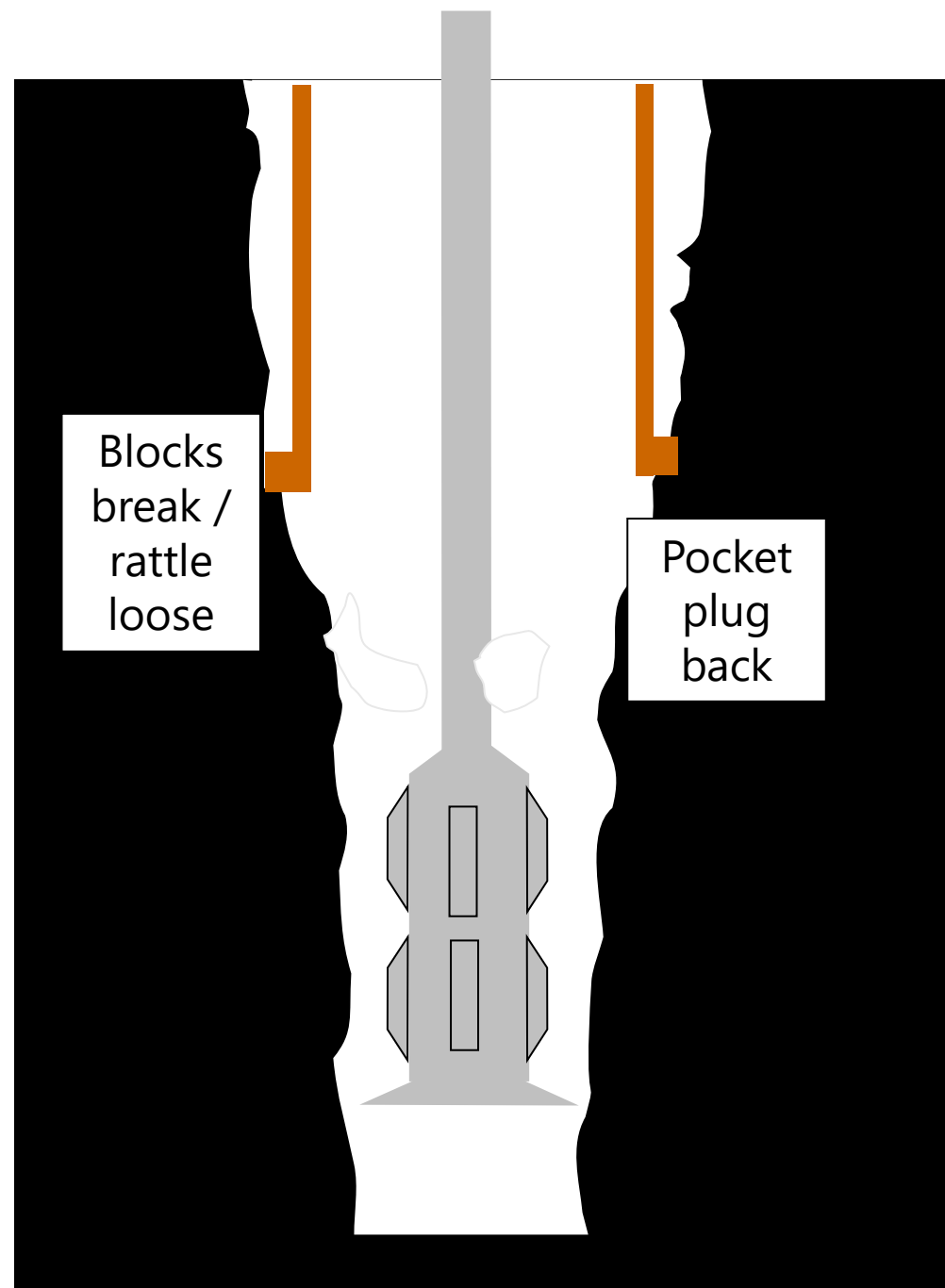
$$\frac{(8.525" \text{ Csg ID} - 8.5" \text{ Stabilizer OD})}{2} = .125"$$

Junk in the Hole

Preventive Actions

- ✓ Keep the hole covered
- ✓ Practice good housekeeping to keep objects from falling in the hole
- ✓ Inspect downhole equipment before it is run in/out of hole
- ✓ Inspect slip and tong dies regularly
- ✓ Install drill strip wiper rubber

Cement Blocks



Causes:

- ▶ Hard cement becomes unstable around the casing shoe, open hole squeeze plugs and kick-off plugs
- ▶ Blocks of cement fall into the wellbore from casing rat holes or cement plugs jamming the drill string

Warnings:

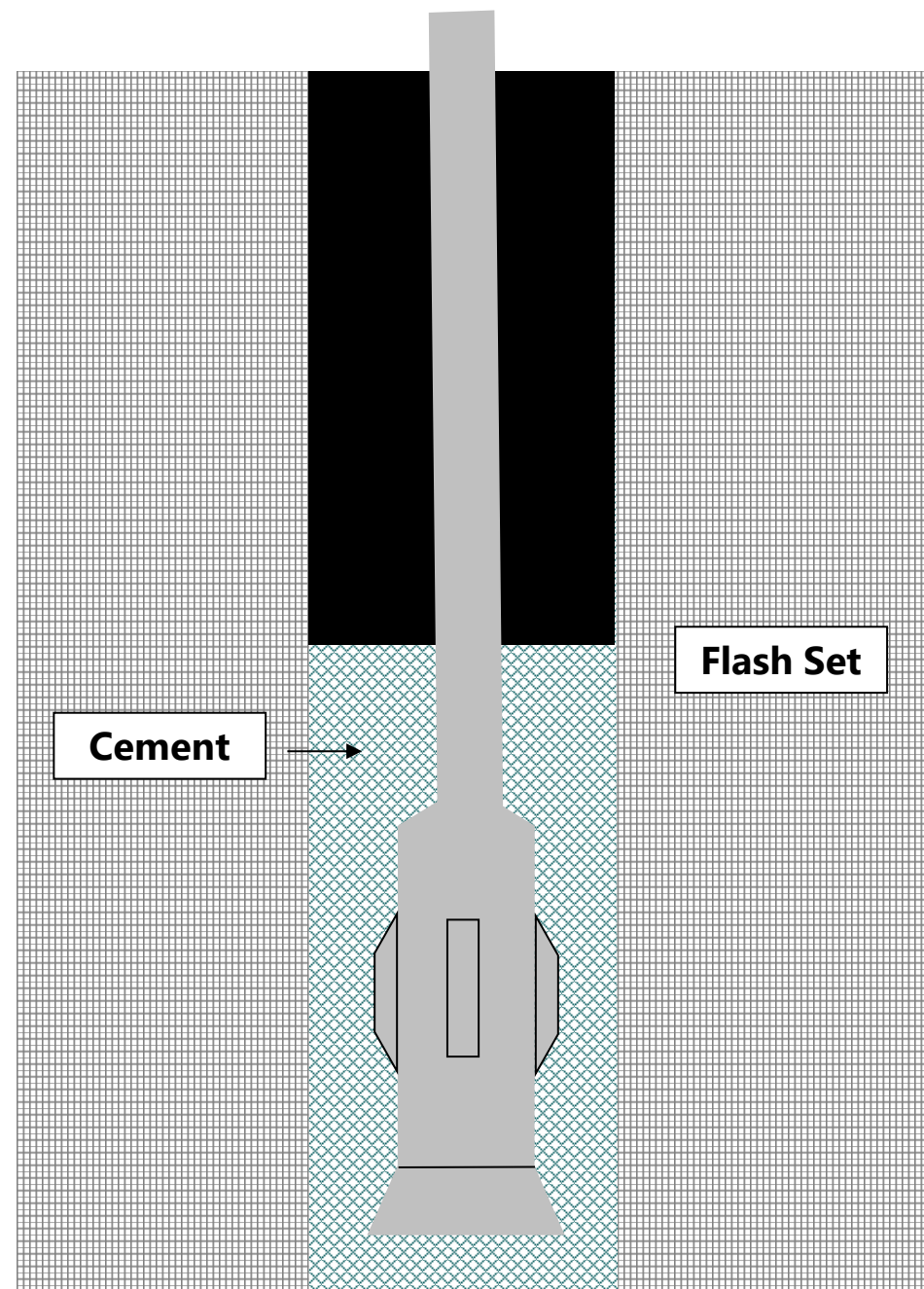
- ▶ Circulation unrestricted
- ▶ Cement fragments
- ▶ Rotation and downward movement may be possible
- ▶ Erratic torque

Cement Blocks

Preventive Actions

- ✓ Minimize rat hole below casing shoe (1-2 meters)
- ✓ Ensure good cement mixing and displacement
- ✓ Trip carefully through rat hole and past cement plugs
- ✓ Consider using fibers in cement to reduce chance of blocks falling

Green Cement



Causes:

- ▶ Drill string becomes planted in soft or “green” cement. This soft cement is often not easily detected when running in and will “flash set” when pressure is applied.
- ▶ Severely contaminated mud can stick the pipe in a similar manner.

Warnings:

- ▶ Increase in pump pressure leading to inability to circulate
- ▶ Loss of string weight
- ▶ Sudden decrease in torque
- ▶ Green cement in mud returns, discoloration of mud

Green Cement

Preventive Actions

- ✓ Don't try to drill out before cement is set
- ✓ Know where top of cement (TOC) is. If channeling has occurred or too much excess pumped, TOC will be higher.
- ✓ Start circulation above TOC
- ✓ Drill out cement with high circulation rate and low weight bit on rate
- ✓ If in doubt, circulate bottoms-up and check shakers before continuing

Effective Length Concept

	Section Length Factor			
Hole Angle	17.5	12.25	8.5	6
0-10	1.5	1.3	1.3	1.3
10-30	1.7	1.4	1.4	1.4
30-60	2.5	1.8	1.6	1.5
60-90	3.0	2.0	1.7	1.6

- Divide well into sections per hole angle
- Multiply actual length by section factor (above) to obtain effective length

$$\text{Volume required} = \frac{\text{Total Effective Length} * \text{Bottoms Up Volume}}{\text{Measured depth}}$$

Effective Length Example

Well RKB to 2,000ft vertical 2,000 to 4,000ft build to 60° , 4,000 to 6,000 ft build to 90° , 6,000 to 12,000 ft horizontal. Annular volume = 5,000 strokes

$$2,000 \times 1.3 = 2,600 \text{ ft}$$

$$2,000 \times 1.8 = 3,600 \text{ ft}$$

$$2,000 \times 2 = 4,000 \text{ ft}$$

$$6,000 \times 2 = \underline{12,000 \text{ ft}}$$

$$= 22,200 \text{ ft}$$

$$\frac{22,200 \times 5,000}{12,000} = 9,250 \text{ strokes}$$

$$12,000$$

Stuck Pipe Prevention

Unconsolidated Formations

- Build low permeability filter cake
- Do not exceed flow rate required to clean the hole
- Avoid rotating bits or stabilizers near the problem area
- Trip past problem area carefully
- Clean out hole fill prior to drilling ahead
- Consider soaking viscous pills into formation

Faulted Formation:

- Check the hole condition constantly
- Wash and ream constantly
- Restrict tripping speed
- Limit circulating pressures

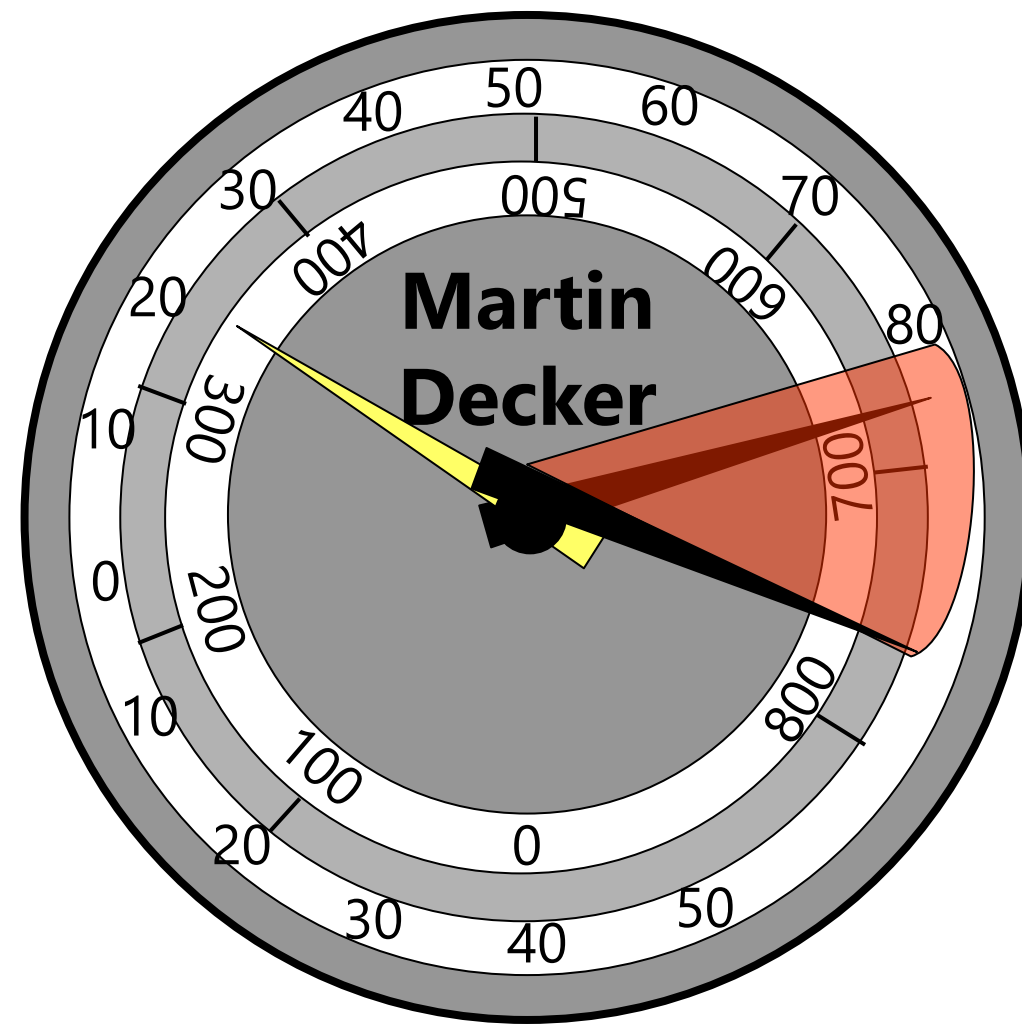
Reactive Formations:

- Drill and case off quickly
- Keep an inhibitive mud in specification
- Monitor the MBT closely
- Minimize the BHA length
- Wipe the hole regularly
- Utilize dump and dilute procedures
- Consider using OBM

Rock Stress:

- Keep hole clean
- Monitor pore pressure
- Increase MW as soon as possible

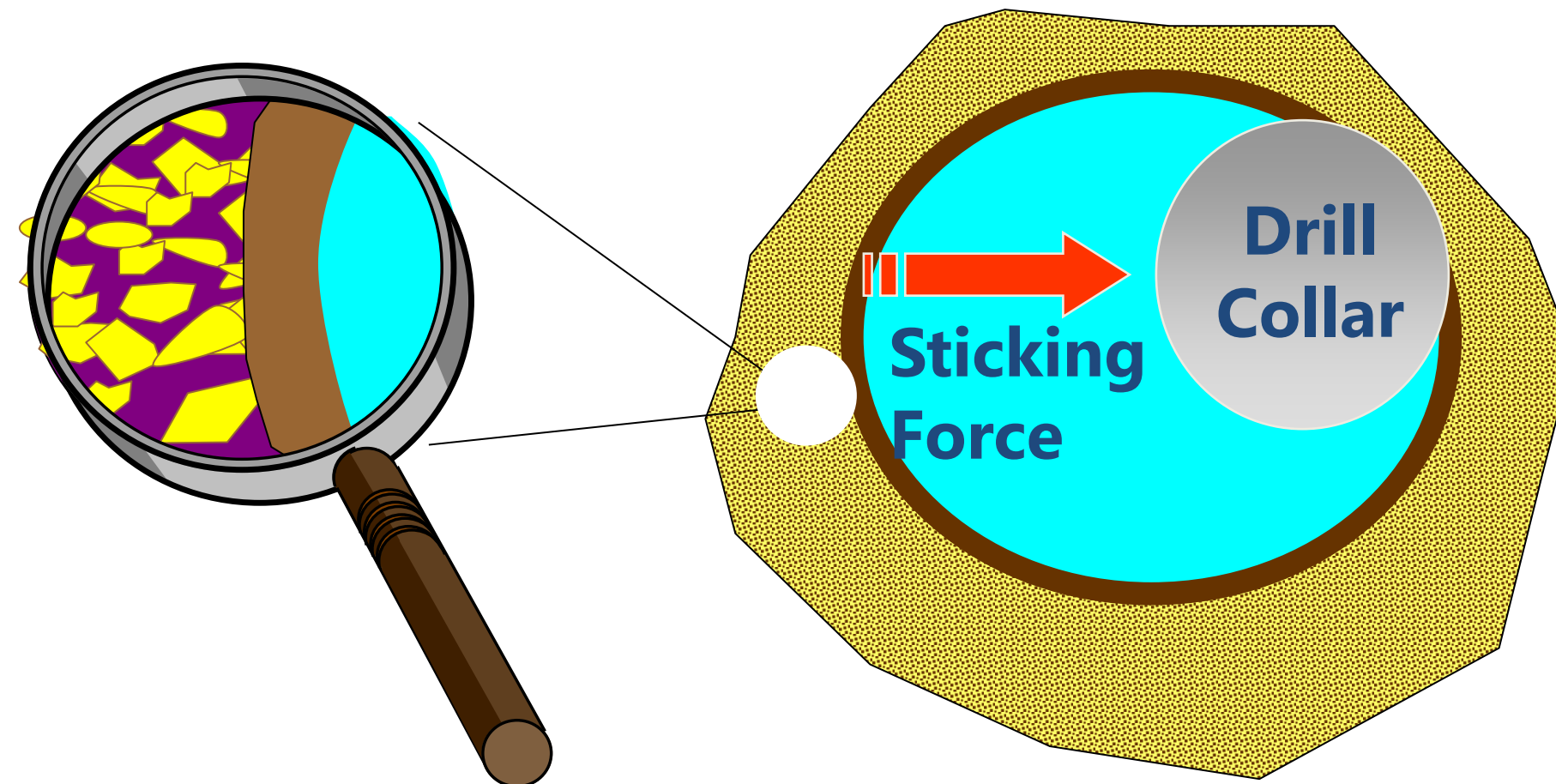
Tight Hole Conditions



Differential Pressure

The pore pressure in a permeable formation is much less than the hydrostatic pressure of the mud column.

This pressure differential creates a force that holds the bottom hole assembly (BHA) against the side of the wellbore.



Hole Pack-off/Bridge

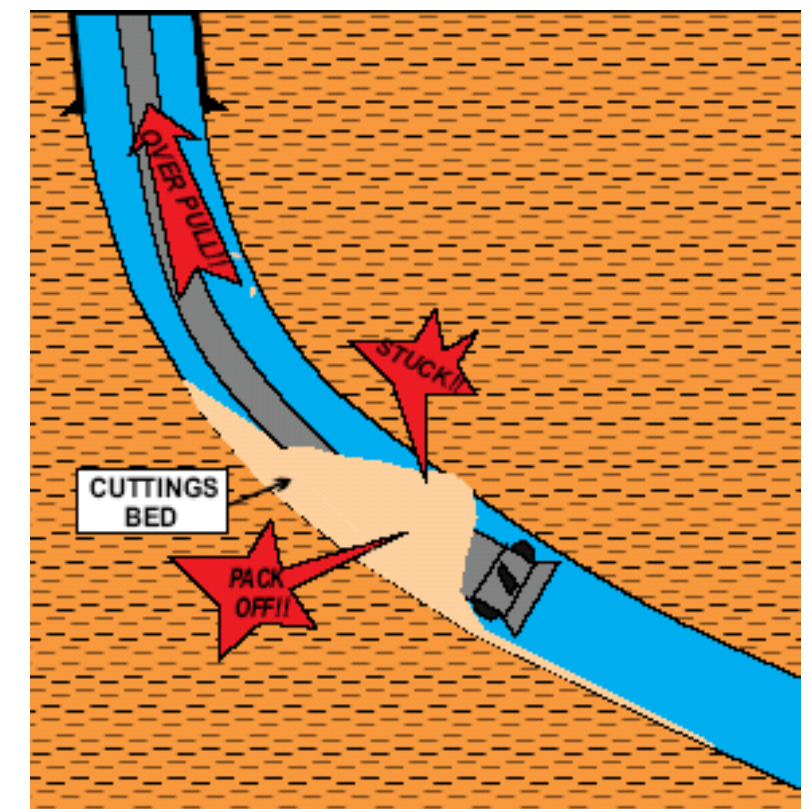
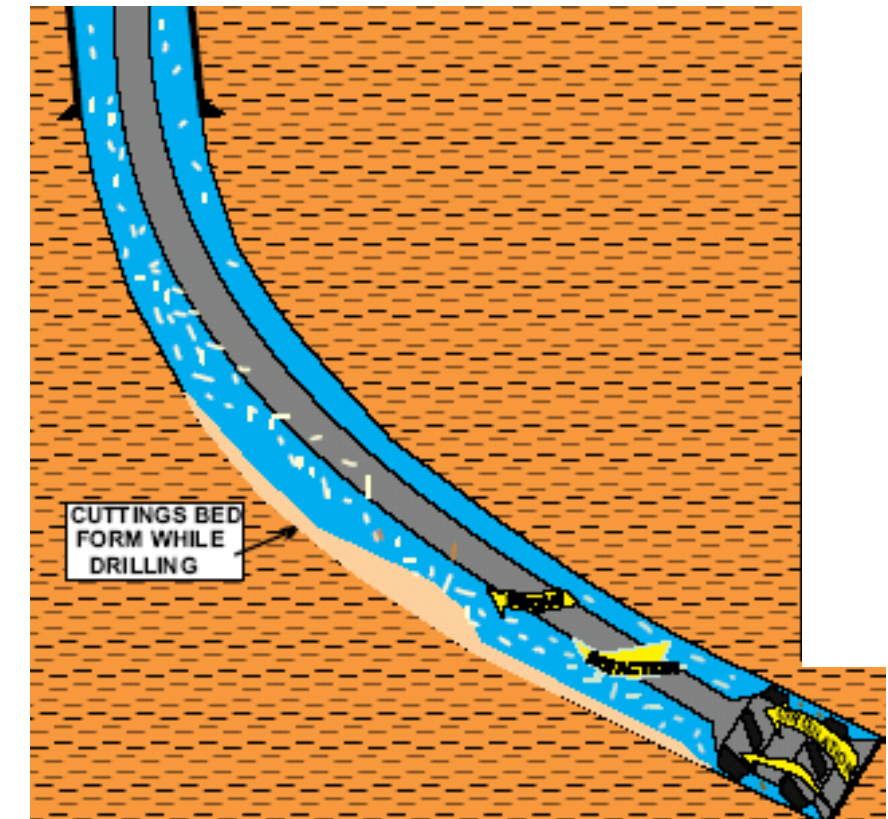
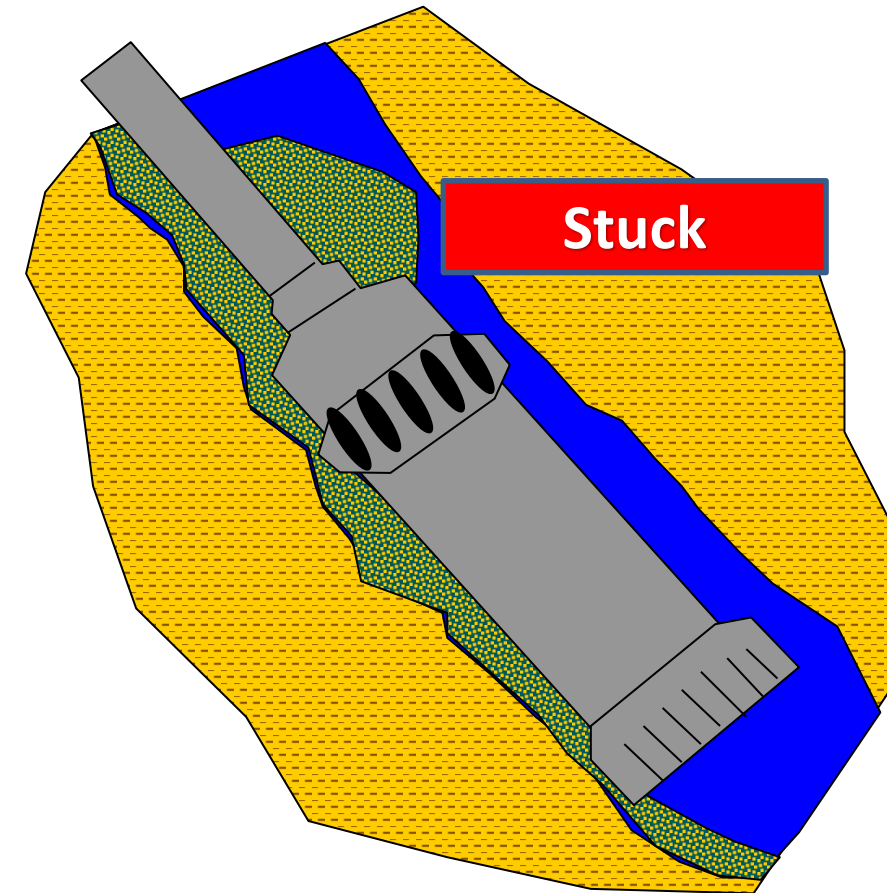
Wellbore totally or partially plugged with solids can result in stuck pipe.

- Wellbore stability problems can result in hole cavings.
- Solids typically are drilled cuttings in the form of cuttings beds in high angle holes.
- Junk may also cause bridging of the hole.
- Deviated wells

Good cleaning practice:

- Clean the hole as fast as drilled
- Ensure mud is within specification
- Circulate clean prior to tripping
- Reciprocate and rotate the pipe continuously
- Plan and perform wiper trips as hole conditions dictate.
- Monitor the shakers
- Keep all circulating and SRE in good working order

Poor hole cleaning



Well-Bore Geometry

Sticking occurs, when changes in the shape of the well bore cause the movement of the string to be impeded.

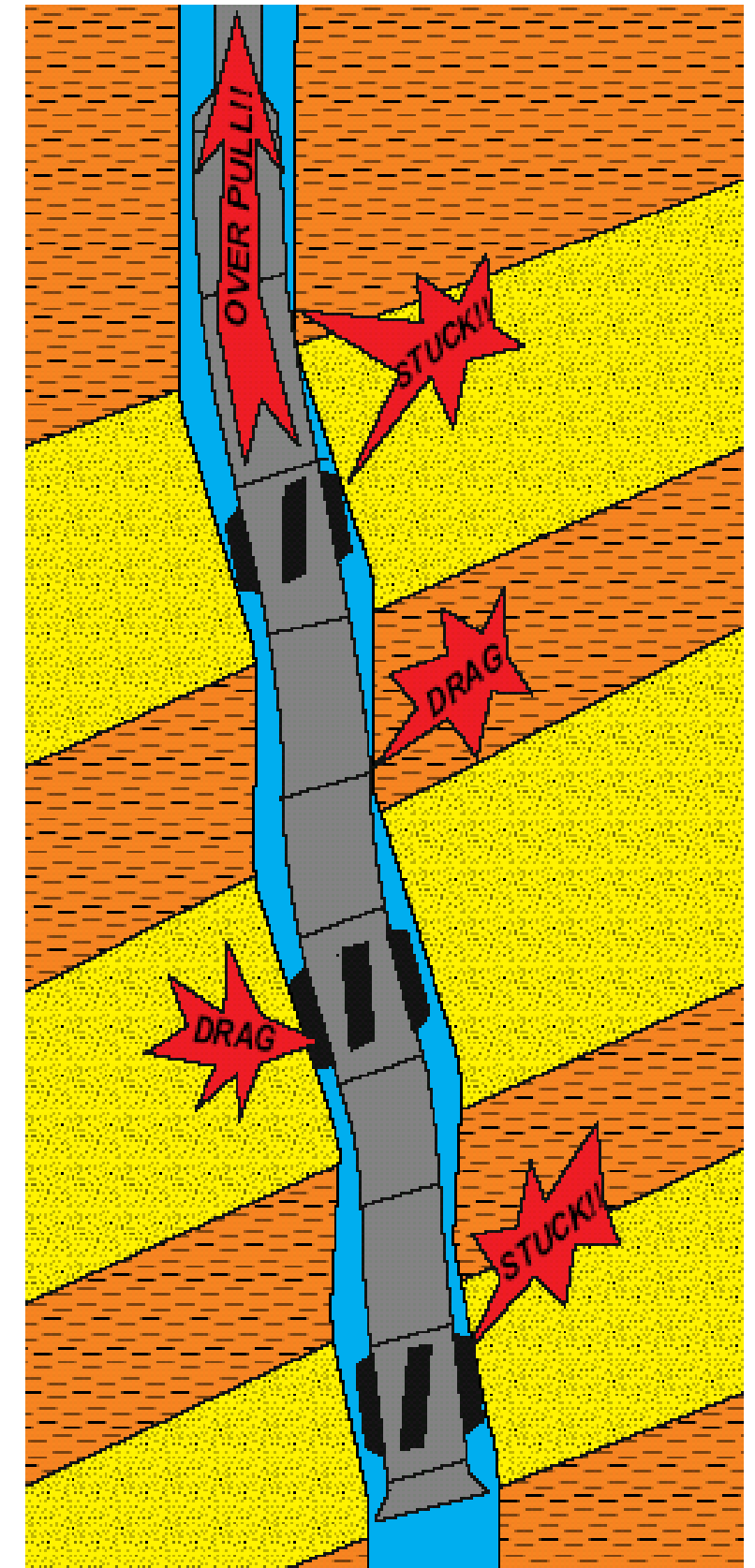
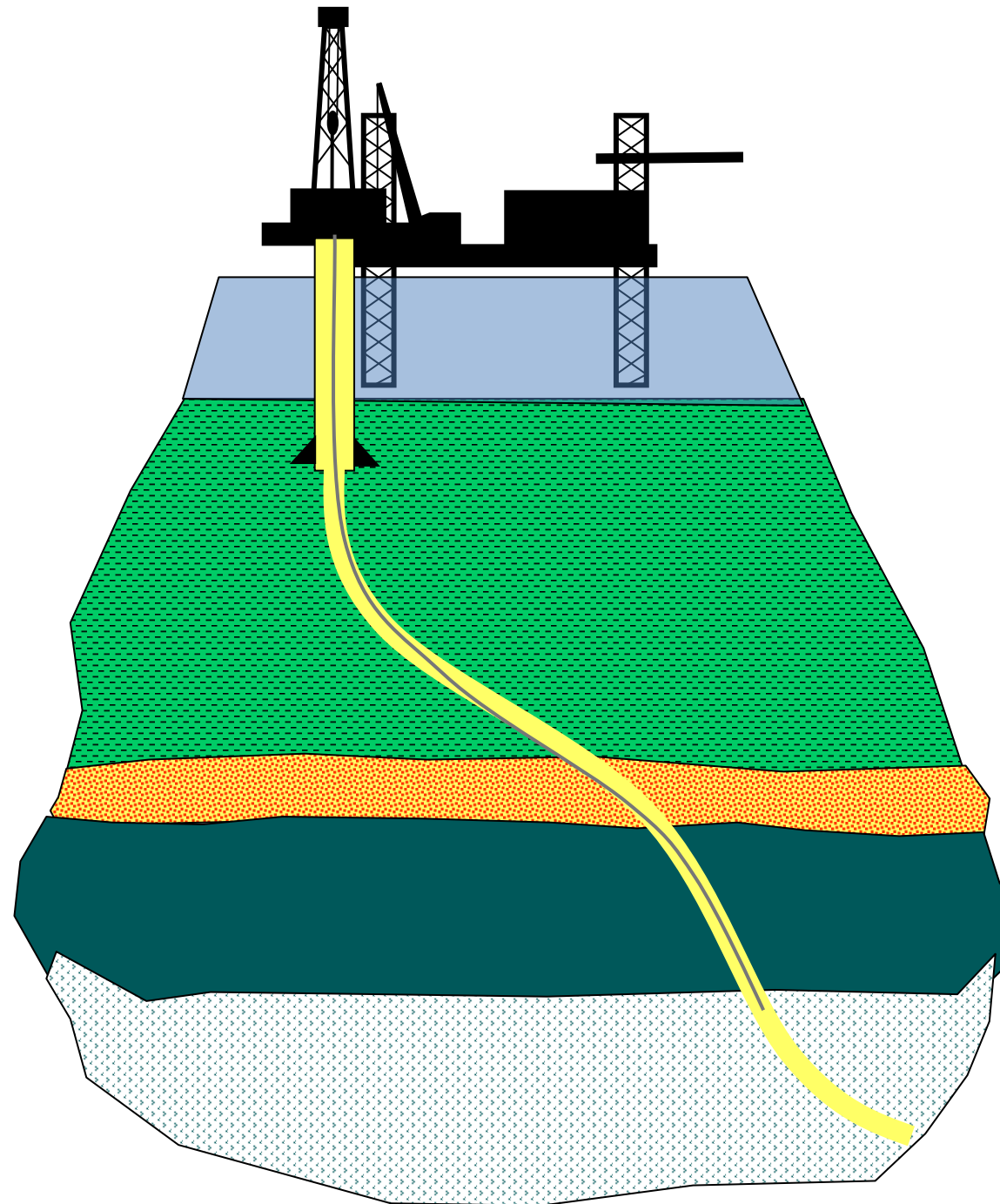
If the make up of the BHA is changed, a similar effect can occur.

Sticking can occur when the string is moving up or down.

Ledges, keyseats, doglegs, squeezing salts, and breakouts are means by which hole shape can change.

Prevention

- Minimize any dogleg severity
- Ream after any BHA change



Simple Tips



Communicate frequently

- Make team members aware about trends and changes
- Get good handover information



Plan ahead

- Discuss stuck pipe hazards in pre-tour meetings
- Be prepared to act quickly



Listen to the hole

- Watch the torque, pump pressure and drag trends
- Monitor the shape and amount of cuttings



Keep the mud in good shape

- Keep mud within spec – density, viscosity, low gravity solids are key



Keep the pipe moving

- Rotate or work the pipe whenever you can in open hole.



Clean the hole as fast as you can drill it

- Over-gauge and deviated holes require faster circulation to prevent cuttings build-up
- Make sure the hole is clean before tripping



Take action early

- Most stuck pipe incidents get worse with time. Take the right action, fast

”

Fundamentals of Teamwork and Team

What is Teamwork?

Teamwork is the efforts of two or more people working together toward a **common goal.**

Teamwork improves identification and resolution of problems.

What does it take to be a successful Stuck Pipe Observation Team?

- ✓ Clear Goals
- ✓ Clear Communication
- ✓ Listen
- ✓ Ask questions

Prevention Team

Who is the Stuck Pipe Prevention Team?

- ▶ Driller
- ▶ Floorhand
- ▶ Derrickman
- ▶ Shakerman
- ▶ Mud Engineer
- ▶ Mudlogger
- ▶ Toolpusher
- ▶ Company Man
- ▶ Directional Driller

Driller

ROLE:

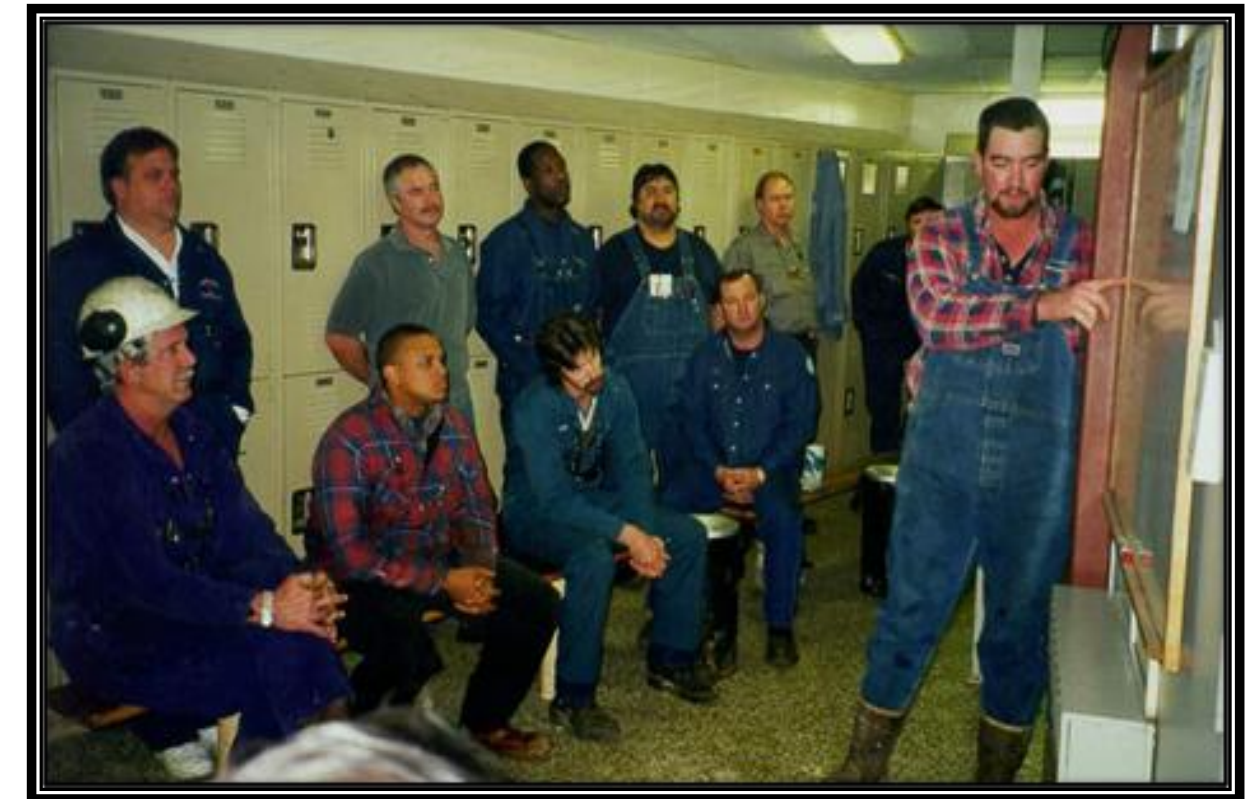
- Recognize and interpret early warning signs and take appropriate action.

RESPONSIBILITIES:

- Communicates with the crew and ensure that information is relayed properly
- Coordinates efforts of crew to work safely and efficiently
- Limits overpull and set down
- Thorough handover notes

Understands and controls drilling parameters

- ✓ Weight on bit
- ✓ Torque
- ✓ Pump pressure



Tight hole problems are usually detected by changes in these important parameters! It is important that the Driller understand the relationship between these functions and stuck pipe mechanisms.

Floorman

ROLE:

- Prevent delays by making sure all tools are available and rig floor is clean and safe

RESPONSIBILITIES:

- Junk prevention
- Tong maintenance
- Draw works maintenance
- Watch condition of string
- Look for slip and tong damage
- Trip and make connections quickly

“Driller’s sixth sense”



Information must be a two way flow!

Derrickman

ROLE:

- Monitor and maintain mud and solids control system

RESPONSIBILITIES:

- Recognize and report stuck pipe warning signs
- Recognize and report different types of cuttings
- Carry out basic mud checks
- Investigate unexpected changes in rheology



Shakerman

ROLE:

- Monitor the mud and solids control system

RESPONSIBILITIES:

- Recognize common shaker problems
- Communicate with Driller & Mud Engineer
- Report and log MWs in and out
- Handle gas cut mud
- Maintain mud cleaning equipment



Changes in the cuttings indicate changes down hole

Mud Engineer

ROLE:

- Keep drilling mud within specifications

RESPONSIBILITIES:

- Control mud to specs
- Recognize problems and suggest remedies
- Ensure adequate store of mud supplies
- Tests, advises and reports

Mud is the **KEY** to successful drilling.



Mud Engineer's Test Kit

Mud Logger

ROLES:

- Analyze the cuttings and gas to create downhole picture

RESPONSIBILITIES:

- Monitor changes and record trends
 - Hook loads
 - Pump pressure and strokes
 - Torque and RPM
- Identify formations and pressures
- Communicate with driller!!



Eyes and Ears of the Rig Team!
Meaningful trend analysis =
early indications

Toolpusher

ROLE:

- Responsible for the safe and smooth running of the rig, crew and equipment

RESPONSIBILITIES:

- Plan ahead – work with team to decide best way to carry out each job safely and effectively
- Understand the drilling program
- Have all tools and equipment available – right place, right time
- Work with crews to have a clear picture of what is happening downhole



Company Man

ROLE:

- Ensure the safety of personnel, the environment, the rig and the well.

RESPONSIBILITIES:

- Make sure that the objectives of the well are met.
- Discuss aspects of the drilling program with team members so they understand the importance and hazards associated with their actions.
- Evaluate and explain the relevance and interaction of all the rig data.



Directional Driller

ROLES:

- Control the path of the well

RESPONSIBILITIES:

- Monitor changes in well direction (inclination and azimuth)
- Instruct the Driller as to correct drilling parameters to correct directional problems
- Communicate with driller!!



Q & A Session 1





Irwin Wardiman

Technical Advisor

Drilling Fluids

Mud Properties

Measurement Device

Solid Control Equipment

Virtual Hydraulic

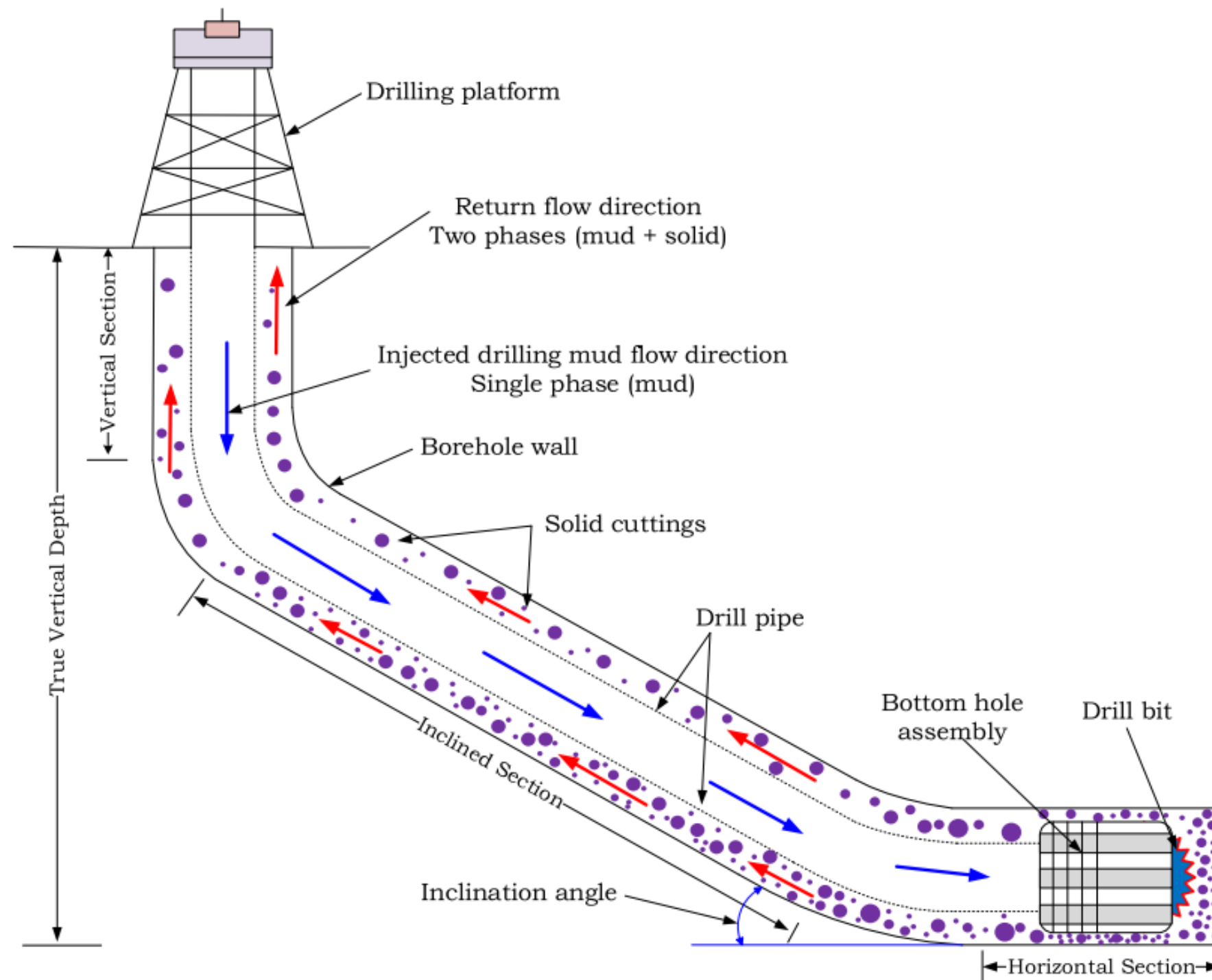
Equipment (Drill String, Drill Collars, Stabilizers, Reamers, Drilling Jars, Accelerator)

Differential Pressure

Hole Pack-off

Salt Drilling Practice

Drilling Operation



A stuck pipe in drilling operations is a costly condition where the drill string becomes immobilized, disjoint movement and rotation.

Type of Stuck Pipe:

- Differential sticking
- Mechanical sticking
- Hole collapse

(Huque et al, 2022)

Drilling Fluids

The main functions of drilling fluids include:

- **Pressure control** → providing [hydrostatic pressure](#)
- Prevent [formation fluids](#) from entering into the well bore
- **Transport cuttings, hole cleaning,** and **suspending** the drill cuttings while drilling is paused and when the drilling assembly is brought in and out of the hole
- **Wellbore stability**/wall support → prevent collapse or caving
- Keeping equipment **cool, lubricated,** and **clean** during drilling
- **Transfer hydraulic energy** while drilling
- **Logging medium** (LWD, MWD, and or WL)

[Liquid drilling fluid](#) is often called **drilling mud**. The main categories of drilling fluids are :

1. [Water-based muds \(which can be dispersed and non-dispersed\)](#),
2. [Non-aqueous muds, usually called oil-based mud or synthetic-based mud](#).

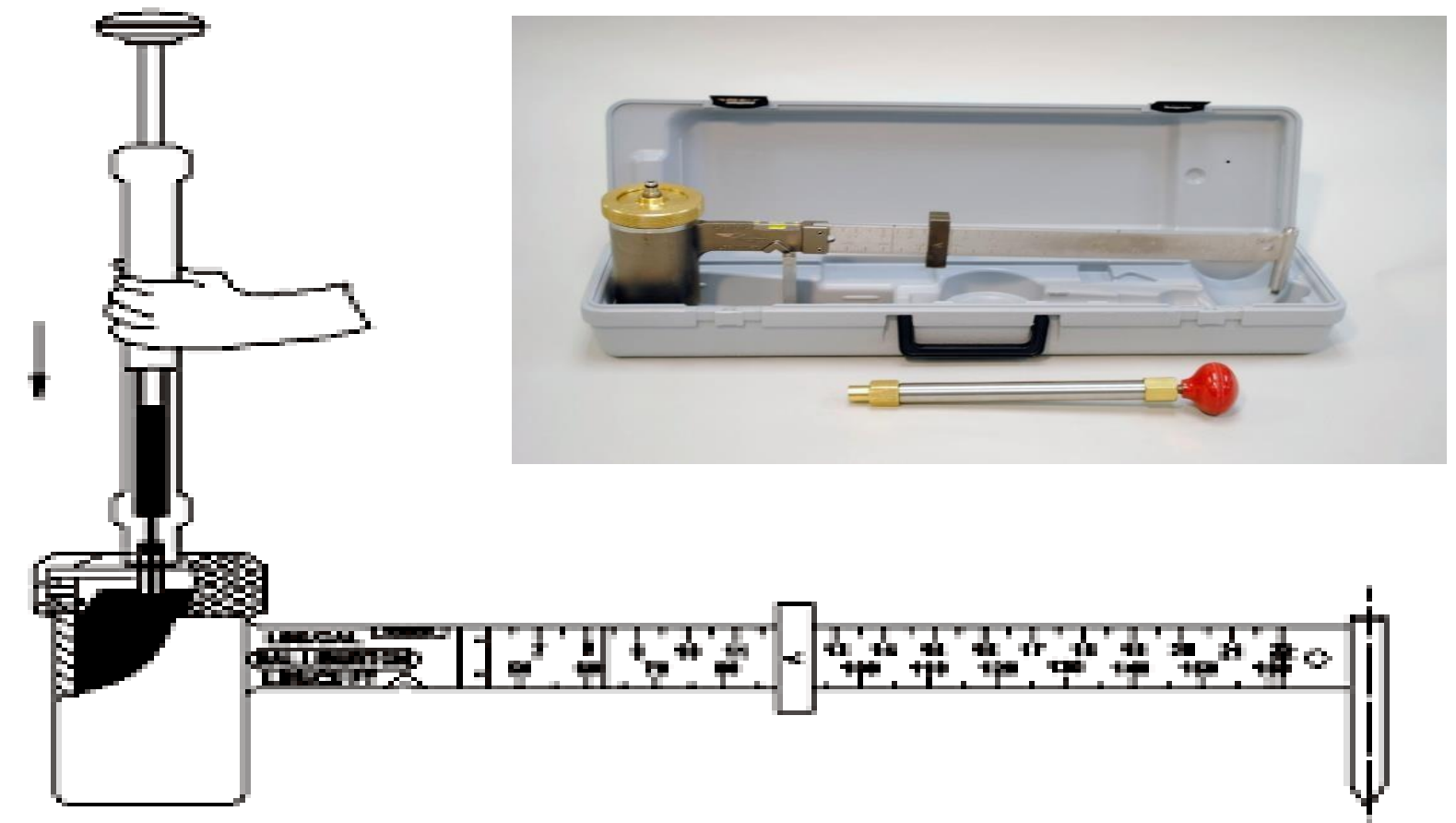
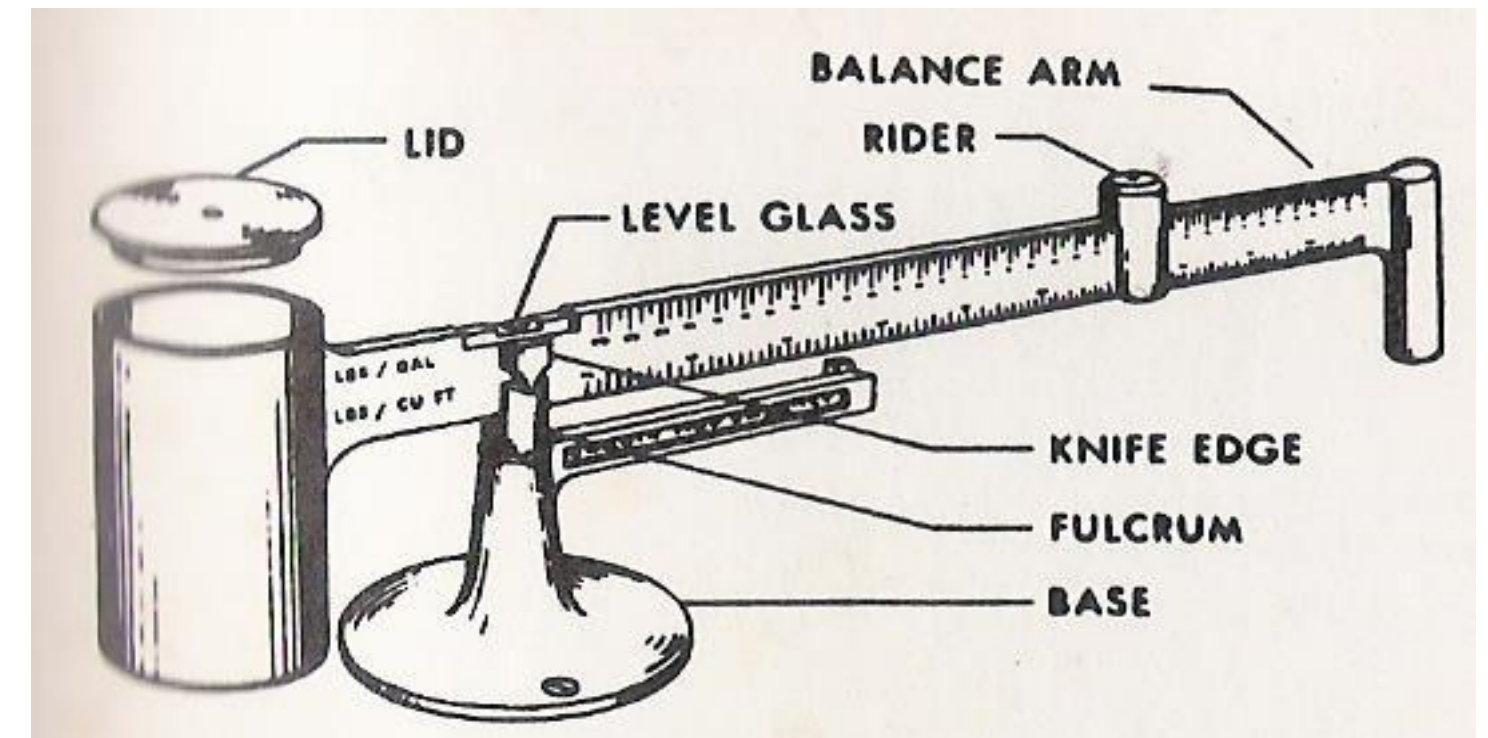
The drilling fluid used for a particular job is selected to avoid [formation damage](#) and to limit corrosion.



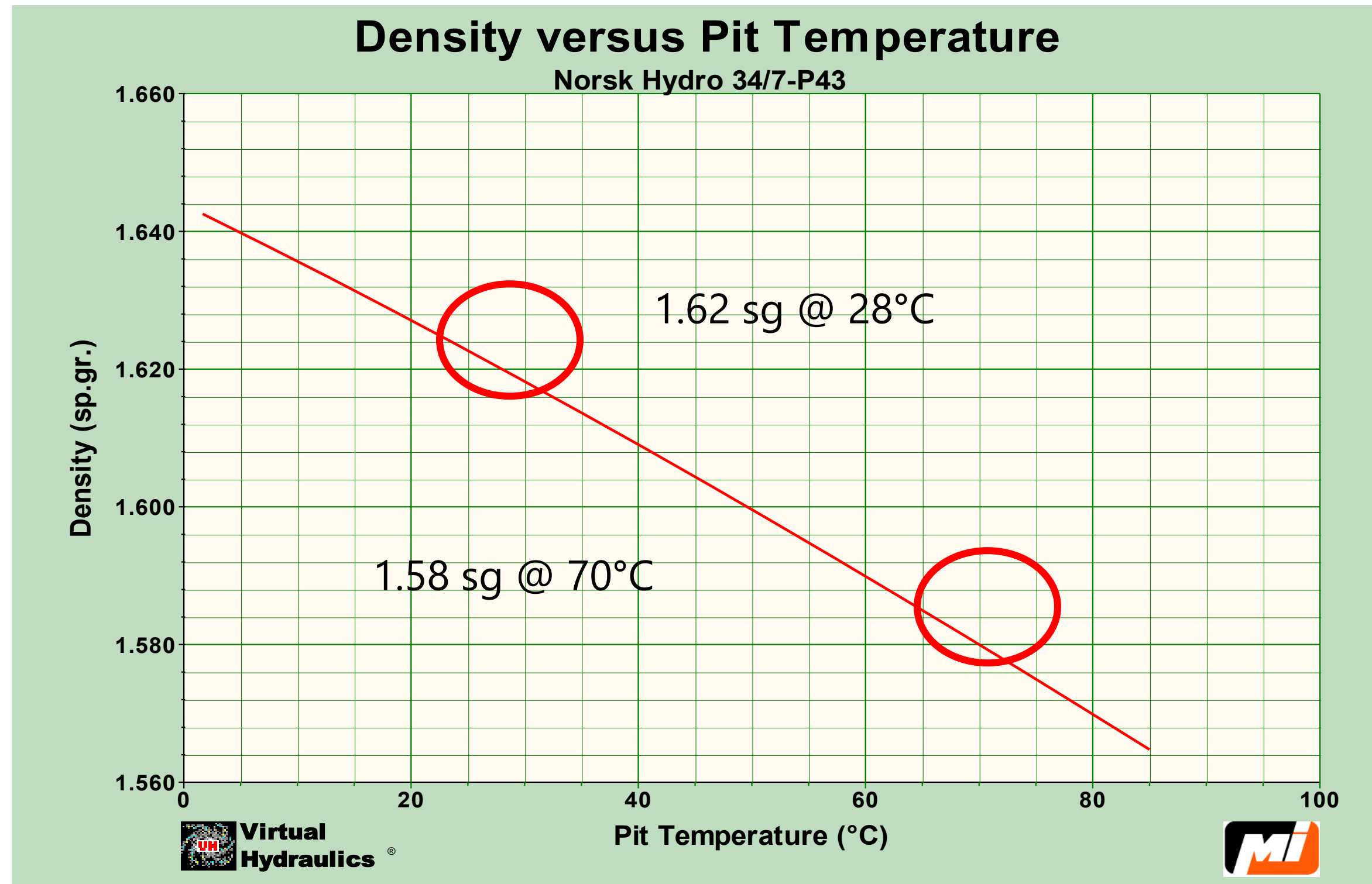
Mud Properties

WBM vs OBM

- OBM stability in high-temperature and high-pressure environments, WBM can break down and lead to loss of viscosity
 - OBM superior shale inhibition, lubricity, faster penetration rates, and not to leach out salty formations.
 - OBM are generally relatively more expensive (synthetic)
-
- ✓ **Low mud weight**
 - Well control (kicks)
 - Hole collapse (caving)
 - Mobile formations
 - Pressure shale
 - ✓ **High mud weight**
 - ~~Differential sticking~~
 - Surges and excessive mud loss

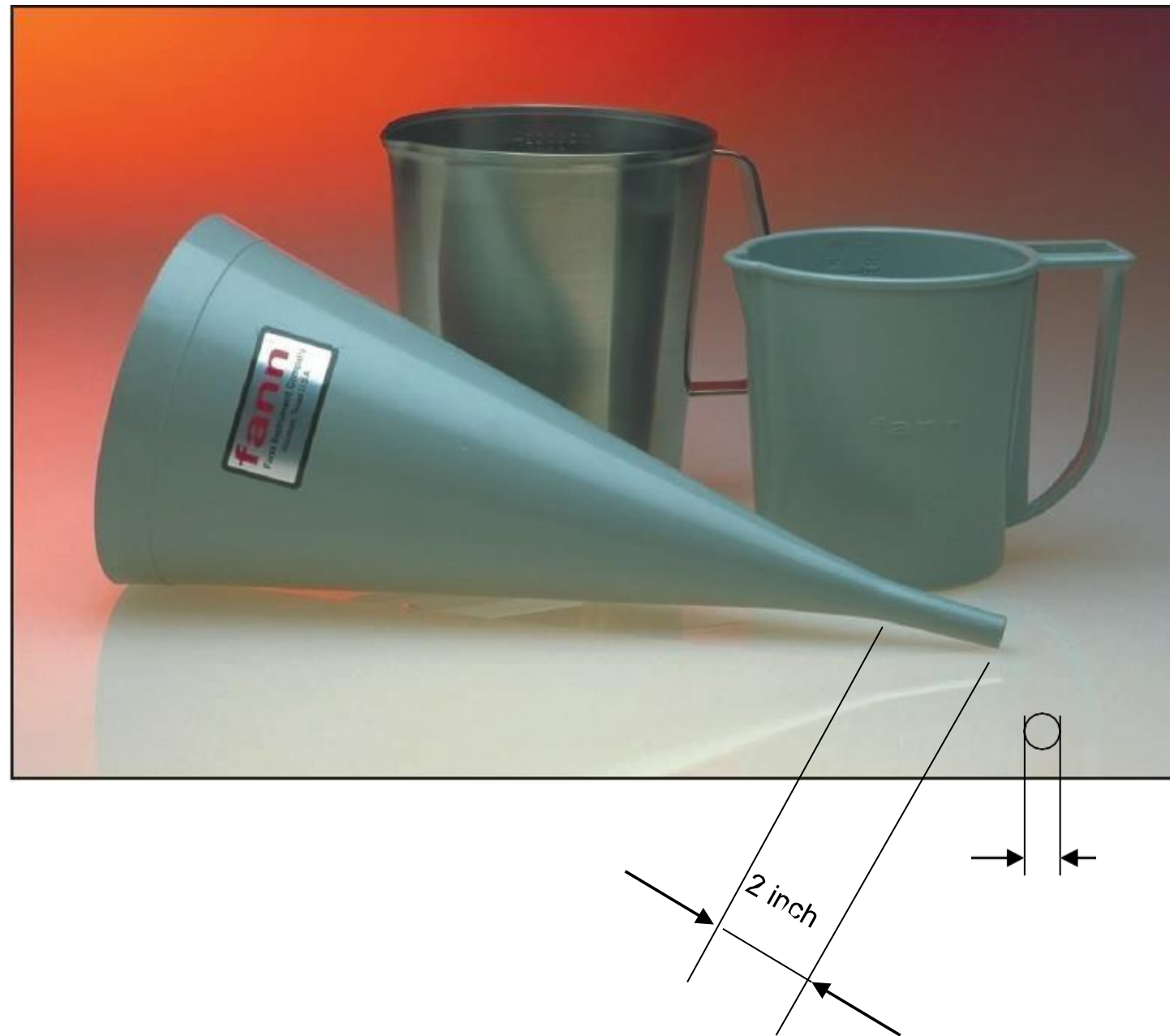


Actual Mud Weight



Marsh Funnel

To calibrate the Marsh funnel for the standard API test, fill the funnel with 1,500 cc of clear water at a temperature between 70° and 80° F (22° and 27° C), and note the time that is required for one quart (0.946 l) to drain from the funnel. The time for fresh water should be 26 seconds, with a tolerance of a half second, more or less.



- Use to measure viscosity Its **qualitative measurement** unit in seconds
- Time is measured the fluid flow as much as **1 quart (= 0.25 gals = 0.946 liter thru 3/16 inch ID x 2 inch**
- **Calibrated** by fresh water will take **26 second**.
- Viscosity < 38 second, difficult to transport drill cutting.
- Viscosity > 45 second will damage the pump because the viscous mud due by solid.

Viscosity measures the mud's **resistance to flow**

- Funnel viscosity is an early indicator of potential mud problems
- Increasing funnel viscosity may indicate poor hole cleaning and poor solids control

Mud Rheology

- Viscometer is used to measure Rheology fluid parameter Plastic Viscosity (PV); Yield Point (YP) and Gel Strength.
- Read RPM_{300} and RPM_{600}
- $PV = R_{600} - R_{300}$ centi poise, cp (Friction between solid and liquid)
- $YP = R_{300} - PV$ lbs/100ft² (Friction among molecules in the liquid)
 - Gel Strength or thickening or thyratrophic measured in 10 seconds and 10 minutes after the fluid moving.
 - Before cutting is presented (New Mud), $PV/YP = 1$
- Rule of Thumb : $PV = 2 \leq MW$ and $PV/YP = 1.5 - 2$
- Gel Strength 3/5 is okay
3/8 is not okay because progressive mean the mud will becoming thick after pump off, and difficult to break circulate.
- Viscosifier : Benex; Thinner : SAPP, Vibraco



- **Plastic viscosity**
 - A measure of the solids in the mud
- **Yield point**
 - Relates to the type of solids in the mud
 - The higher the yield pt the better the carrying capacity of **the mud**

Filtration Properties

■ Fluid loss

- A measure of the fluid lost from the mud to the formation.
- A permeable formation must be present

■ Filter cake

- Results from buildup of solids as filtration progresses. High fluid loss means a thick filter cake.
- Ideal filter cake:
 - Thin, tough, and impermeable



Filtration Loss

- Performed by **100 psi** in **30 minutes**.
- The result is , a filtrate measured by glass measurement
- Beside Mud filtrate, also recovered mud cake measured by 1/32 unit inch.
- Less filtrate, is a better mud, because less formation damage, but more expensive mud, \$/ barrel will be increased
- Fluid loss between 15 – 20 cc is enough (<10\$/bbl) Fluid loss < 10 cc become expensive (>20\$/bblnya).

Solids Control Equipment

Shale Shaker

The shale shaker is designed to remove larger particles brought to the surface – basically visible cuttings. The screens should be sized to remove the smallest possible solids without blanking off and allowing fluid to fall through the screens.



Degasser

Available in atmospheric and vacuum models. Responsible for removing gas from the drilling fluid before it passes through the desanders, desilters, centrifuge, and mud cleaner.



Desander/Desilter

Removes particles from the drilling fluid via centrifugal force. They are available in a wide range of sizes and cut points. Cut point refers to the minimum sized particle which can be removed by the desander or desilter cone.



Gumbo Buster

The “gumbo buster” is a device designed to reduce highly hydratable clays commonly found in the surface part of the hole. Ineffective can result in almost total failure of solids control equipment downstream of the gumbo buster. Shaker screens will blank off, hydrocyclones will plug and become useless and centrifuges, if being used, become overloaded

Solids Control Equipment

Centrifuge

A centrifuge removes fine solids from a fluid by centrifugal force imparted by the rotation of an internal auger. They are commonly used in both weighted and unweighted mud systems.

Mud Cleaner

Mud cleaners are a combination of hydrocyclones and fine screen shale shakers. They are designed to remove fine solids and are placed in the system upstream of centrifuges.

Centrifugal Pump

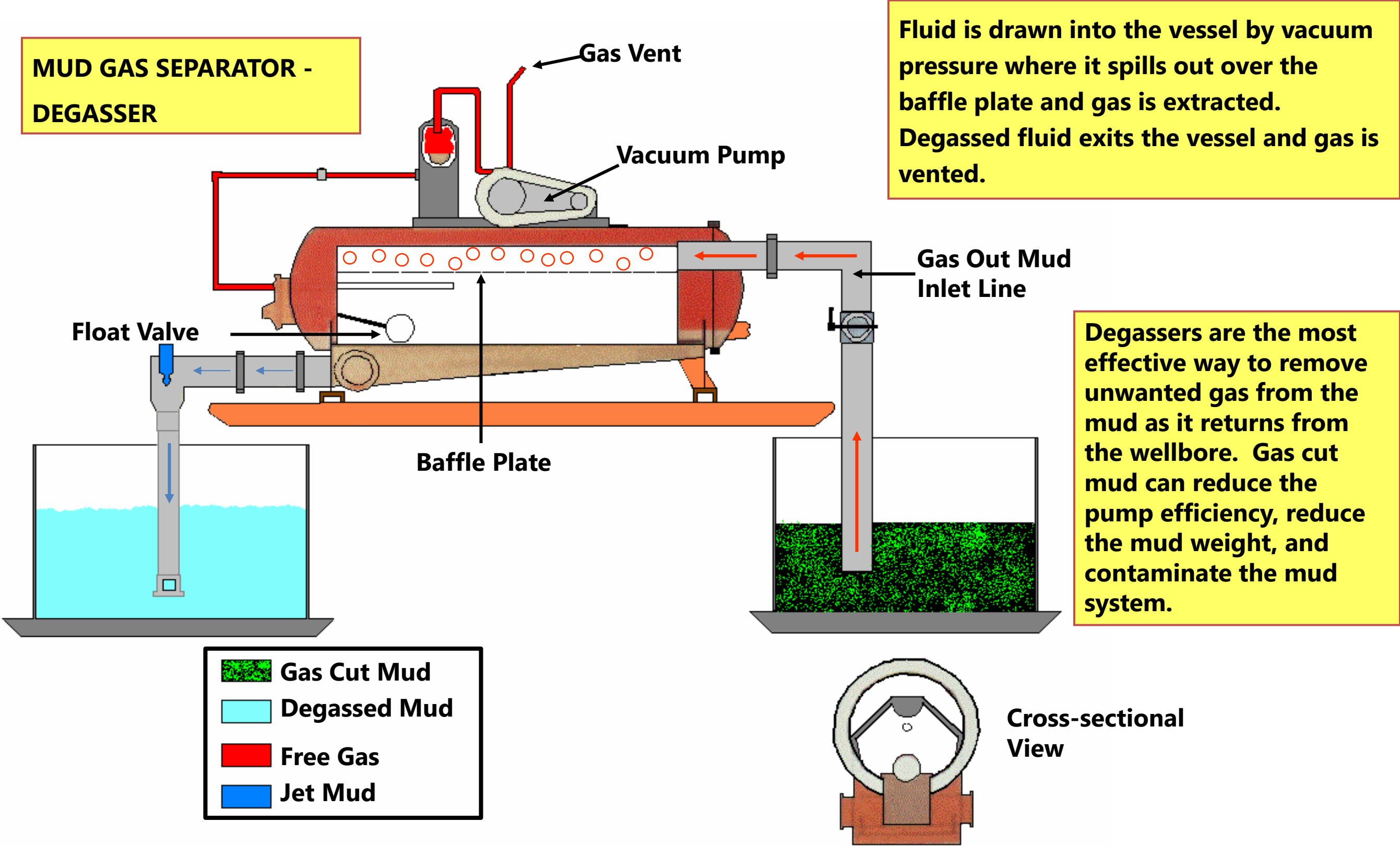
Pumps which move fluid by means of impellers as compared to pistons in positive displacement pumps.

Mud Agitator

Mud agitators are used to keep the mud stirred up in the pits of the solids control system as well as the mixing/suction pit. They are available in two configurations.



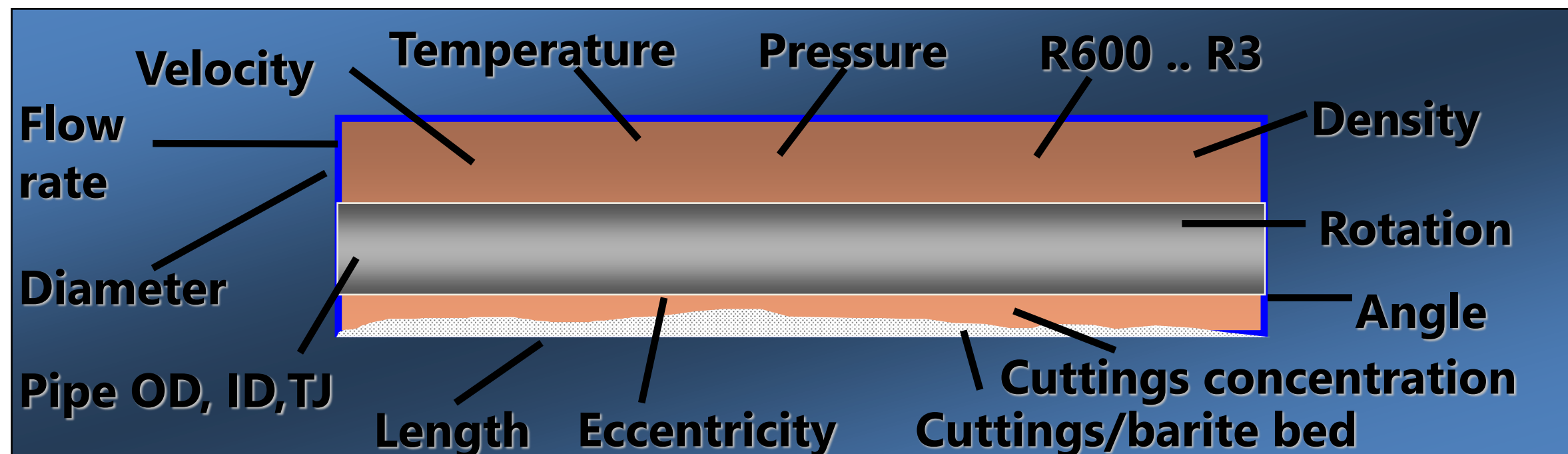
Solids Control Equipment



What is Virtual Hydraulics?

Integrated suite of computer programs which targets hydraulics modeling of synthetic and oil-based drilling fluids

- Technique for predicting downhole rheology using “data cubes” which combine data from different rheological measurements
- Subdivide the well into short segments, each with own set of properties

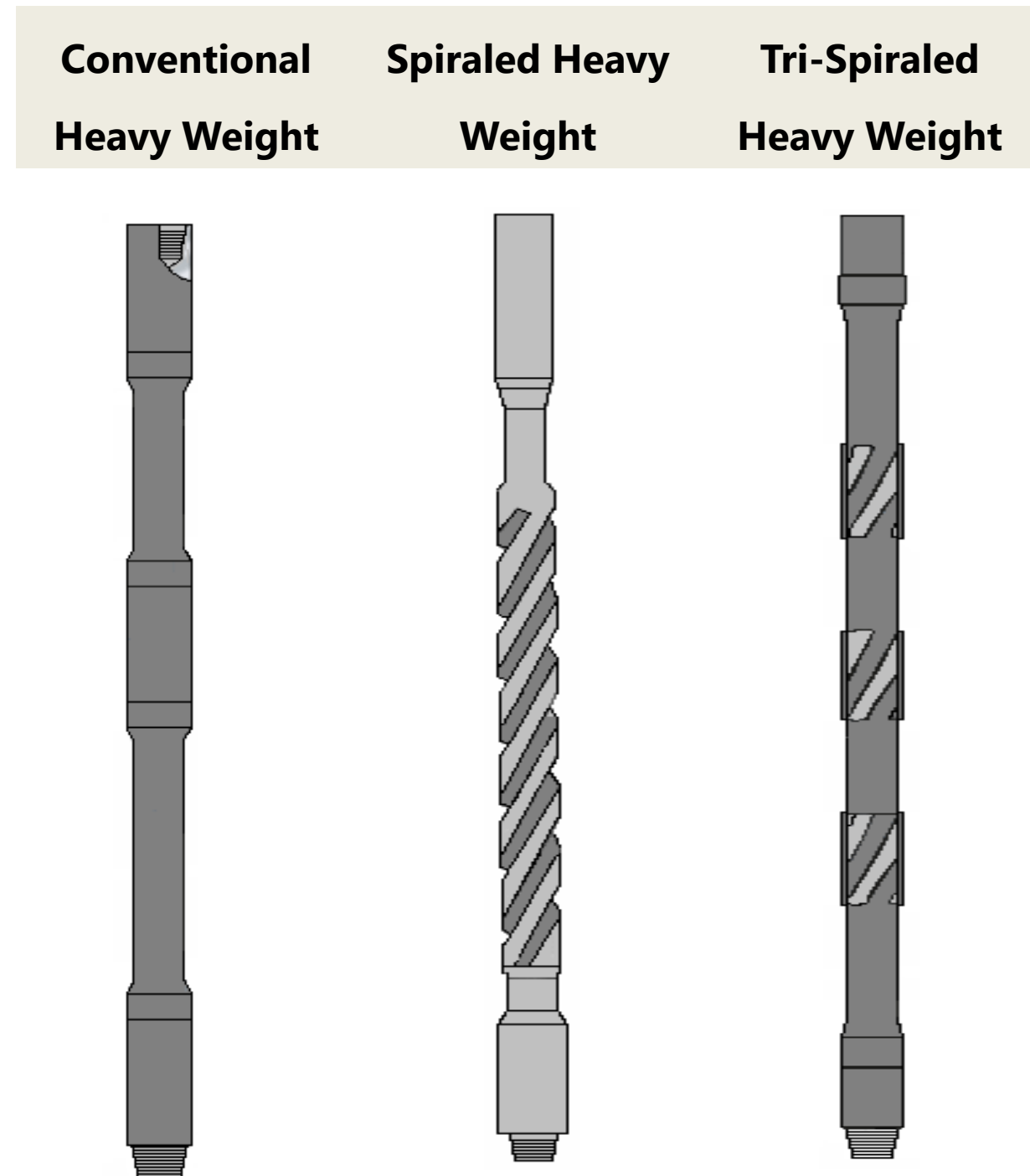


Heavy Weight Drill String

- Provide a smooth size transition between drill collars and drill pipe
- Suited for directional drilling owing to its flexibility and strength
- Reduces the concentrated stress at the top of the collar section
- Allows for higher RPM drilling while reducing torque and minimizing differential sticking
- Used to apply bit weight in drill strings which do not use any drill collars

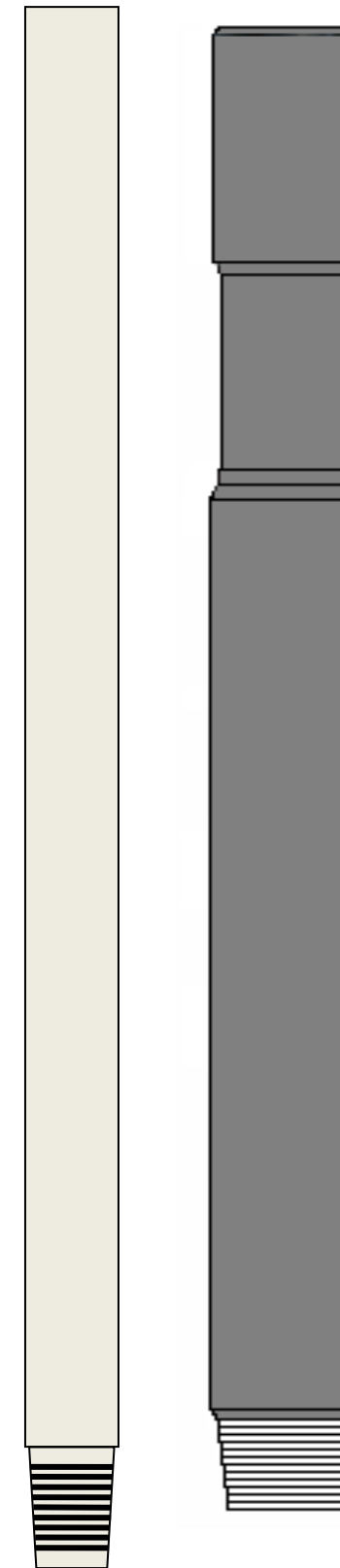
The spiraled versions further reduce the chance of differential sticking and improve circulation around the BHA.

Courtesy of Grant Prideco



Drill Collars

- Heavy-walled, heavy-weight pipe used as bottom hole assembly components
- Prime function: apply weight to the bit while drilling
- Types: non-spiraled and spiraled
- Available with hard facing to minimize excessive wear when drilling hard, abrasive formations



Spiral Drill Collars



Spiral drill collars may be used in wells where there is an appreciable risk of differential sticking. Spiral collars present less contact area with the wall of the hole as compared to non-spiralled collars.



Stabilizers

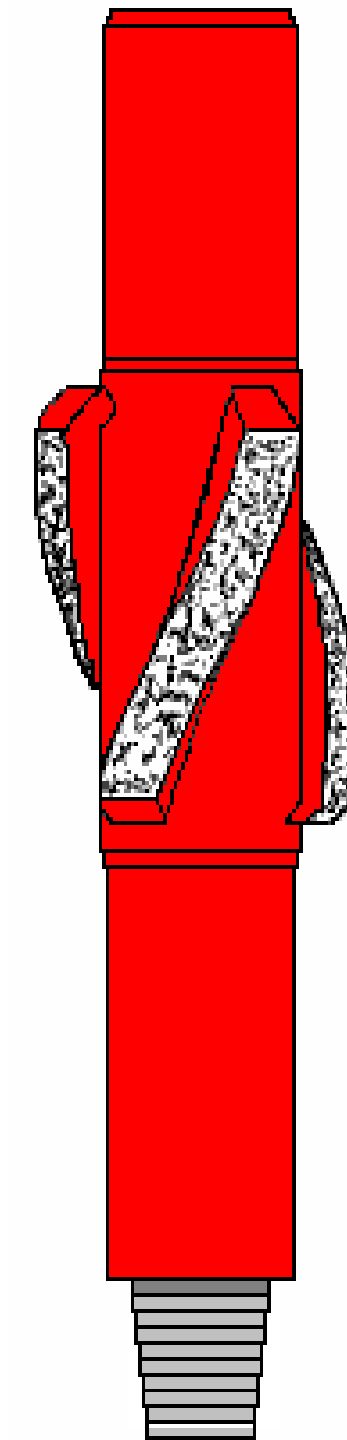
Welded blade stabilizers - used in soft to medium hard formations

Integral blade – one piece units, used in hard to very hard formations

Sleeve type – can be used in varying hole sizes

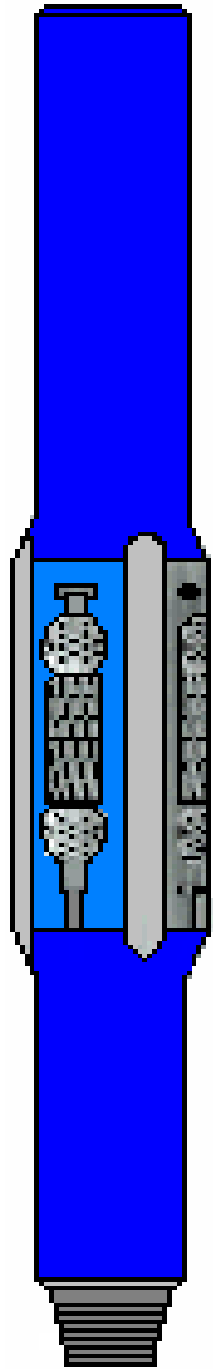
Replaceable blade - Used as a near-bit stabilizer for keeping the hole in-gauge in harder, more abrasive formations

Non-rotating rubber sleeve - used in relatively hard formations such as limestone and dolomite



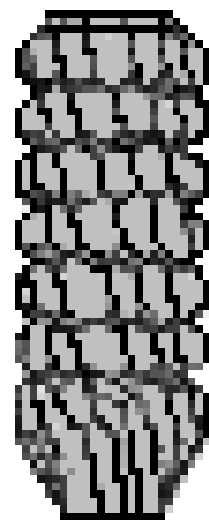
Welded Blade Stabilizer

Reamers

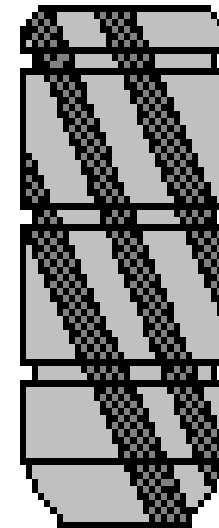


Roller reamers run between the bit and drill collars

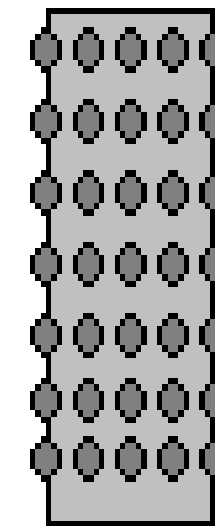
- maintain the hole in-gauge when drilling hard formations
- little area contact reduce chance of ticking
- selected based on the hardness and abrasiveness of the formations to be drilled



**Hard faced teeth for
soft formations**



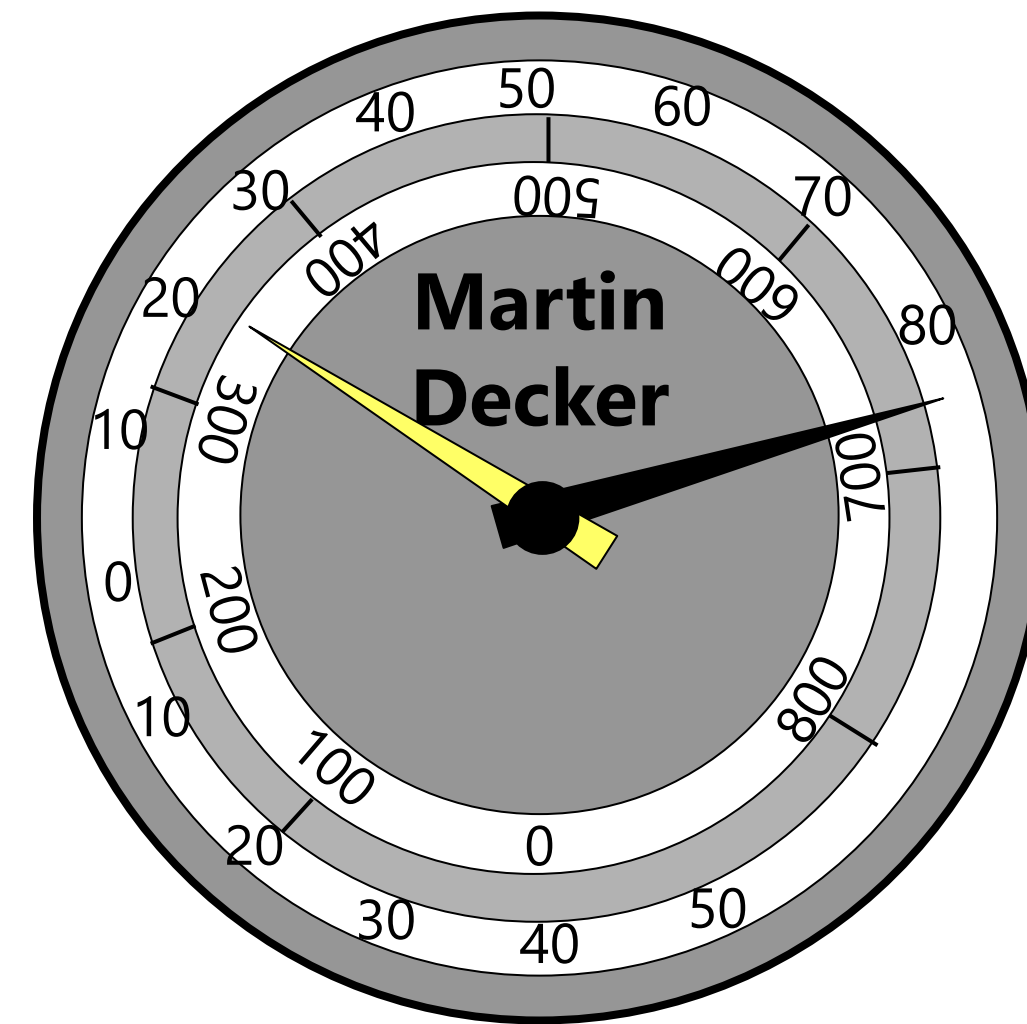
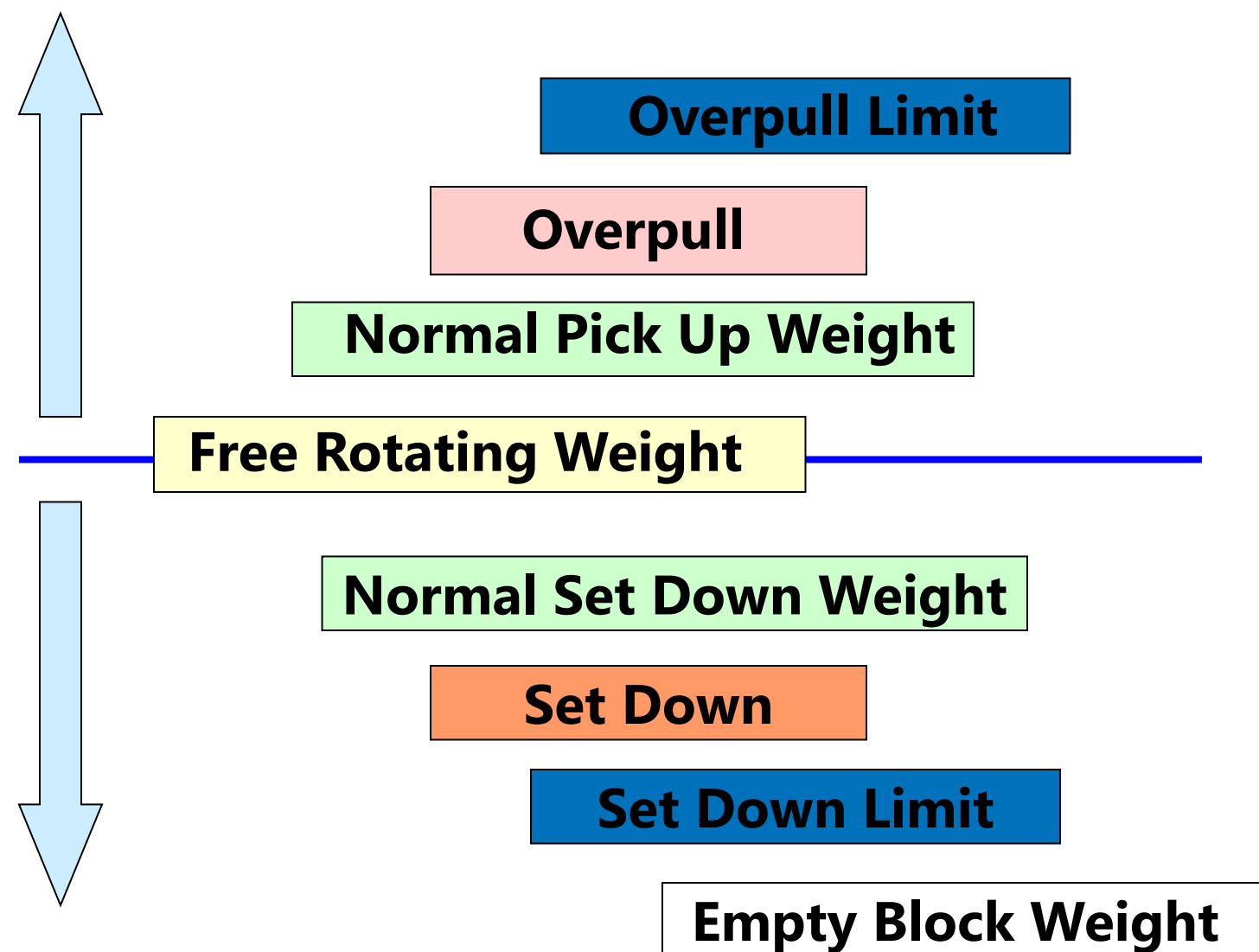
**Hard facing for
medium-hard formations**



**Tungsten carbide inserts
for hard formations**

Weight Indicator Readings

Know and Use Common Terminology



Drilling Jars

Friend or Foe: Drilling jars can get you out of a sticking situation or if used in the wrong direction or too “heavily” can permanently stick a string.

Three types of drilling jars:

- Hydraulic
- Mechanical
- Hydro/Mechanical



Performance Problems: Jars – Not Firing Up

- ▶ **Stuck above jars** – when jars are run low in the bottom hole assembly it's possible for the jars to be below the stuck point – if this is the case, the jars usually can not be activated.
- ▶ **Cock jars** – make sure adequate weight has been added or adequate pull has taken place in order to cock the jars – take into account pump open force if circulation is taking place.
- ▶ **Time to fire** – all too often drillers are too impatient when it comes to allowing the jar to fire – they're not instantaneous devices – they take several minutes to go off
- ▶ **Cycle the jars** – if the jar is not responding, re-cycle the jar – this may mean apply enough weight to close the jar, or apply adequate pull to re-open the jar.

Performance Problems: Jars – Not Firing Down

- ▶ **Stuck above jars**– when jars are run low in the bottom hole assembly it's possible for the jars to be below the stuck point – if this is the case, the jars usually can not be activated.
- ▶ **Effect of pump** – calculate pump open force and add this force to the required weight to activate the jars
- ▶ **Effect of drag** – the last recorded drag should also be taken into account when applying set-down weight
- ▶ **Time to fire**– all too often drillers are too impatient when it comes to allowing the jar to fire – they're not instantaneous devices – they take several minutes to go off
- ▶ **Cycle the jars**– if the jar is not responding, re-cycle the jar – this may mean apply enough weight to close the jar, or apply adequate pull to re-open the jar

Performance Problems: Running Jar in Compression

- Achieve optimum hammer weight above jar with HWDP usage
- Add weight slowly, bleeding through detent
- Open jar before coming out of the hole

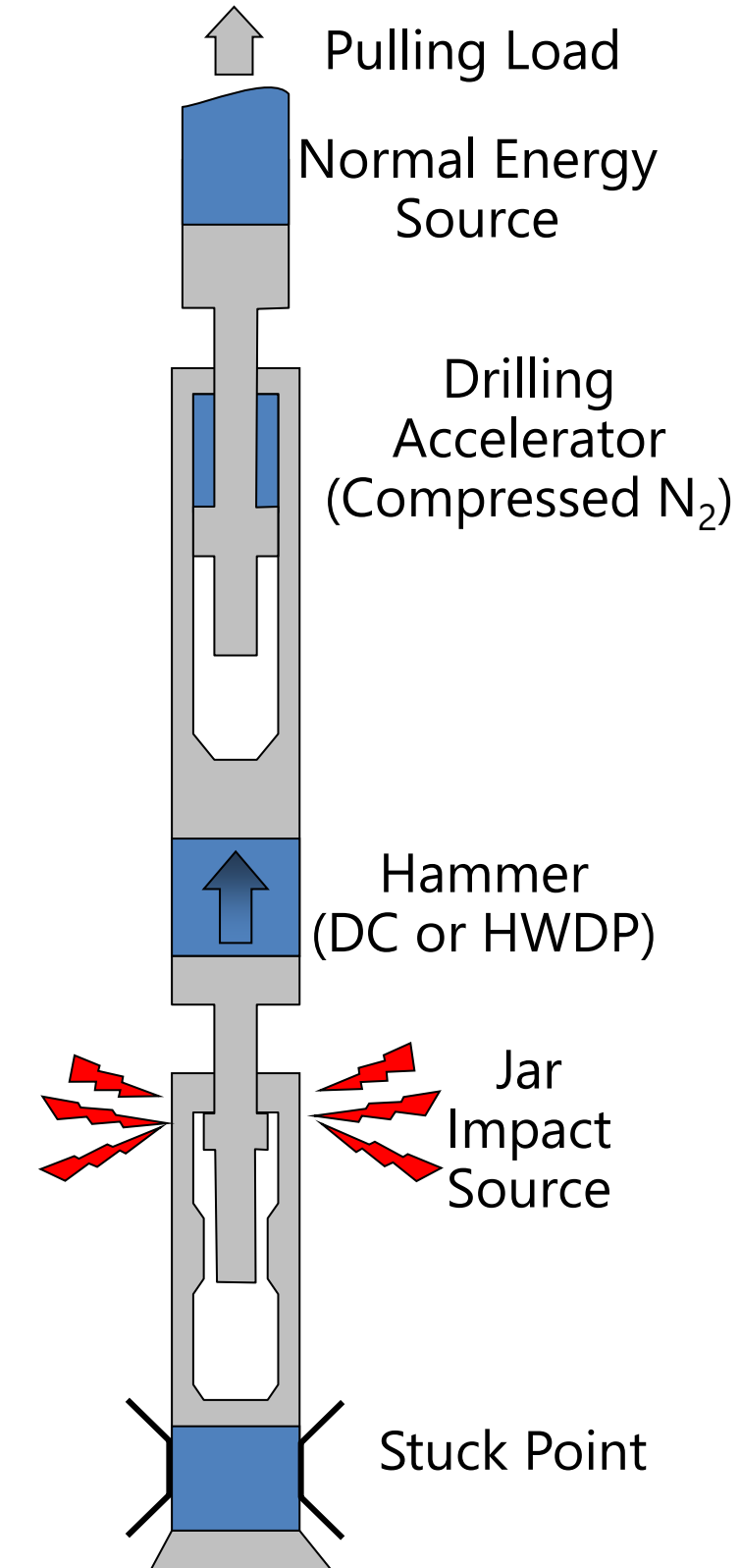
Accelerators

Features and Benefits:

- ✓ Fluid compression inside the tool compensates for limited drill pipe stretch in shallow wells, providing stored energy to create maximum impact
- ✓ Full bore design minimizes pressure losses and provides wireline tool compatibility
- ✓ Can create up to twice as much impact force for jarring up, down, or up and down
- ✓ Operates automatically with the jar, increasing reliability and ease of use
- ✓ Design ensures full over-pull force can be delivered to the hammer section, creating maximum positive impact
- ✓ Protects the drillstring and surface equipment from damaging shock waves

Drilling Accelerator

- The accelerator is run between the DP and the BHA.
- The impact force of the jar is directly related to the acceleration of the drill collar (HWDP) mass above the jar. The stretch of the DP string is the normal energy source to accelerate this "hammer". This same force that stretches the DP string also compresses the N₂ gas in the accelerator. When the jar releases, these energy sources combine to provide faster acceleration of the DC mass and the resulting higher impact.
- Should be used in crooked and horizontal type wells or for shallow wells that don't allow for much stored string energy.



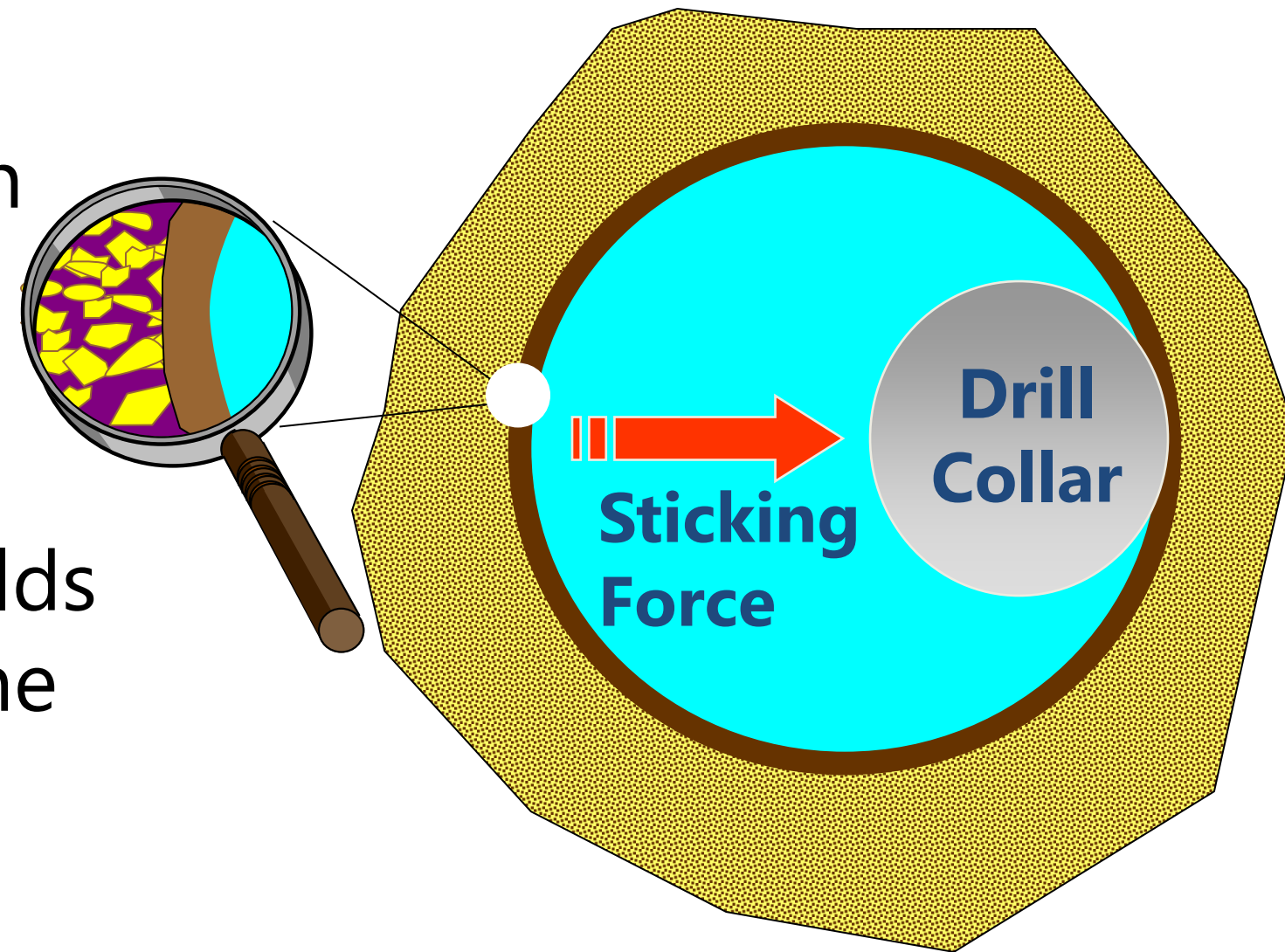
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Differential Pressure

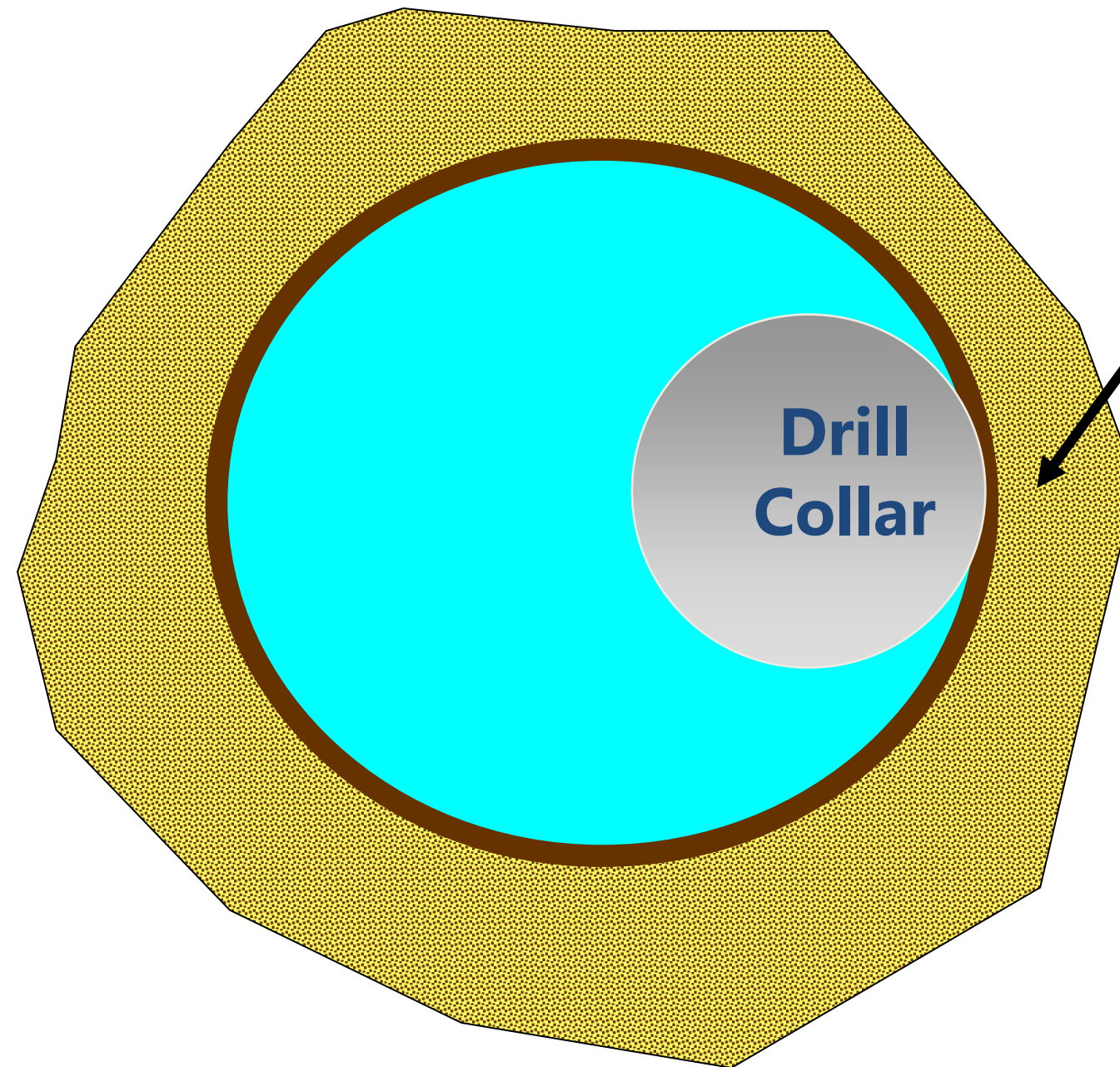
Differential Pressure

The pore pressure in a **permeable** formation is much less than the **hydrostatic** pressure of the mud column.

This difference in pressure holds the BHA against the side of the wellbore.



Differential Sticking



As the contact area increases the filter cake dries and a differential pressure is created between the wellbore and the permeable formation.

Key Conditions

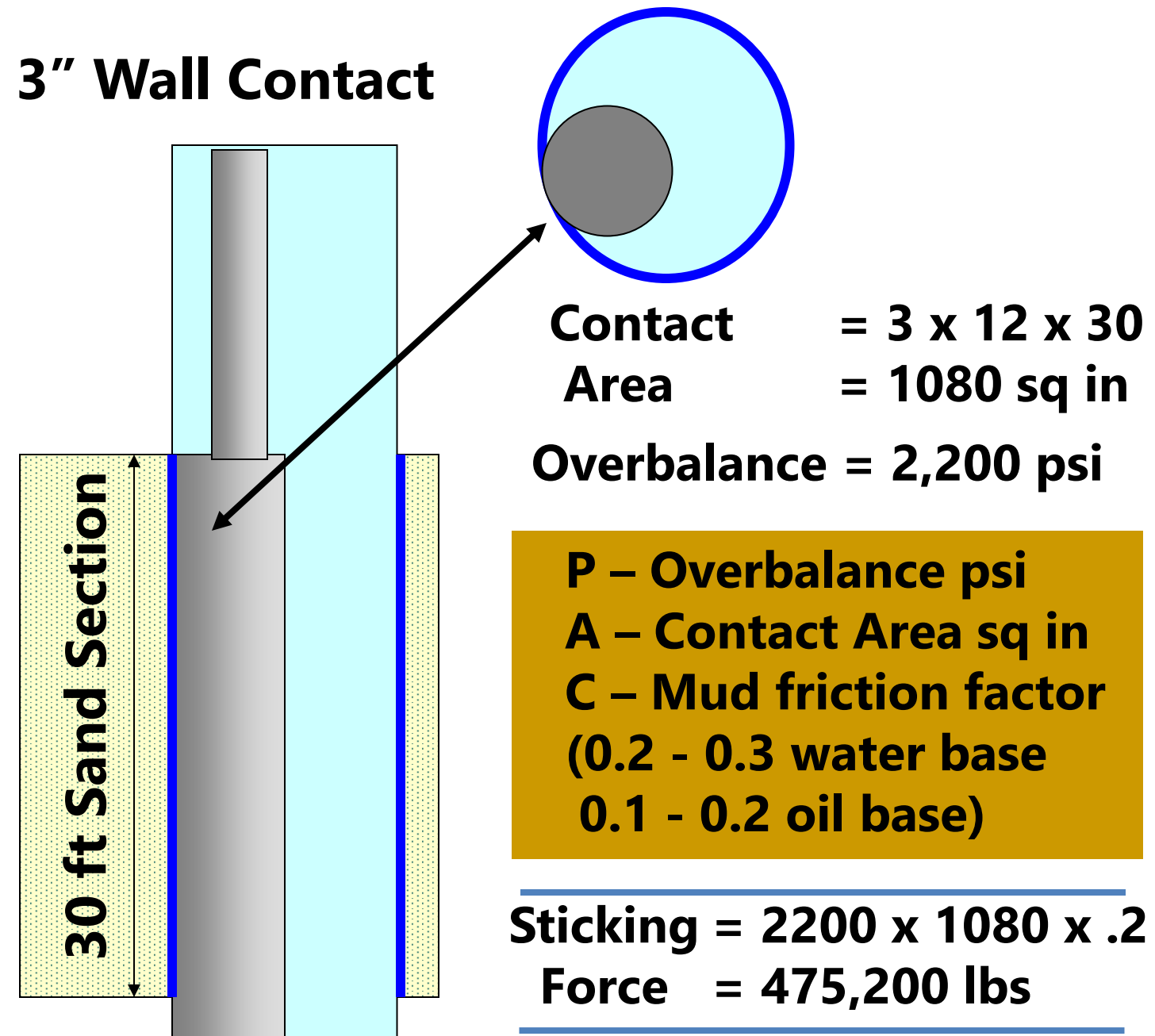
- 1. Permeable Formation** – allows for filtrate movement from the wellbore to the formation
- 2. No Pipe Movement** – static pipe against a permeable formation provides the time required for filter cake to build around the pipe and, the filter cake to thicken and dry, creating a seal between the wellbore and the formation and allowing a differential pressure to be created
- 3. String in Contact** – the pipe must be in contact with the wall of a permeable formation
- 4. Thick Wall Cake** – a thick wall cake increases the contact area – this can be minimized with efficient solids control
- 5. Overbalance MW** – mud hydrostatic pressure excessively greater than the formation pressure of a permeable zone

Sticking Force

The amount of force holding the pipe against a permeable formation is based on the degree of overbalance, the contact area and the friction factor of the mud in the hole.

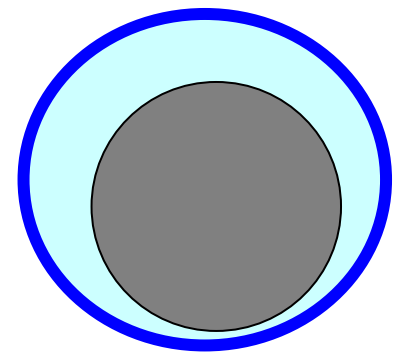
$$\text{Sticking Force} = P \times A \times C$$

3" Wall Contact

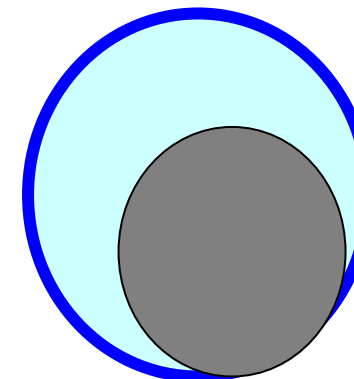


Wall Cake

The behaviour of wall cake can increase the required amount of freeing force as time passes.



Initial sticking – minimal contact area



Wall cake “creep” increases the contact area

Differential Sticking

The differential pressure is based on the difference between mud hydrostatic pressure and the formation pressure of the zone in question. Differential pressure can be estimated by:

$$((MW \times .052) - \text{Formation Gradient}) \times \text{Vertical Depth}$$

The contact area is based on the area of the pipe in contact with the formation in two dimensions: length and width. It can be estimated:

$$\text{Contact}_{\text{Width}} \times 12_{\text{In / Ft}} \times \text{Contact Length}_{\text{Feet}} = \text{Contact Area}_{\text{Sq In}}$$

Differential Sticking

Example:

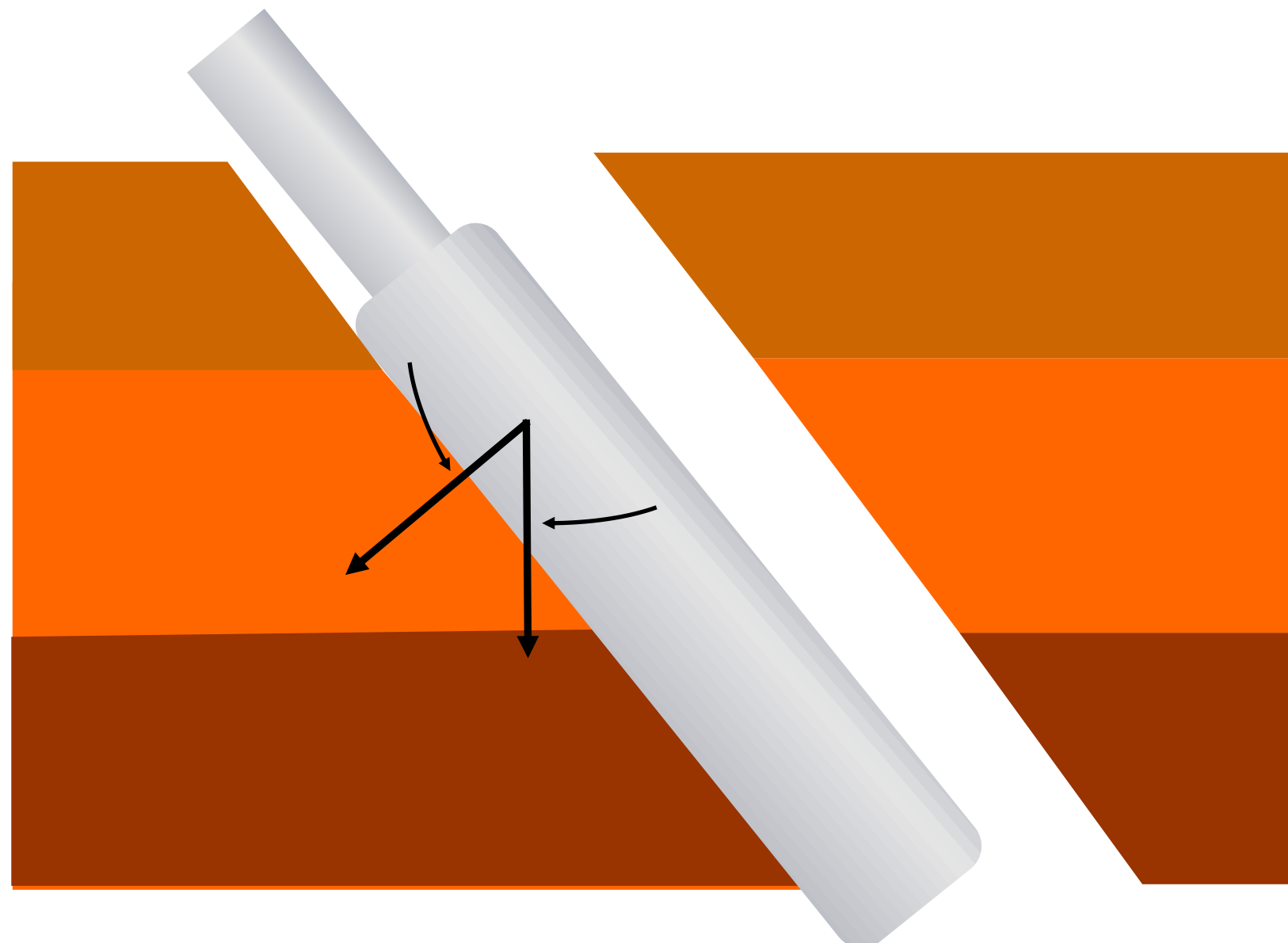
A 30 foot drill collar is in contact with a permeable formation at 8000 feet that is normally pressured (.465 psi/ft). The mud in use is water-based and has a density of 11.5 ppg.

Differential Pressure
$$\left((11.5 \times .052) - .465_{psi/ft} \right) \times 8000 = 1064_{psi}$$

Contact Area
$$3''_{Width} \times 12_{In / Ft} \times 30_{Feet} = 1080_{Sq In}$$

Differential with High Angle

Potential differential sticking increases when drilling high angle well. In a deviated well, there will be a sidewall force proportional to the weight of the BHA and the deviation angle.



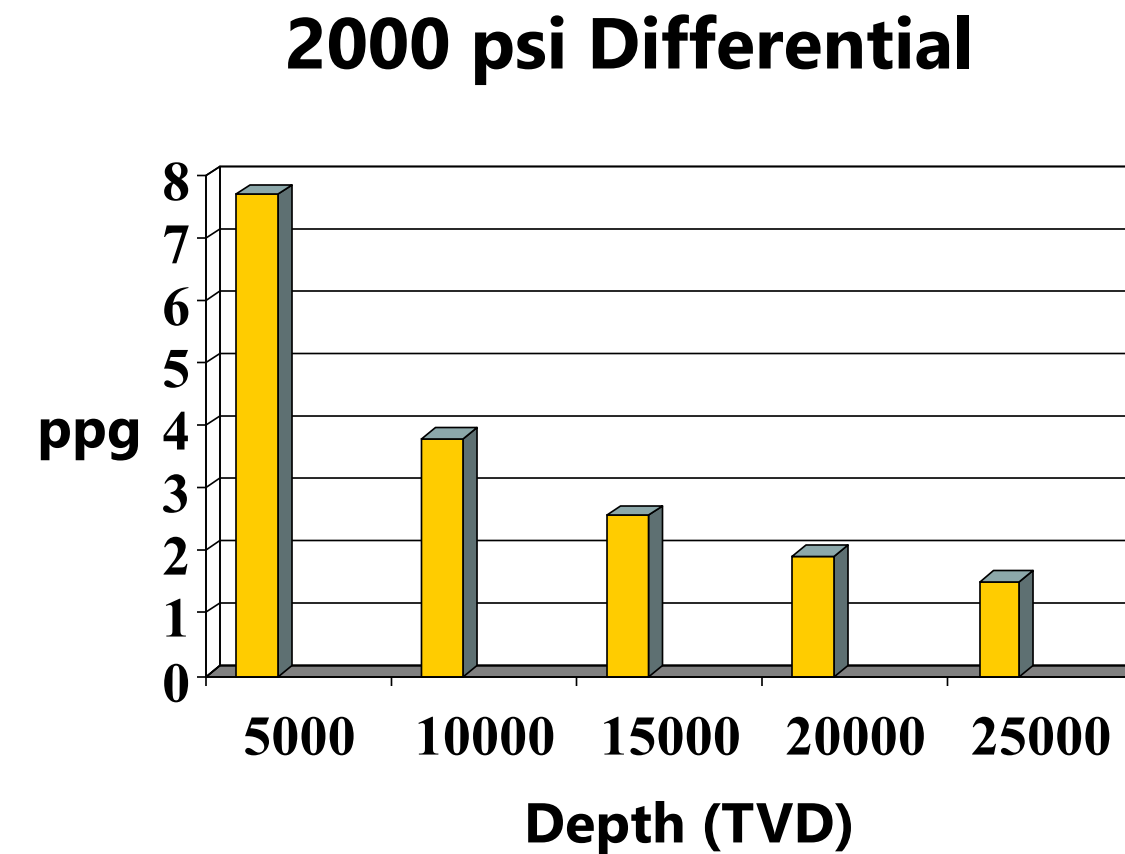
$$\text{Sidewall Force} = \text{Weight} \times \cos \text{Angle}^*$$

***90 degree hole angle**

Differential Pressure

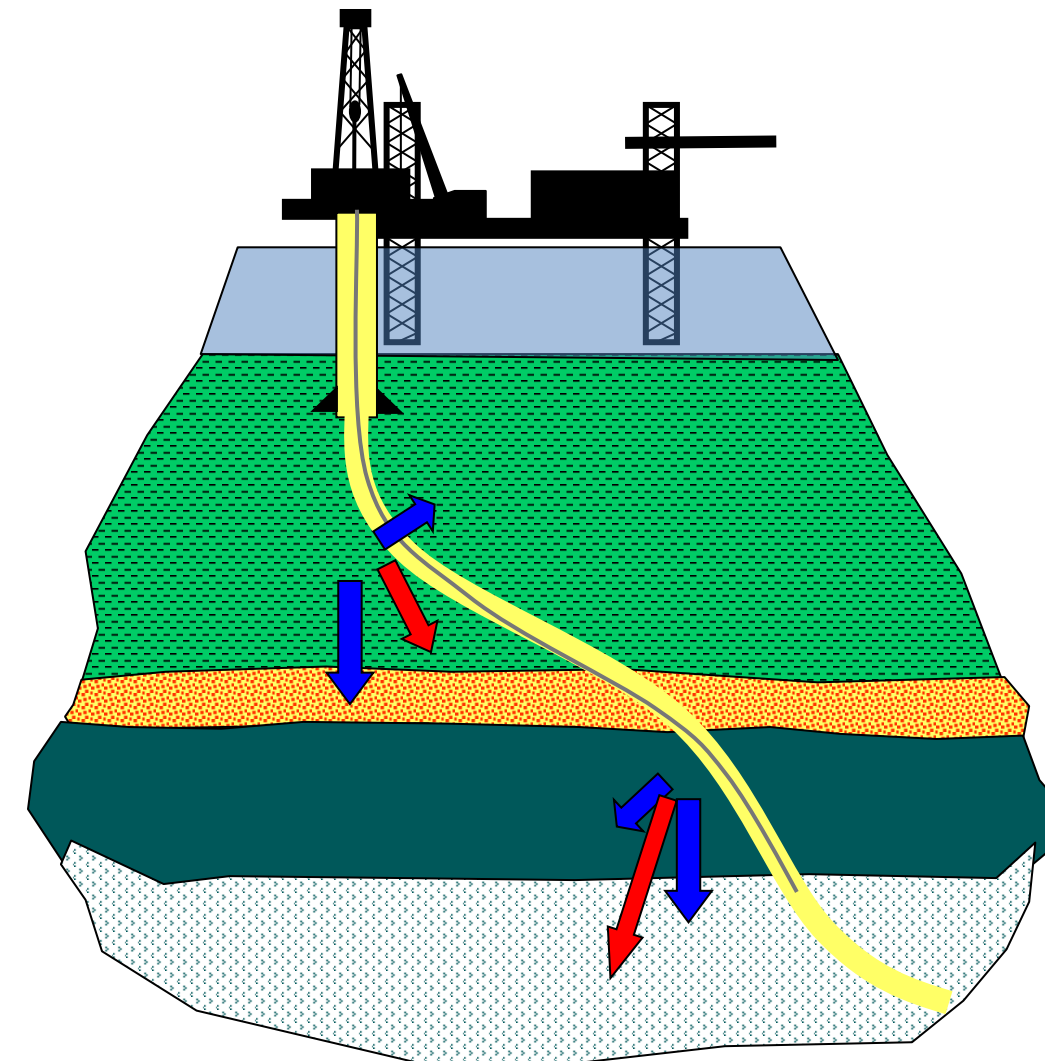
High Risk Indicators

- Depleted Sands
- High Surge/Swab/ECD
 - Extended Reach
 - Deep Water Subsea
- Increasing Mud Weight
- Fluid Loss
- Permeable Zones Exposed
- Surveys (Static Times)
- Connections (Makes & Breaks)



Torque & Drag

- **Avoid drop sections**
- **Horizontal well will have two bottoms**
 - End of well
 - Beginning of horizontal section



Sliding vs. Rotating

- **Rotation is the most effective pipe motion to reduce torque & drag**
- **Can slide with pipe buckled**
 - May be necessary to get WOB
- **Cannot rotate with pipe buckled**
 - Causes rapid wear and failure of the drill string
- **Reduce WOB when going from sliding to rotating**

Preventing Differential Sticking

- ✓ **Hand over information**

- Monitor, record, and handover key hole info

- ✓ **Be prepared!!**

- Be aware of problem formations
- Have spotting agents on board
- Know how to apply freeing force to stuck point
- Torque and slump procedure can be applied immediately after differential sticking identified



Preventing Differential Sticking

- ✓ **Minimize pipe static time**
 - Keep the pipe moving
 - Brainstorm w/driller and crew for operations at hand
 - Minimize BHA and maximize string movement
 - Use under gage stabilizer at BHA top
 - Prevent wall contact with stabilizers and drill collars
 - Rotate the string on connection before tripping out of hole
 - Placing the jar in HWDP can help

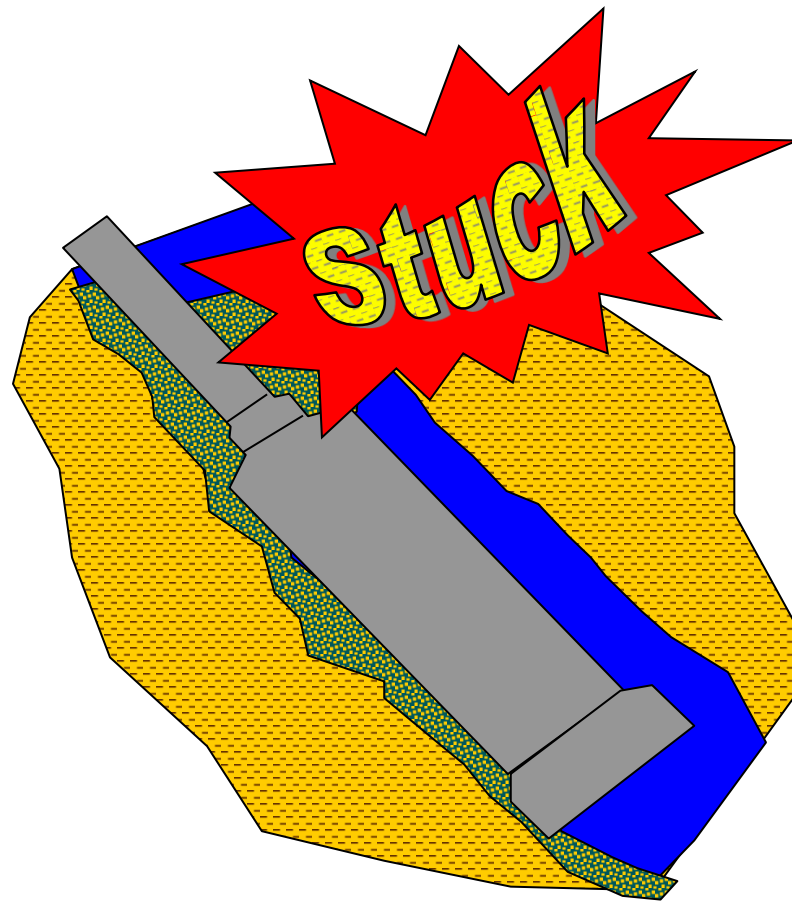
- ✓ **Mud management is key**
 - Monitor cuttings returns, mud properties, and the functioning of the solids control equipment
 - Keep circulation going
 - Plug permeable zones
 - Limit penetration & allow sealing time
 - Lower ROP to reduce cuttings



”

Hole Pack-Off

Hole Pack-Off/Bridge



- **Hole Pack-off:** the actual mechanism works through a combination of contact between the string and a build up of material in the annuls and the effect of compaction caused by the subsequent restriction of flow.
- **Bridge:** Bridging often refers to the hole being blocked by material off bottom when the string is out of the hole.

Hole Pack-Off

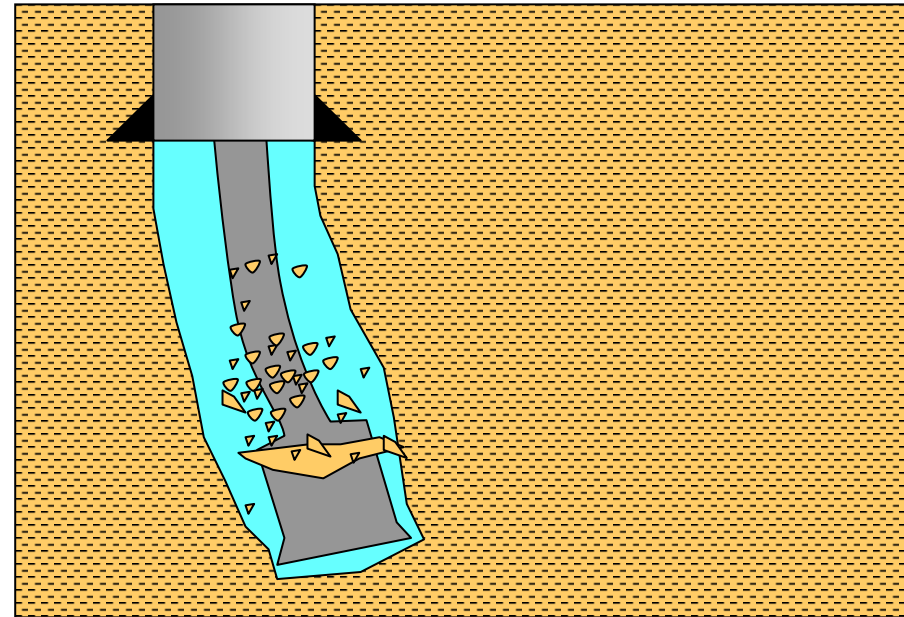
- Settled Cuttings
 - Poor Hole Cleaning
 - Excessive ROP
- Settled Cavings
 - Wellbore Instability
 - Cement Related

Hole Bridging

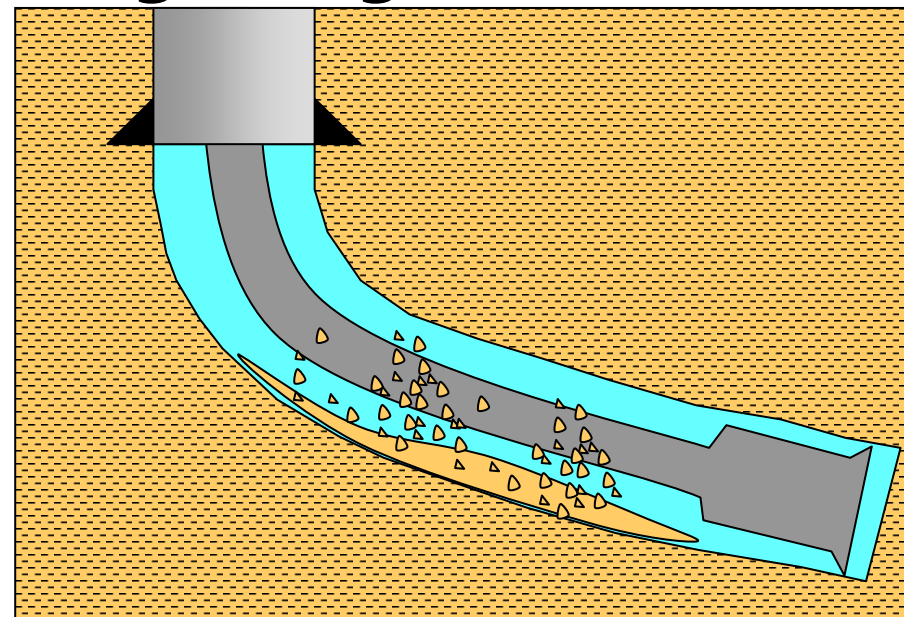
- Settled Cavings
 - Wellbore Instability
- Unconsolidated Formations
- Fractured Formations
- Cement Blocks
- Junk in Hole

Causes of Settled Cuttings

Low Angle Holes $< 35^\circ$



High Angle Holes $> 35^\circ$



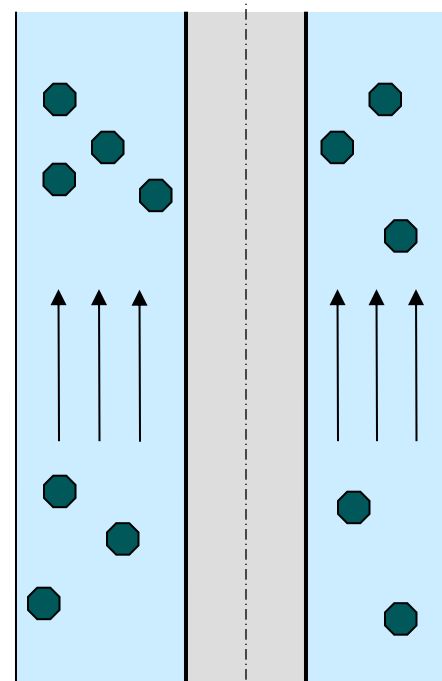
- Excessive ROP
- Insufficient Annular Velocity
- Poor Mud Parameters
 - YP, Gels
- Insufficient Circulating Time
- High Hole Angle
 - 55° Most Difficult

Inadequate Hole Cleaning Signs

- Erratic pump pressure.
- Poor weight transfer to bit.
- Difficulty orienting toolface.
- Absence of returns at shakers.
- Overpulls inside casing.

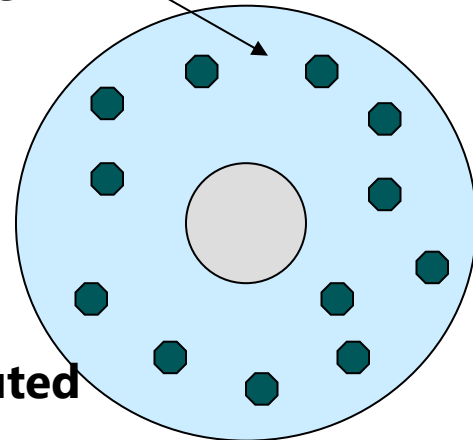
Annular Fluid Movement

VERTICAL WELLBORE



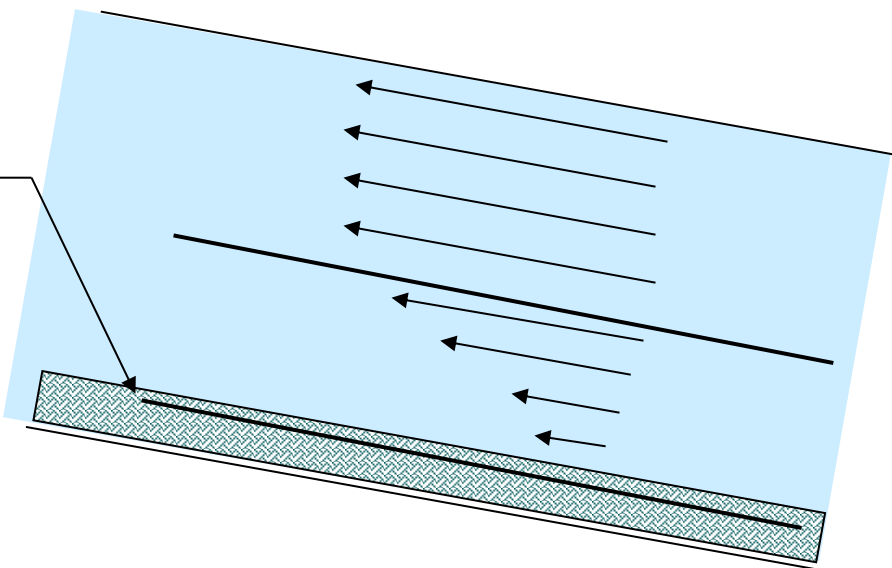
Fluid moves nearly uniformly in annulus of a vertical well

Cuttings are evenly distributed in annulus



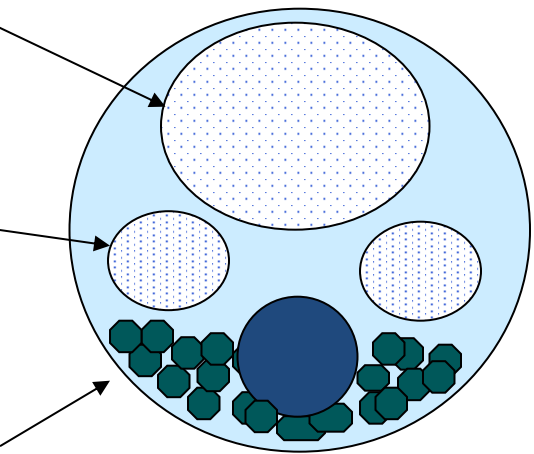
HIGH ANGLE WELLBORE

Cuttings bed



High Velocity Fluid

Low Velocity Fluid



Cuttings on low side of hole – remain stationary unless stirred up by pipe rotation

Hole Cleaning

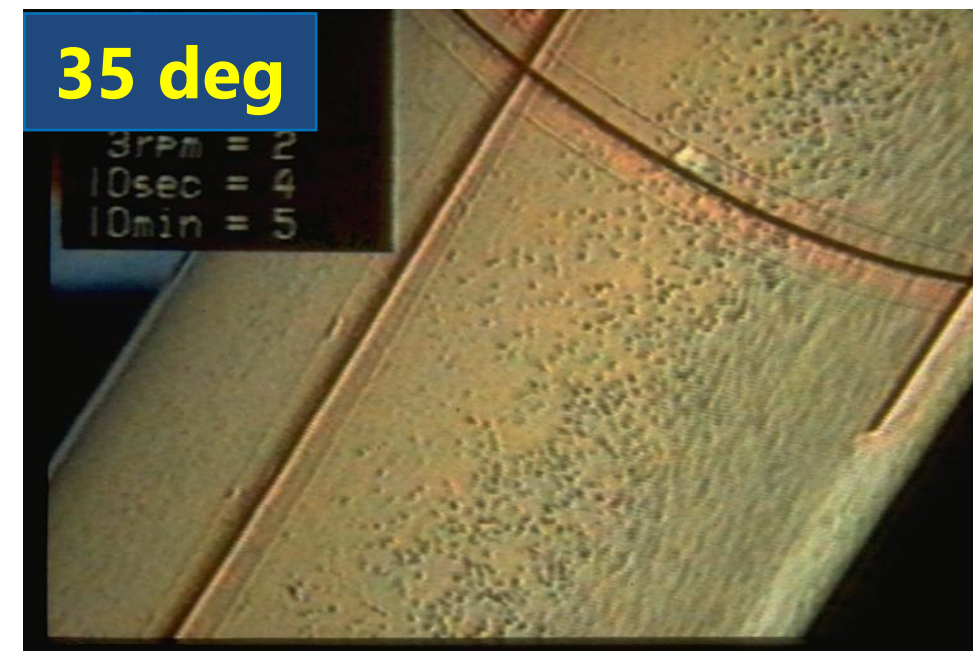
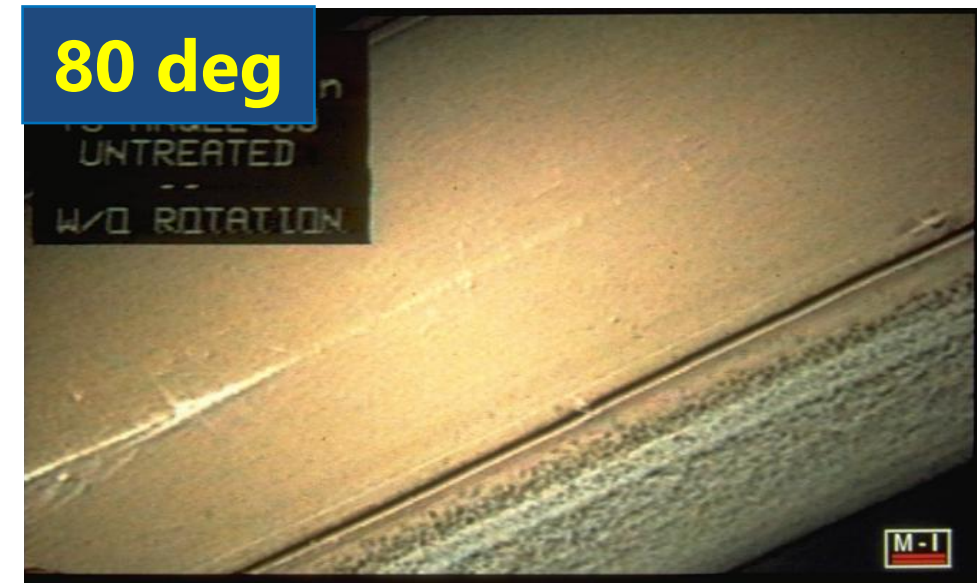
Good hole cleaning means removal of sufficient solids from the well bore to allow the reasonably unhindered passage of the drill string and the casing.

Basic Factors:

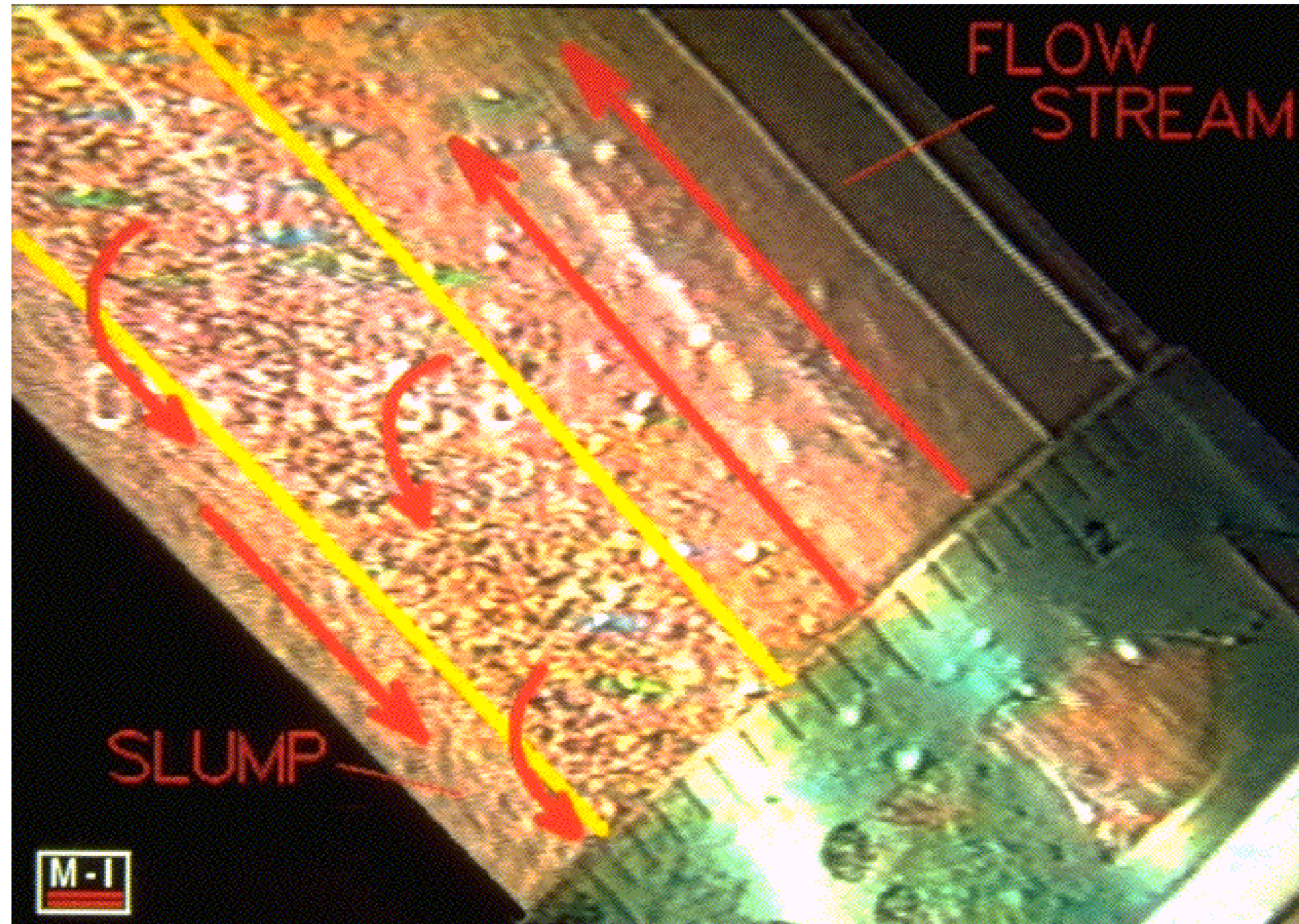
- Rate of Penetration - Determines cuttings volume
- Hole Stability - Cavings load
- Annular Velocity - Lifts the cuttings
- Mud Rheology - Suspend and carry the cuttings
- Circulating Time - Transport cuttings to the surface
- Hole Angle - Reduces the ability to clean the hole

Hole Cleaning – Well Profile

- 4 hole-angle ranges:
0-10, 10-30, **30-70**, 70-90
- Optimum hole cleaning for one interval may be inadequate in another
- Build sections of directional wells most difficult to clean

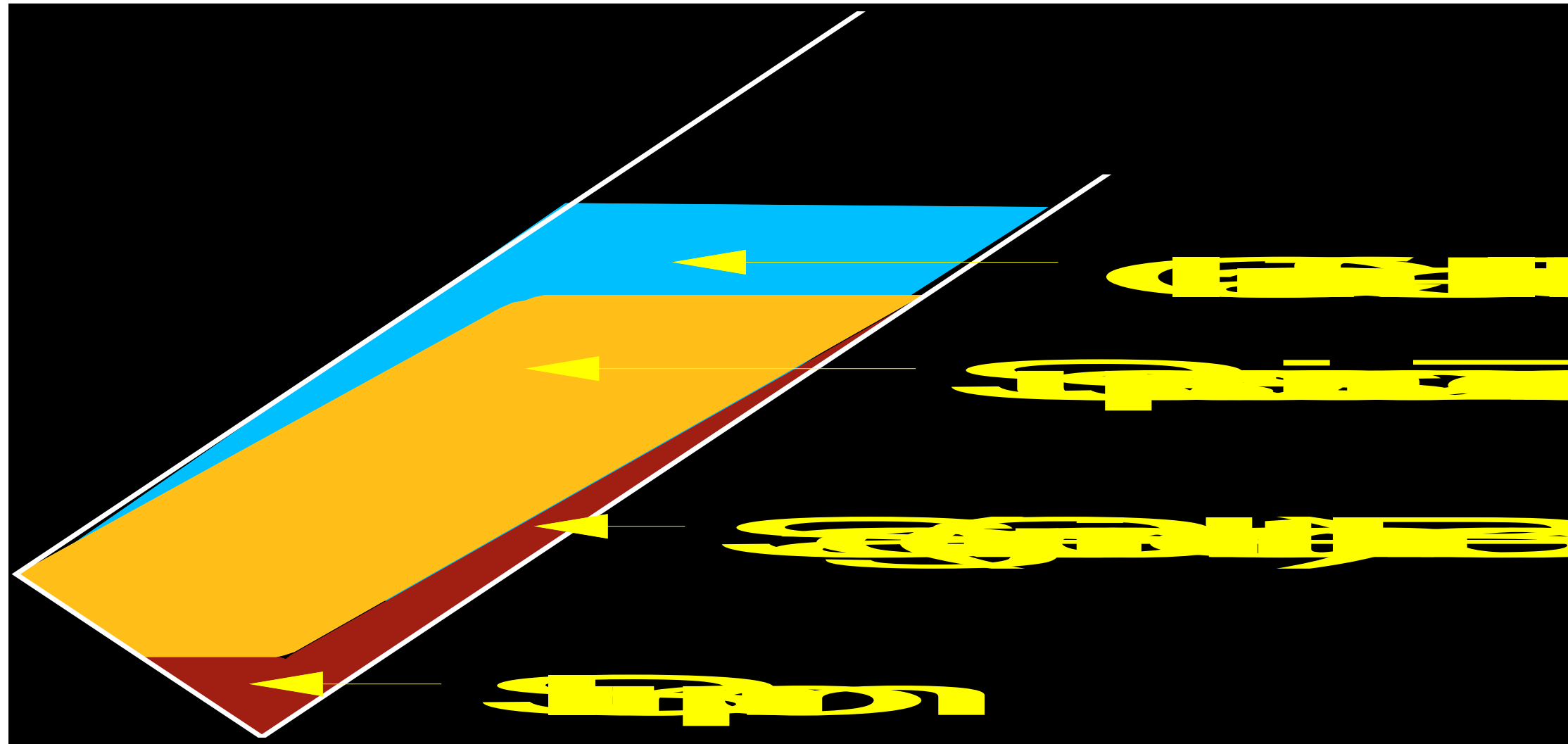


Flowstream & Slumping



Boycott

“Boycott” settling accelerates bed formation, especially in the build section.



Cuttings and Cuttings Beds

- Near-vertical and low ranges: cuttings concentration (little to no bed)
- Intermediate range: cuttings concentration, bed thickness, and propensity for slumping
- High range: bed thickness increases as with physical characteristics
- Cuttings beds: easy to deposit, difficult to remove

Cuttings Beds

- Unstable between 35 and 50 degree hole angles
- Resist removal once formed
- Thickness varies with annular velocity
- Removal is assisted by pipe rotation (120+ rpm) and turbulent flow
- Chances of formation minimised if fluid has good suspension characteristics

Flow Parameters

Turbulent Flow:

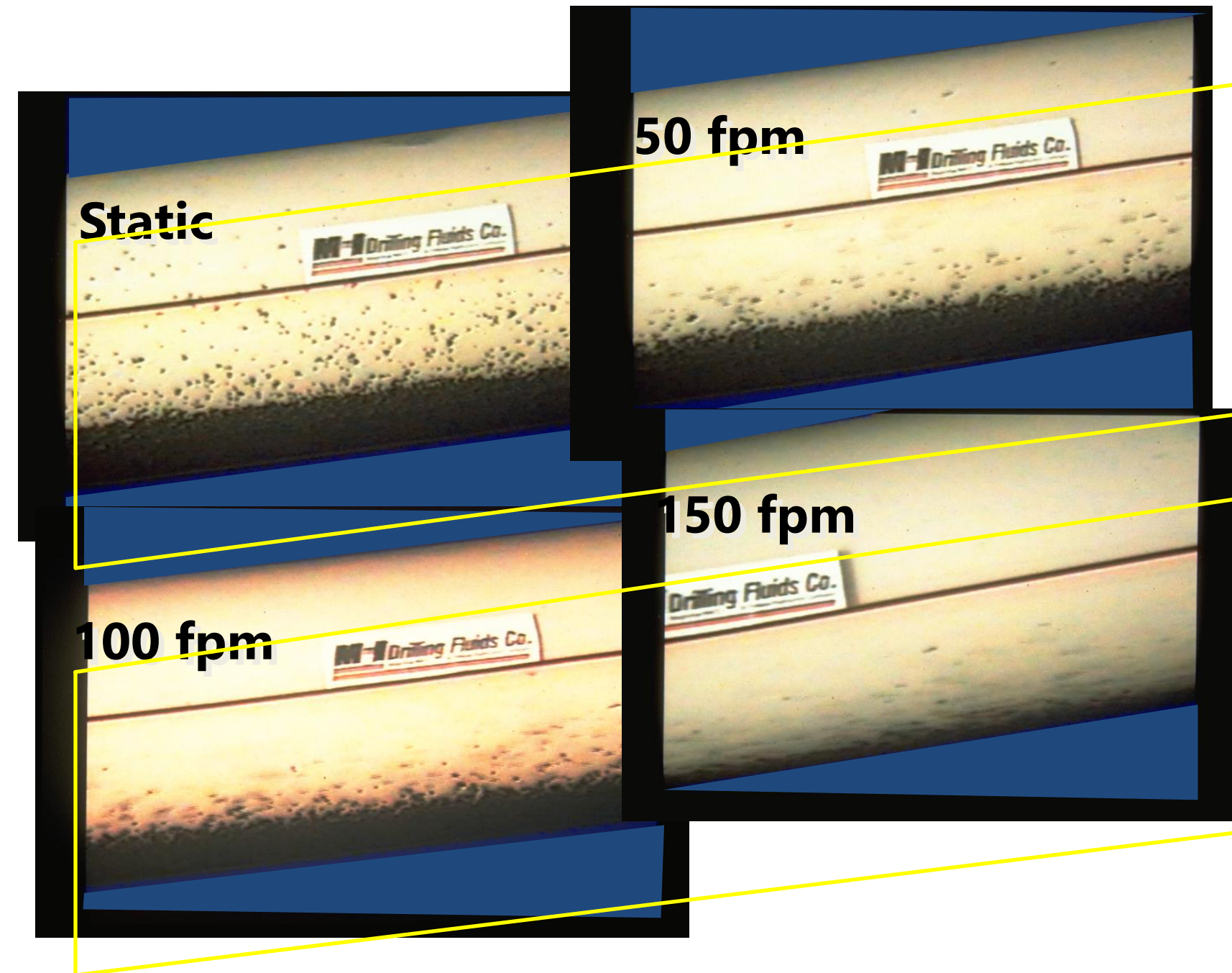
- ✓ High-velocities can erode beds and transport cuttings.
- ✓ Turbulent flow is effective in high-angle, small diameter intervals in competent formations.

Laminar Flow:

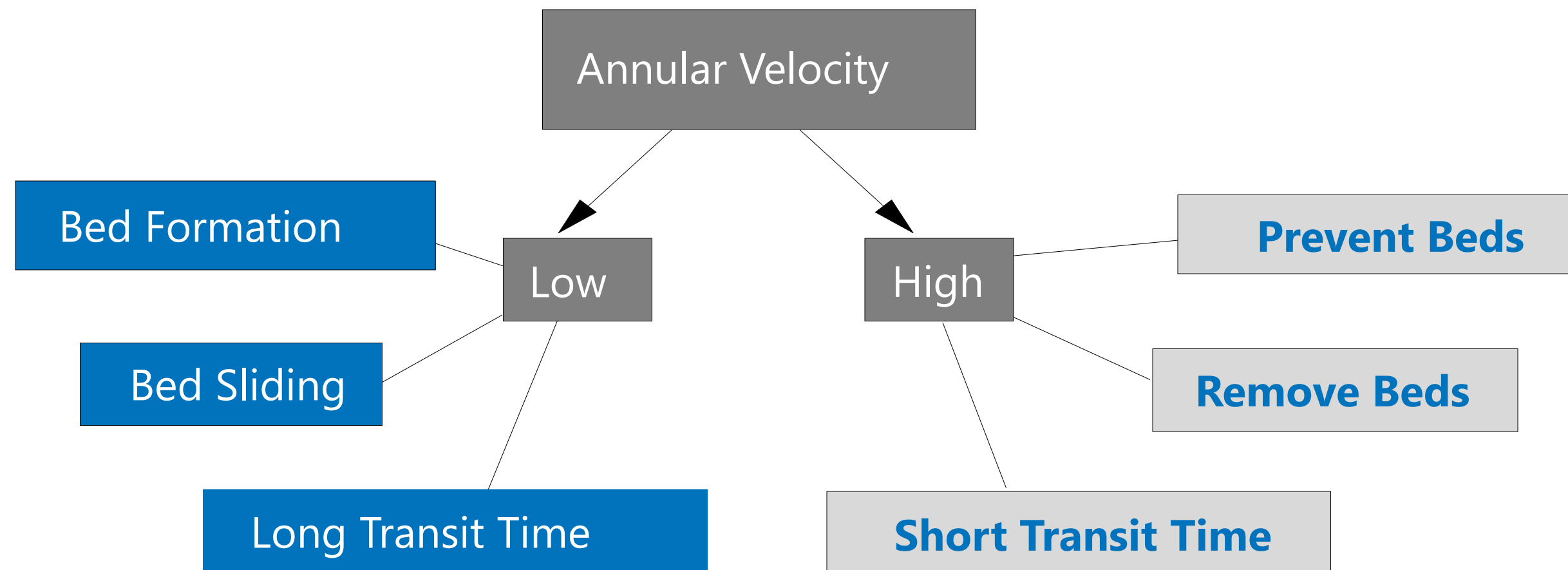
- ✓ Lower flow rates, rheology is critical.
- ✓ Laminar flow is preferred if formations are sensitive to erosion.
- ✓ Hole-cleaning capacity in laminar flow is improved by elevated low shear-rate viscosity and gel strengths.

Hole Cleaning – Flow Parameters

Highly dependent on annular velocity



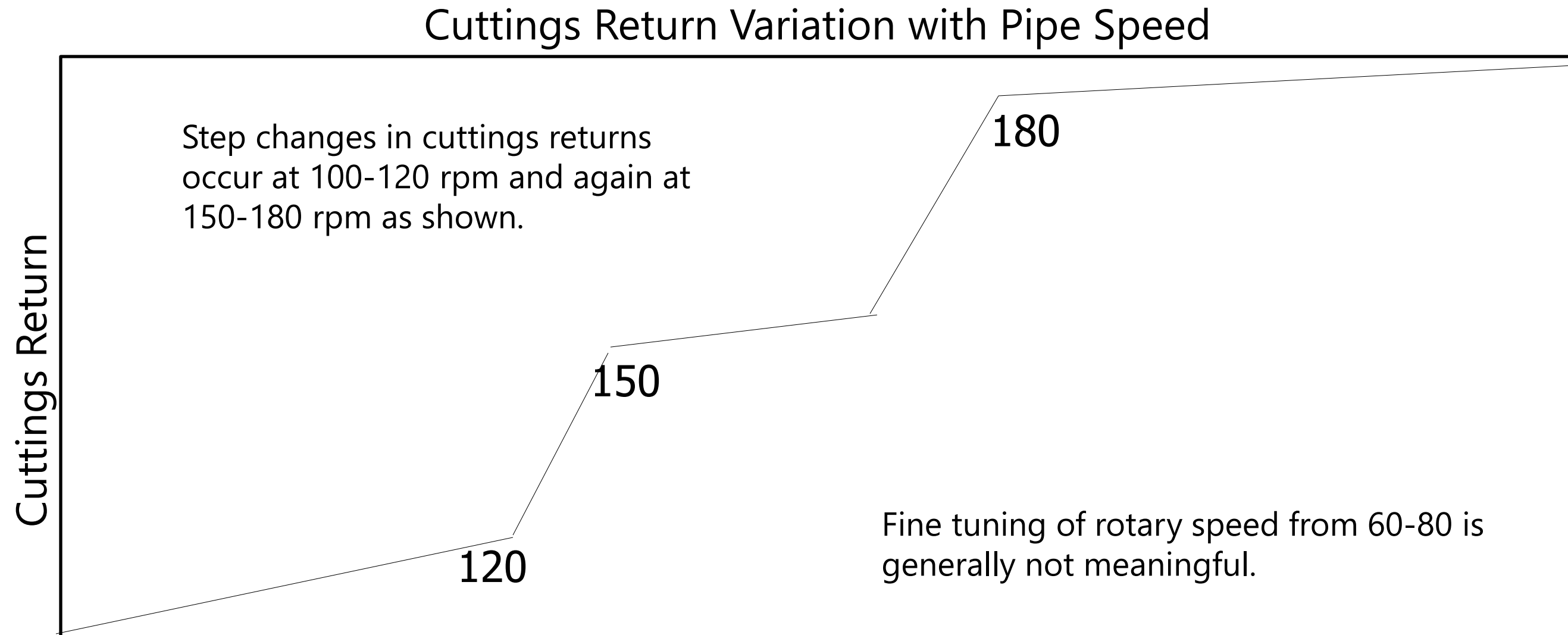
Annular Velocity



Effect of Mud Density

- Hole cleaning is much less problematic in high-density muds due to the increased buoyancy of the cuttings.
 - In some cases, the cuttings may actually float – e.g., 2.1 s.g. mud used to drill salt section.
 - ROP tends to be lower when heavy muds are used.
- A 1ppg increase in mud weight has more benefit on hole cleaning than any change in mud rheology.
- Above 15 ppg it is very unusual to have hole cleaning problems.

Pipe Rotation and Hole Cleaning



High RPM recognized as a key factor in getting good hole cleaning in high angle hole sections.

Hole Cleaning Guidelines

**ROP
Max (ft/hr)**

Angle Interval	Hole Size			
	26"	17 1/2"- 16"	12 1/4"	8 1/2"
0° - 35°	60	110	155	240
35° - 55°	40	75	85	125
55° +		60	75	100

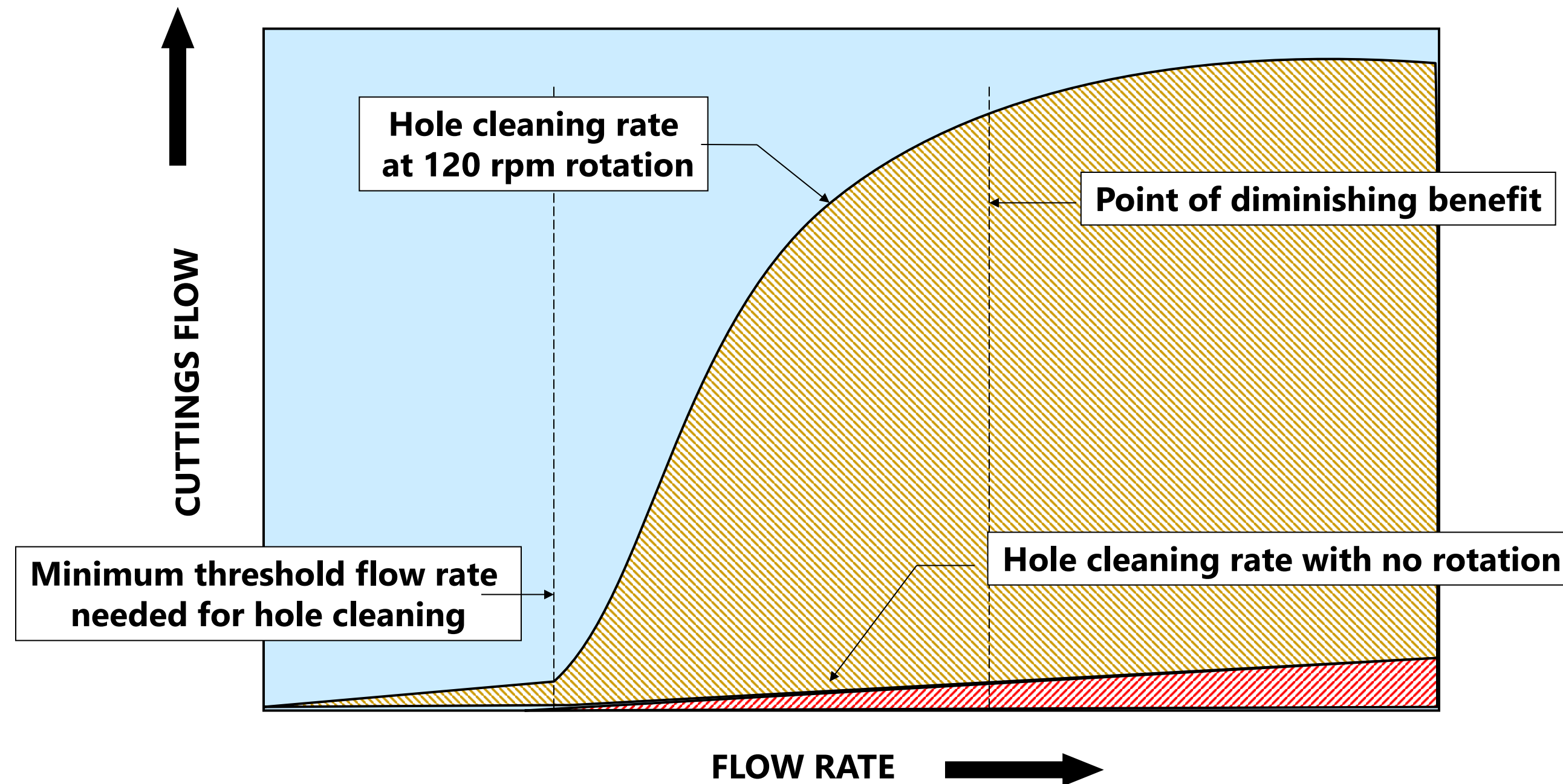
**Circulating
Rate
Min (gpm)**

0° - 35°	700	500	400	300
35° - 55°	1250	950	650	450
55° +		1100	750	500

**Circulating
Strokes Factor
(CSF)**

0° - 35°	2	1.7	1.4	1.4
35° - 55°	2.5	2.5	1.8	1.6
55° +		3	2	1.7

Hole Cleaning vs. Flow rate (> 35°)



The illustration above points out the benefit of rotation in conjunction with circulation when attempting to clean a deviated hole. Rotation should always be used in conjunction with a sweep – especially in

Hole Cleaning Guidelines (> 35°)

- ✓ Maintain sufficient MW for hole stability
- ✓ Circulate at maximum rate for hole size
- ✓ Limit ROP to the max. recommended value
- ✓ Back ream each stand drilled with motor
 - Rotate at high RPM (120+)
 - Continue back reaming if hole conditions dictate
 - Consider wiper trip after long hole section drilled w/motor
- ✓ Consider low viscosity high viscosity tandem sweeps

Salt Drilling Practices

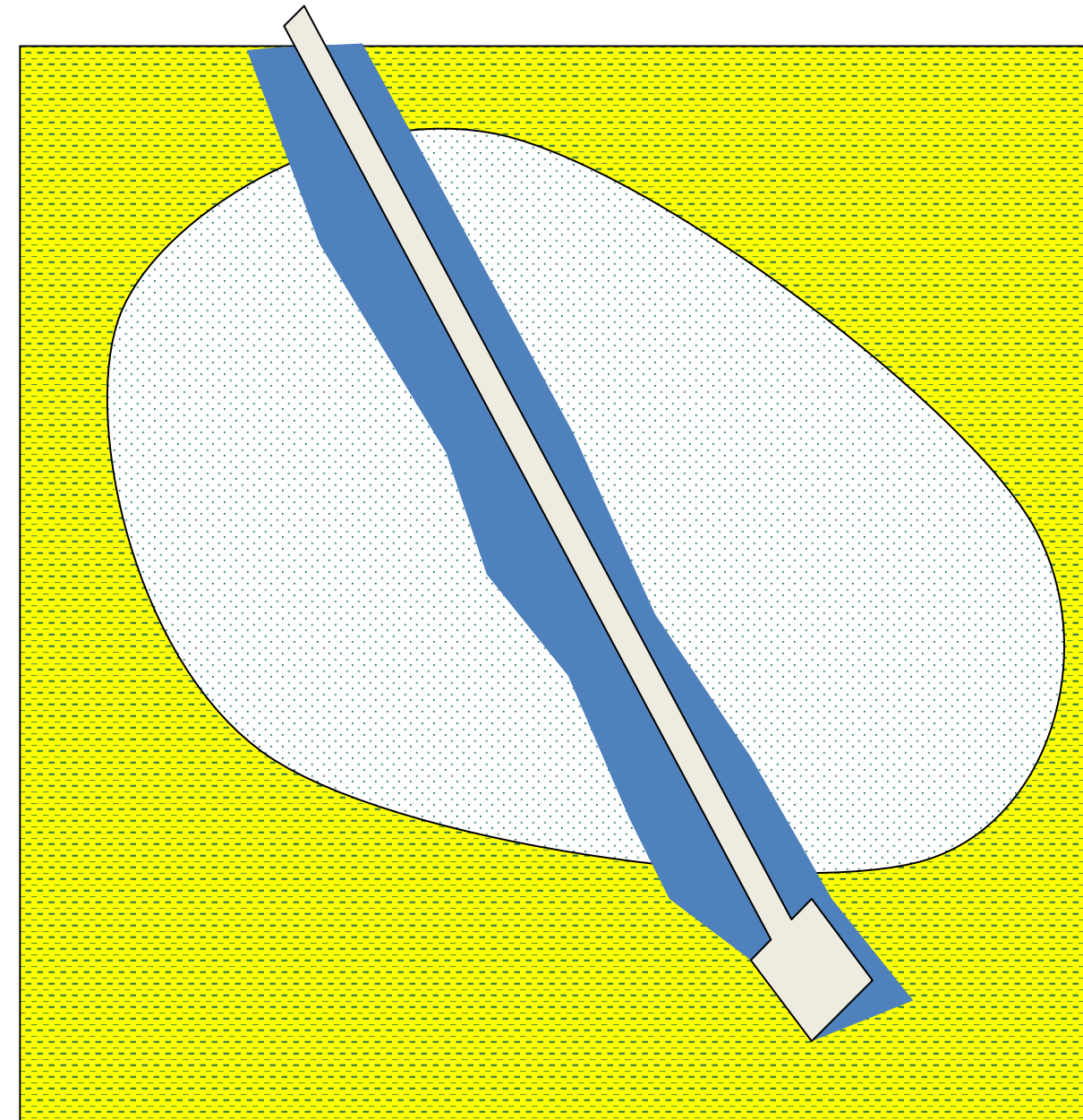
- When first entering the salt, drill no more than one meter before back reaming out.
- Continue to drill in one meter bites, back reaming each bite and taking the bit above the salt for the first 10 meters.
- Pull back above the salt and circulate for 10 minutes , trip in to check for salt movement.(Squeezing)
- If ok, drill normally.
- The above procedure may be required as different salt beds are drilled through.
- The plasticity of salts vary with their make up.

Salt Drilling Practices

- Back ream each stand twice before making connections.
- When drilling out of the salt be prepared for additional squeezing along the bedding plane.
- Have fresh water pills ready for use.
- Some form of lubrication can be added without compromising the washing aspect.
- Limit your jarring loads to avoid buckling the pipe when jarring downwards.
- With water-base mud there is usually wash outs present.
- Limit the jarring and optimize the pill use.
- Check trips are to be considered.
- Back ream on check trips only if necessary.

Salt Exit Strategy

- Hole instability can be expected. Conduct wiper trips every 100' below BOS (for the first 500 ft) to help gauge stability.
- After exiting salt, pickup and space out to drill down an entire stand without the need for a connection.
- Every 10' drilled below the base of salt, continue to rotate and circulate while picking up into the salt.



Q & A

Session 2





Abdullah Adel Abdel Aziz

Country Manager

Stuck Logging Tools

Stuck Wireline

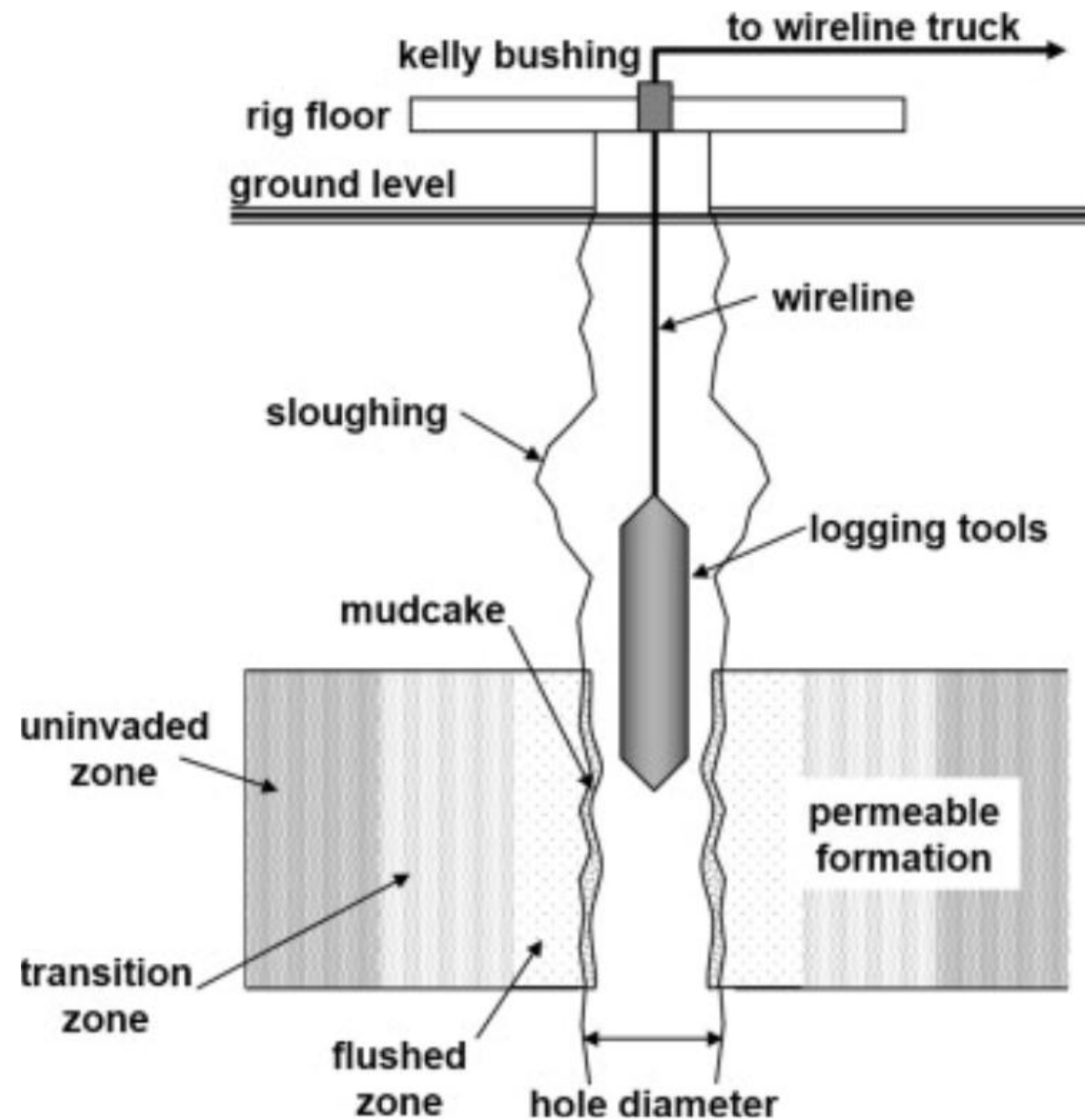
Stuck Mechanism and Recovery Method

Wellbore geometry

Freeing Techniques

Stuck Pipe Prevention

Logging Tools



(Baker et al, 2015)

A logging toolstring is an assembled string of specialized cylindrical downhole sensors and instruments lowered into a wellbore to measure physical properties of subsurface rock formations and fluids

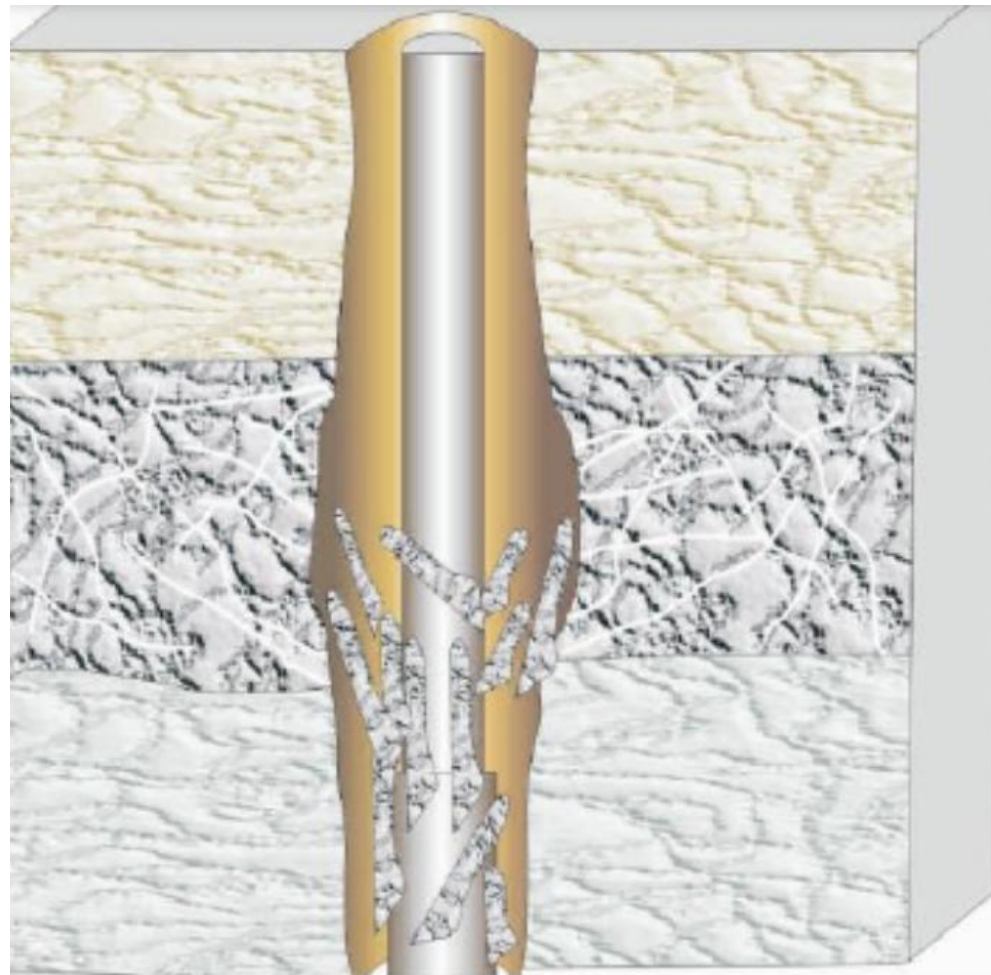
How a Logging Tool Becomes Stuck?

1. Tool runs downhole
2. Tool reaches problematic interval
3. Pulling force increases
4. Tool cannot be recovered.

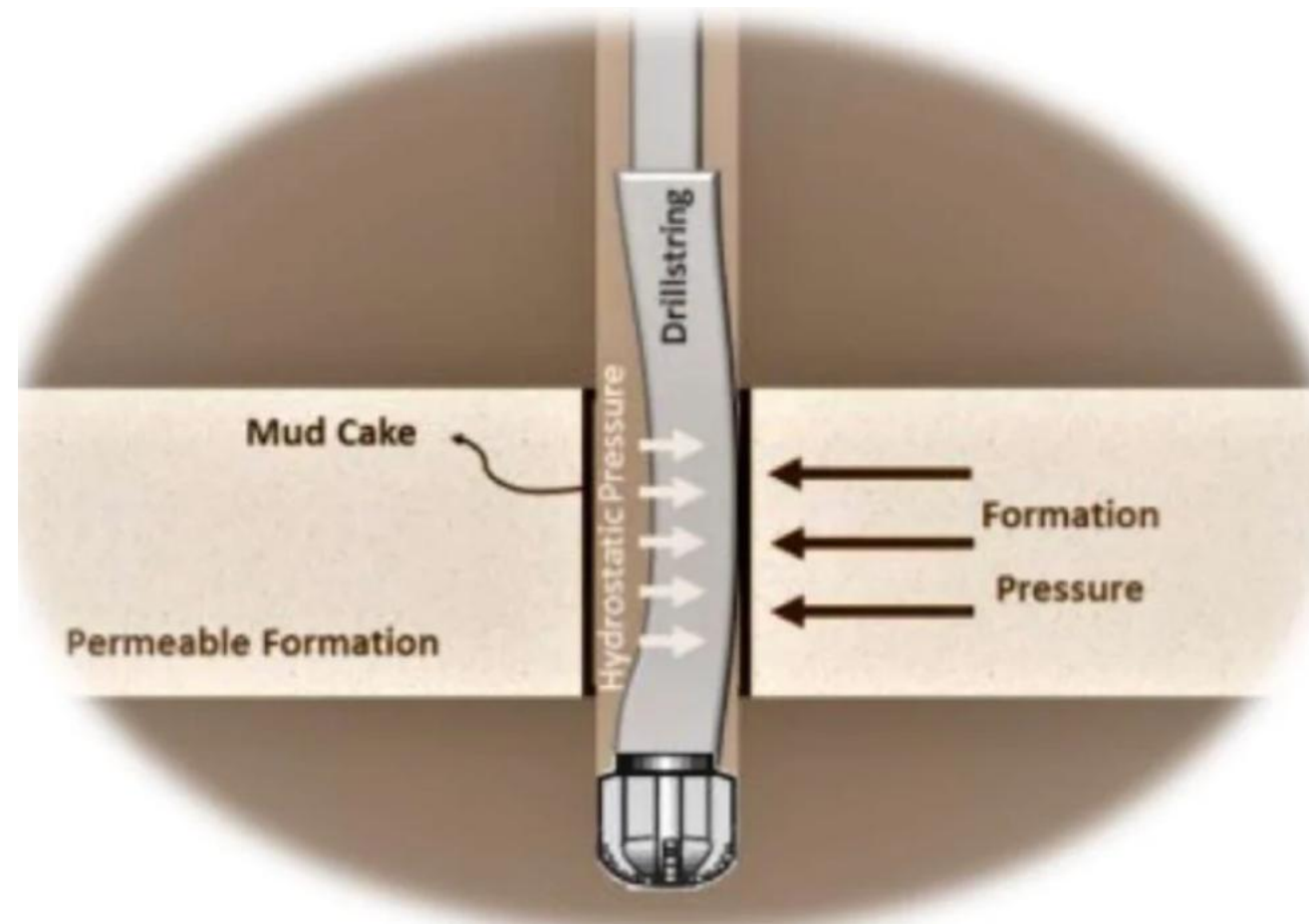
Possible mechanisms: differential sticking, keyseat, restriction, or debris.

Stuck Logging Tools

Logging tools can become stuck because of open-hole instability, differential sticking, keyseating, hole restrictions, debris accumulation, or excessive toolstring contact.



open-hole instability (Nmegbu et al, 2014)

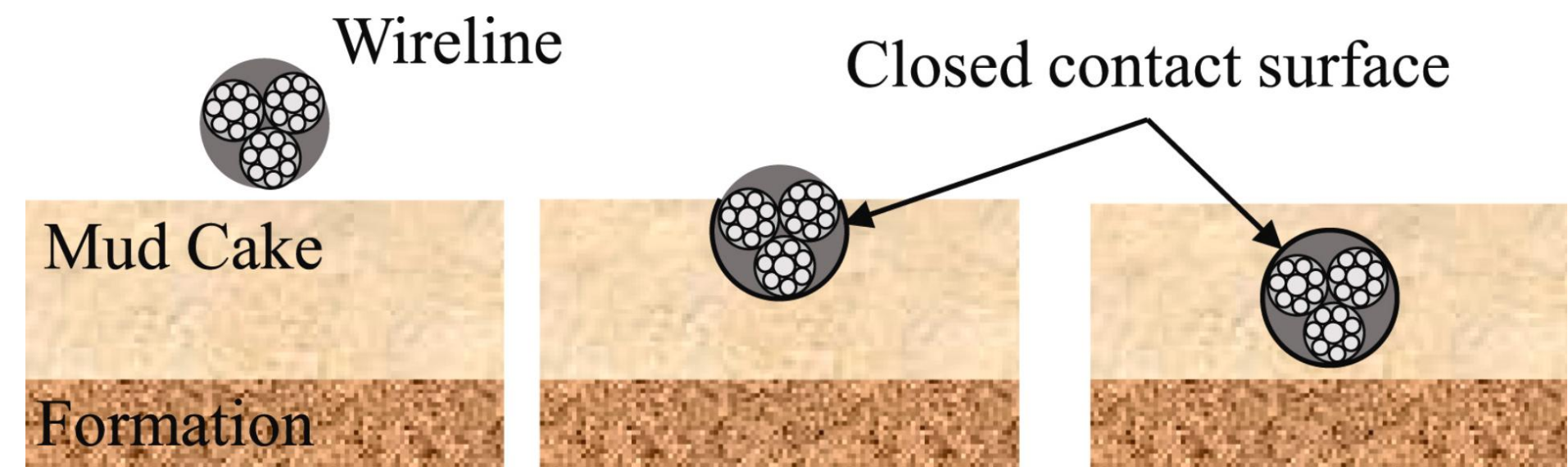
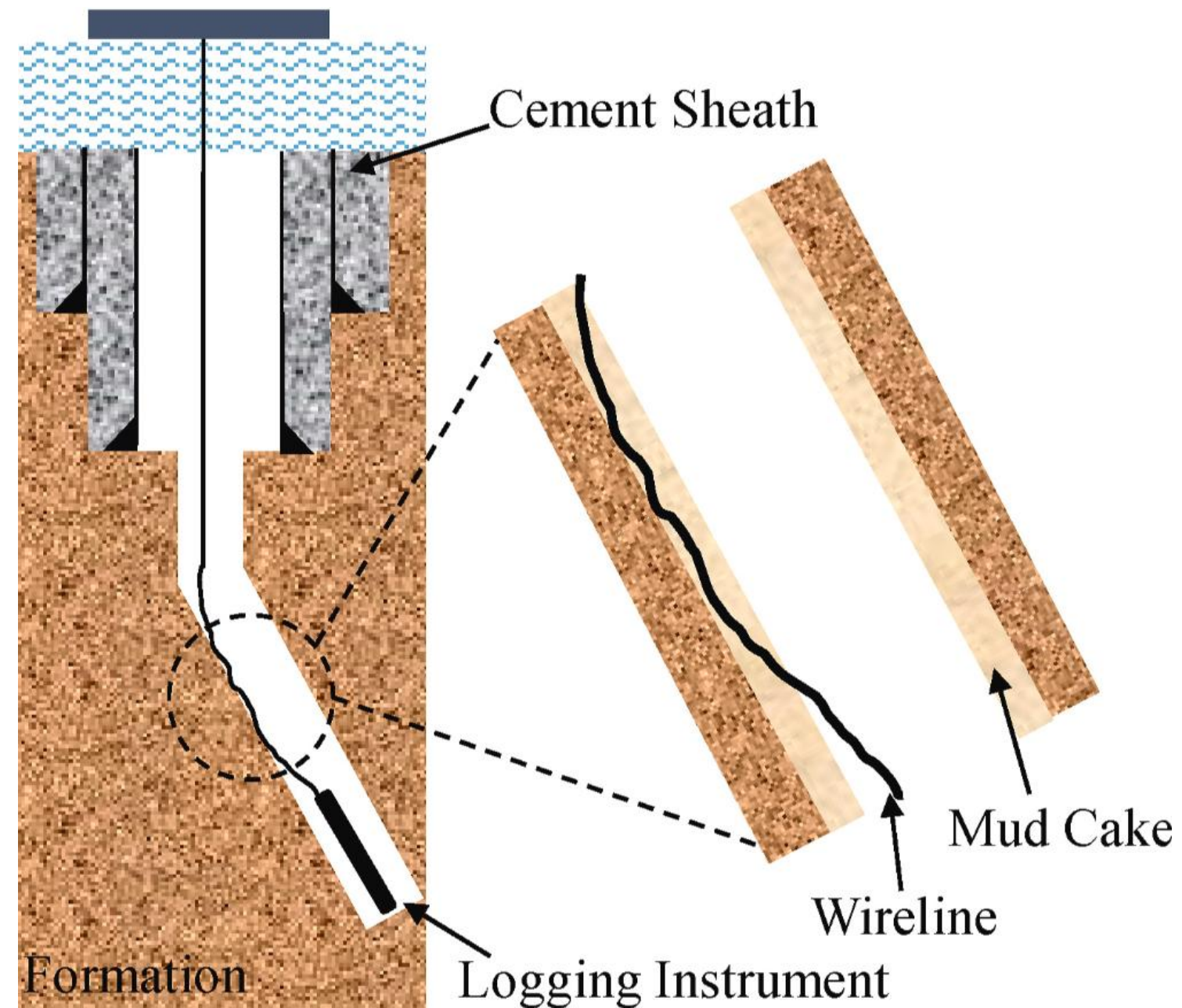


Differential sticking (Abrar et al, 2018)

If the tool is stuck, excessive surface tension can result in **cable failure** or **weak-point activation**.

Stuck Wireline

The variable formation pressure system and complex well structure in these wells result in frequent stuck during the wireline logging operation.



(Qu et al., 2023)

”

Wellbore Geometry

Well bore Geometry

Sticking occurs, when changes in the shape of the well bore cause the movement of the string to be impeded.

If the make up of the BHA. is changed, a similar effect can occur.

Key Seat

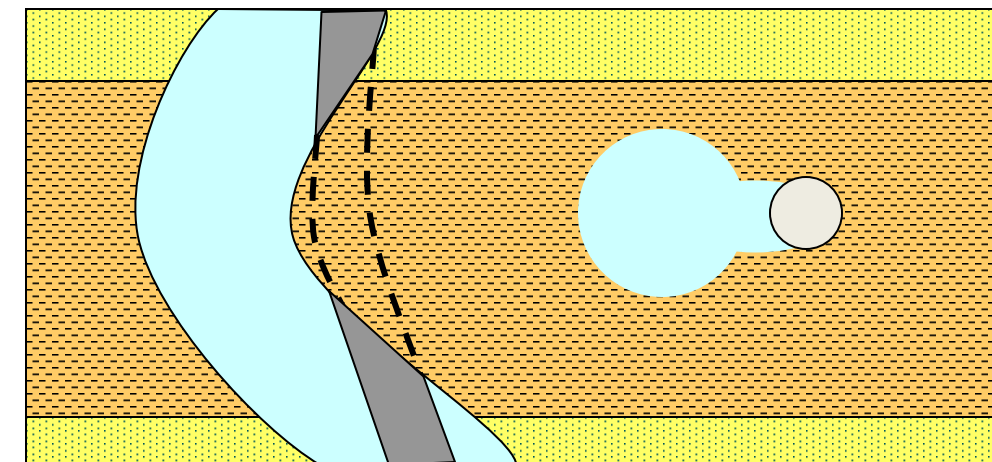
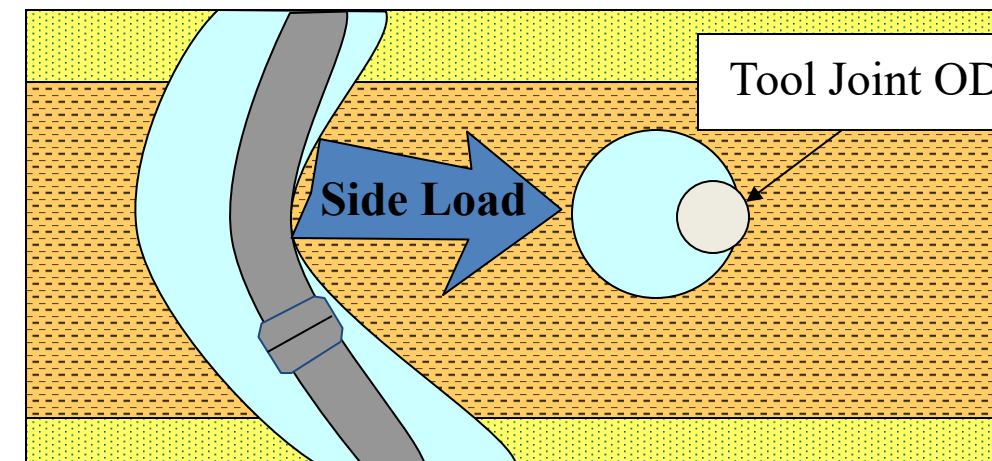
Cause

- ✓ Abrupt change in angle or direction in medium soft formation
- ✓ High string tension and pipe rotation wears a slot into the formation
- ✓ While POOH, the drill collars jam into the slot or “key seat”

Warning Signs

- ✓ High angle dogleg in upper hole section
- ✓ Long drilling hours w/no wiper trips through dogleg
- ✓ Cyclic overpull at tool joint intervals when POOH

Formation of a Key Seat



Key Seat

Preventive Actions

- ✓ Minimize dogleg severity to 3°/100'
- ✓ Limit overpulls through suspected zone
- ✓ Plan reamer/or wiper trip if dogleg present
- ✓ Run string reamer or key seat wiper if suspected
- ✓ Avoid using square-shouldered stabilizers
- ✓ Minimize casing rat holes

Keyseating

In the context of well intervention, a keyseat refers to a mechanical issue that can occur during operations involving wireline, coiled tubing, or other intervention methods in oil and gas wells. A keyseat occurs when a tool or equipment encounters an obstruction or narrow slot in the wellbore, which prevents further downward movement or retrieval of the tool.

Keyseating during well intervention can happen for various reasons:

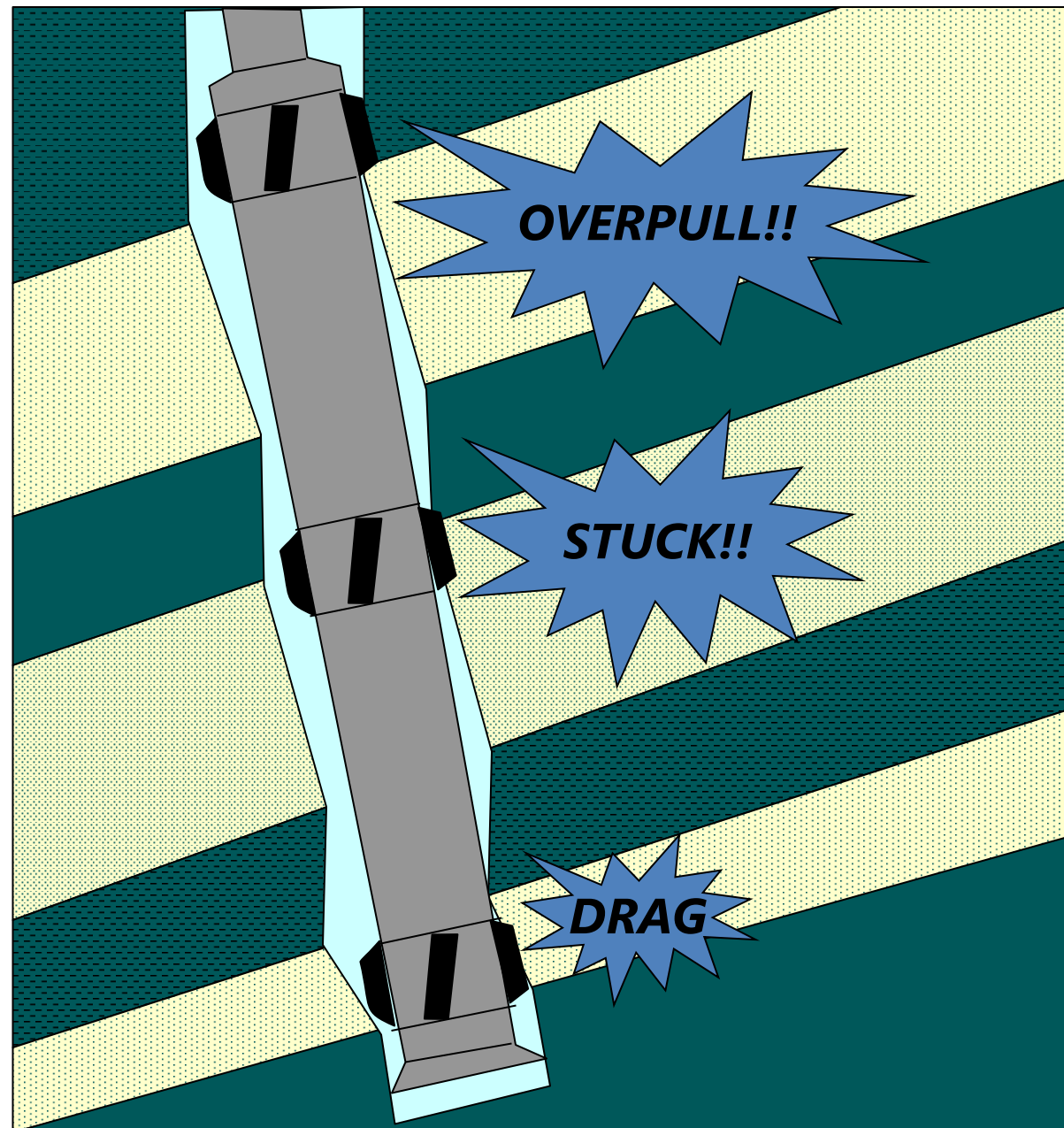
Formation Irregularities: The wellbore may have irregularities caused by the varying hardness of rock formations. If a hard formation ledge is present between softer formations, it can create a narrow groove or keyseat. Tools and equipment can get stuck in these grooves.

Wellbore Deviations: If the wellbore has significant deviations or bends, tools may encounter resistance when they reach those sections, leading to keyseating.

Equipment Interaction: During well intervention, tools such as wirelines or coiled tubing can interact with the wellbore walls, potentially causing abrasion and creating keyseats.

Obstructions: Debris, scale, or other materials within the wellbore can lead to the formation of keyseats as tools try to pass through.

Micro Doglegs and Ledges



Cause

- Hard soft interbedded formations
- Frequent corrections in hole angle or direction
- BHA jams in the successive micro doglegs or pulls into a ledge

Warning Signs

- Prognosed hard/soft interbedded formations
- Frequent angle/direction changes
- Drilling/sliding with downhole motor
- Erratic torque and drag on connections
- Sudden erratic overpull or set down

Micro Doglegs and Ledges

Preventive Actions

- ✓ Minimize BHA changes
- ✓ Minimize direction/angle changes
- ✓ Back-ream frequently when drilling hard/soft formations
- ✓ Slow trip speed through suspected zones
- ✓ Leding reduced by running a packed hole assembly
- ✓ Avoid prolonged circulation across soft interbedded formations
- ✓ Limit initial set-down weight to less than 50% of down drag to minimize effects in a tight zone
- ✓ Do not start angle building operations too close to shoes (start at least 30m below old hole TD)

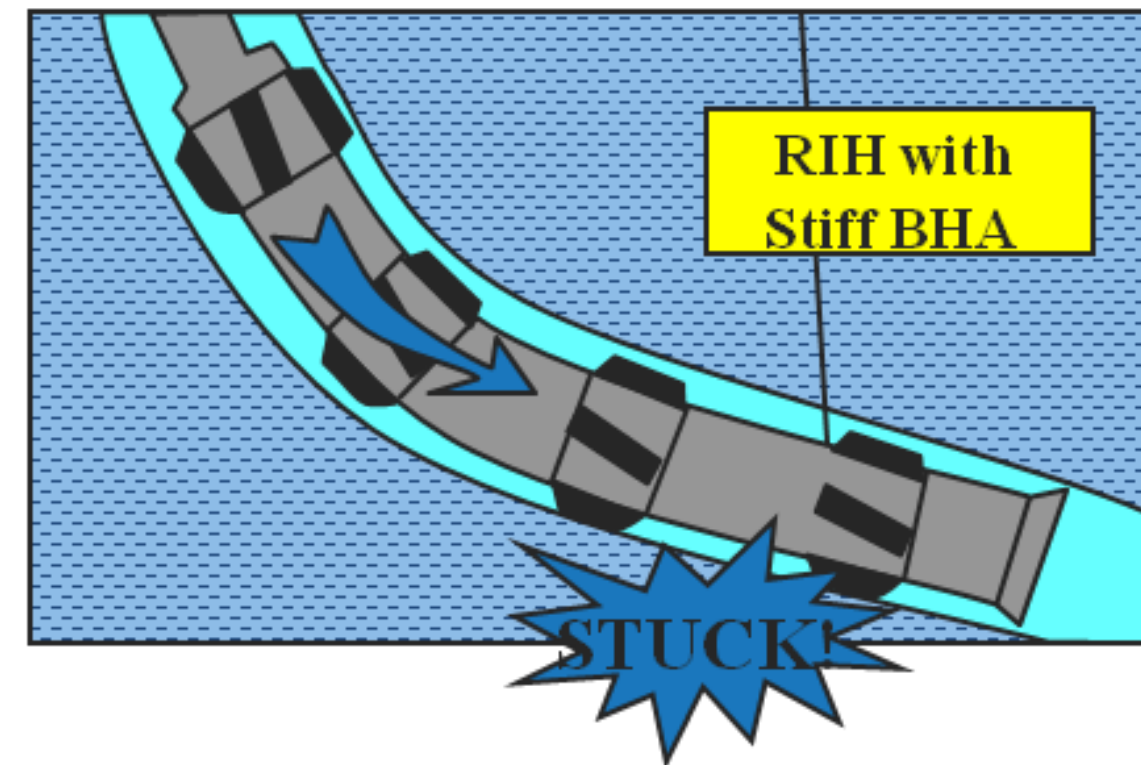
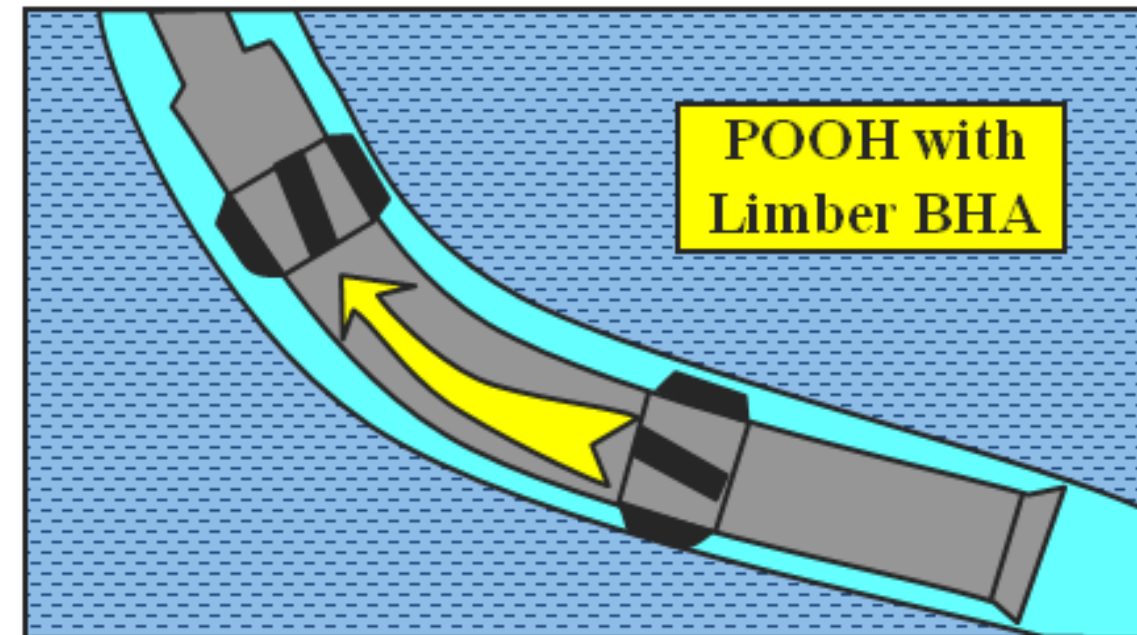
Stiff Assembly

Cause

- ✓ The stiff BHA can not negotiate hole angle/direction changes and becomes jammed.

Warning signs

- ✓ Doglegs present
- ✓ Pulled stabilizers out of gauge
- ✓ New BHA design is picked up
- ✓ Sudden set-down weight

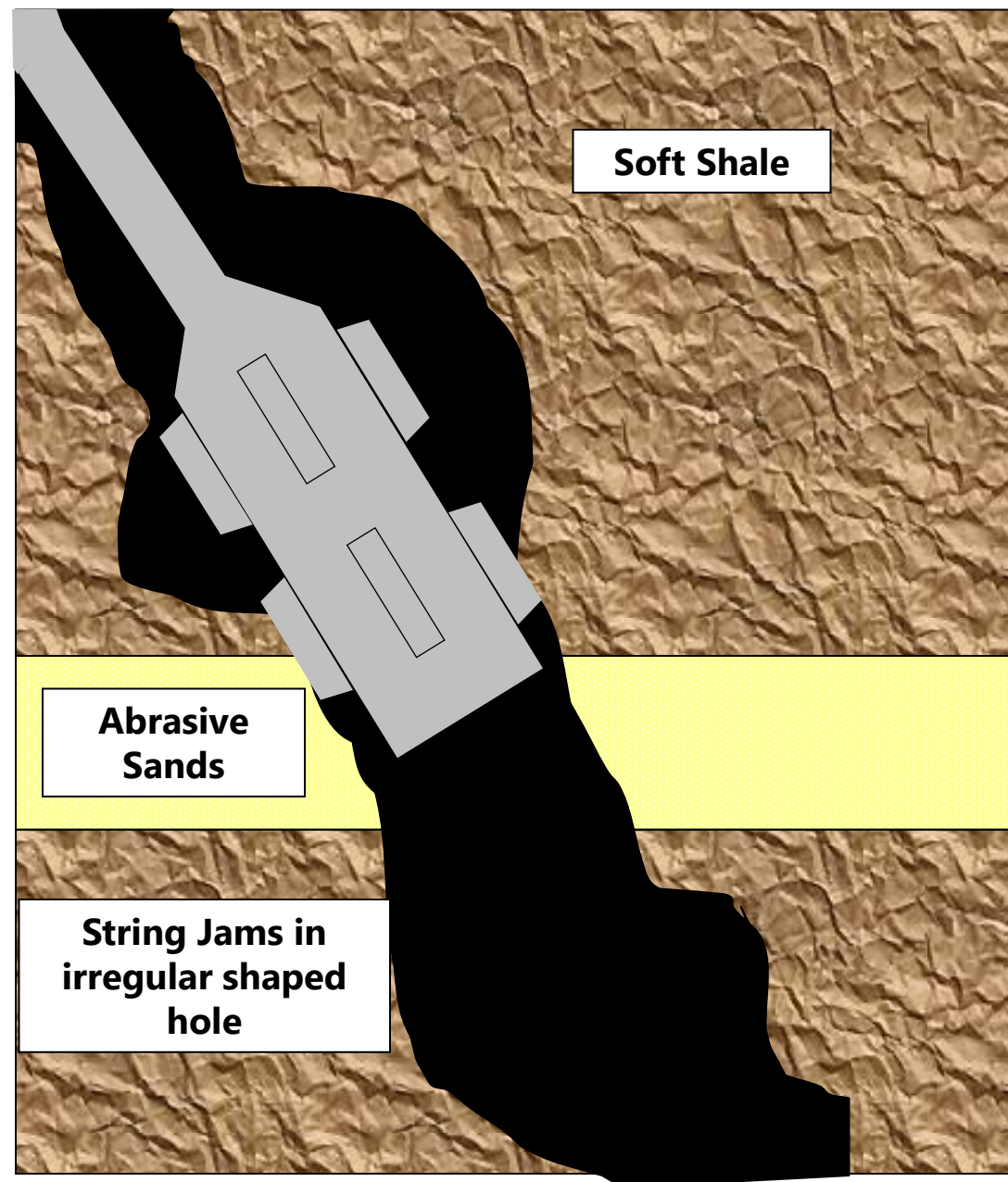


Stiff Assembly

Preventive Actions

- ✓ Minimize BHA changes, consider a reaming trip
- ✓ Limit dogleg severity
- ✓ Slow trip speed before BHA enters suspected zone, plan to ream
- ✓ Limit set-down weight

Undergauge Hole



Causes

- Undergauge bit and stabilizers
- Abrasive formation

Warning Signs

- Tight hole on trip in
- Low ROP due to weight not getting to the bit
- Directional control difficult

Preventive Actions

- Trip in slowly
- Ream to bottom
- Run roller reamer above bit

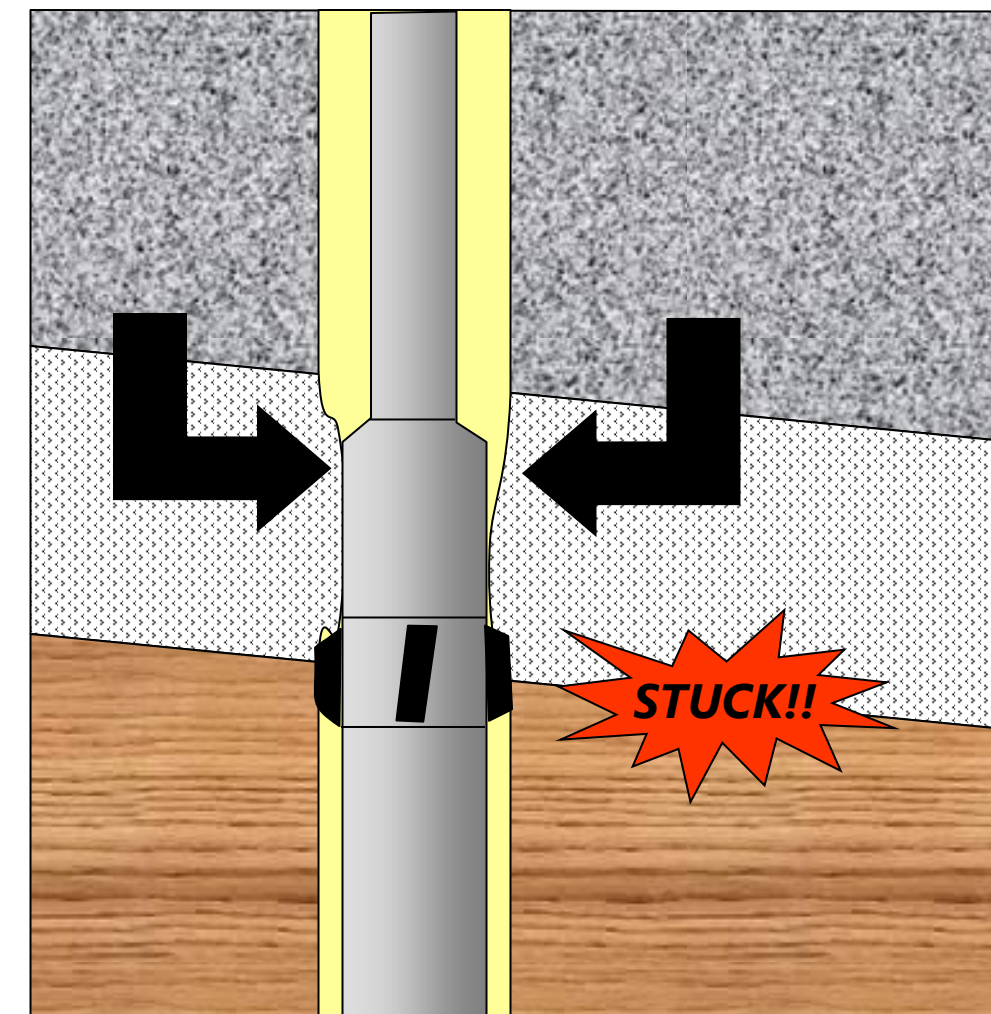
Mobile Formations - Salts

Causes

- Overburden squeezes plastic salt or shale into the wellbore.
- BHA becomes jammed in the under-gauge hole.

Warning Signs

- Prognosed salt or plastic shale
- Sudden increase in overpull or set-down weight
- Sudden torque increase with fast moving plastic formation
- Generally occurs when POOH
- Increase in mud chlorides
- Pump pressure increase

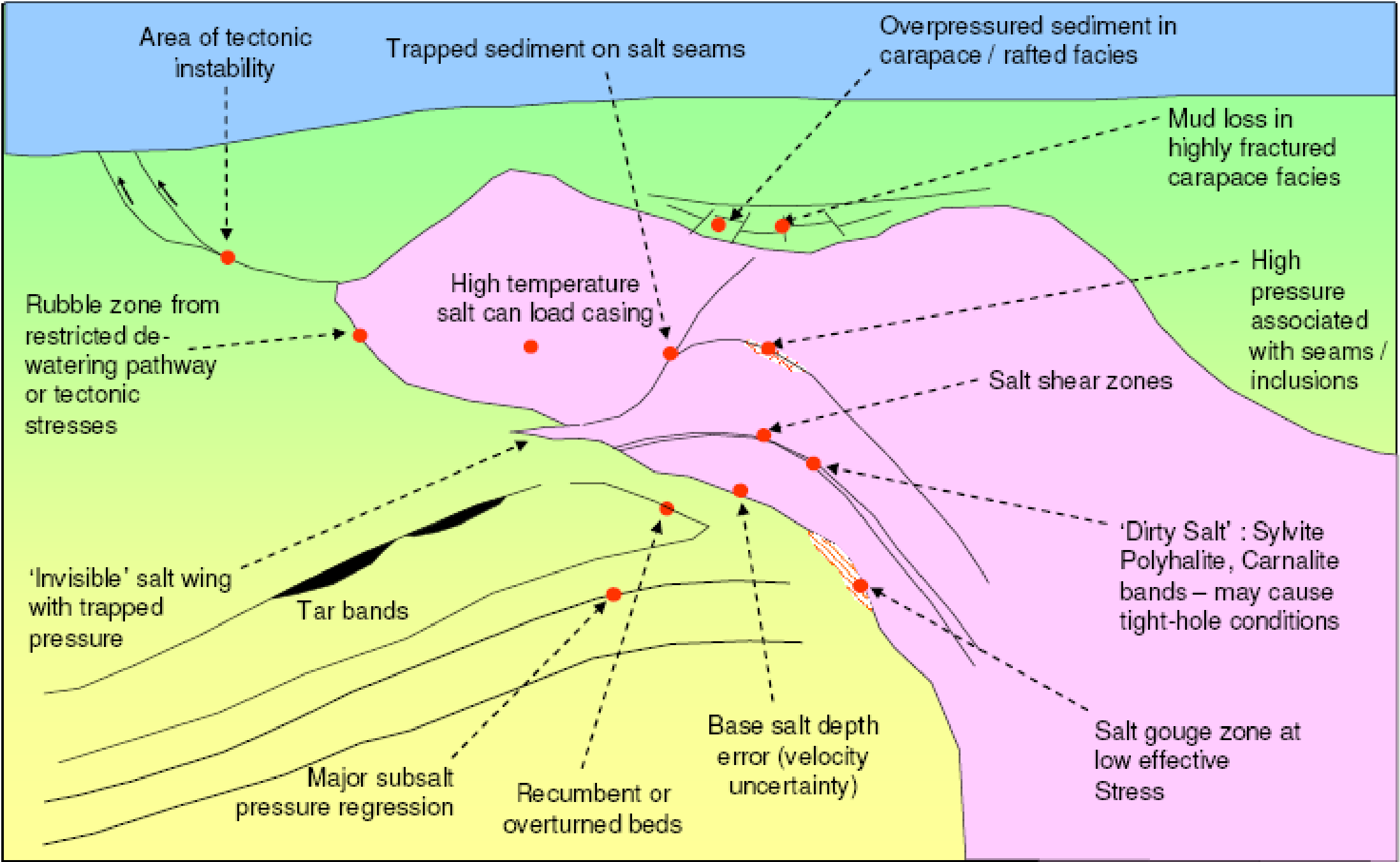


Mobile Formation Salts

Preventive Actions

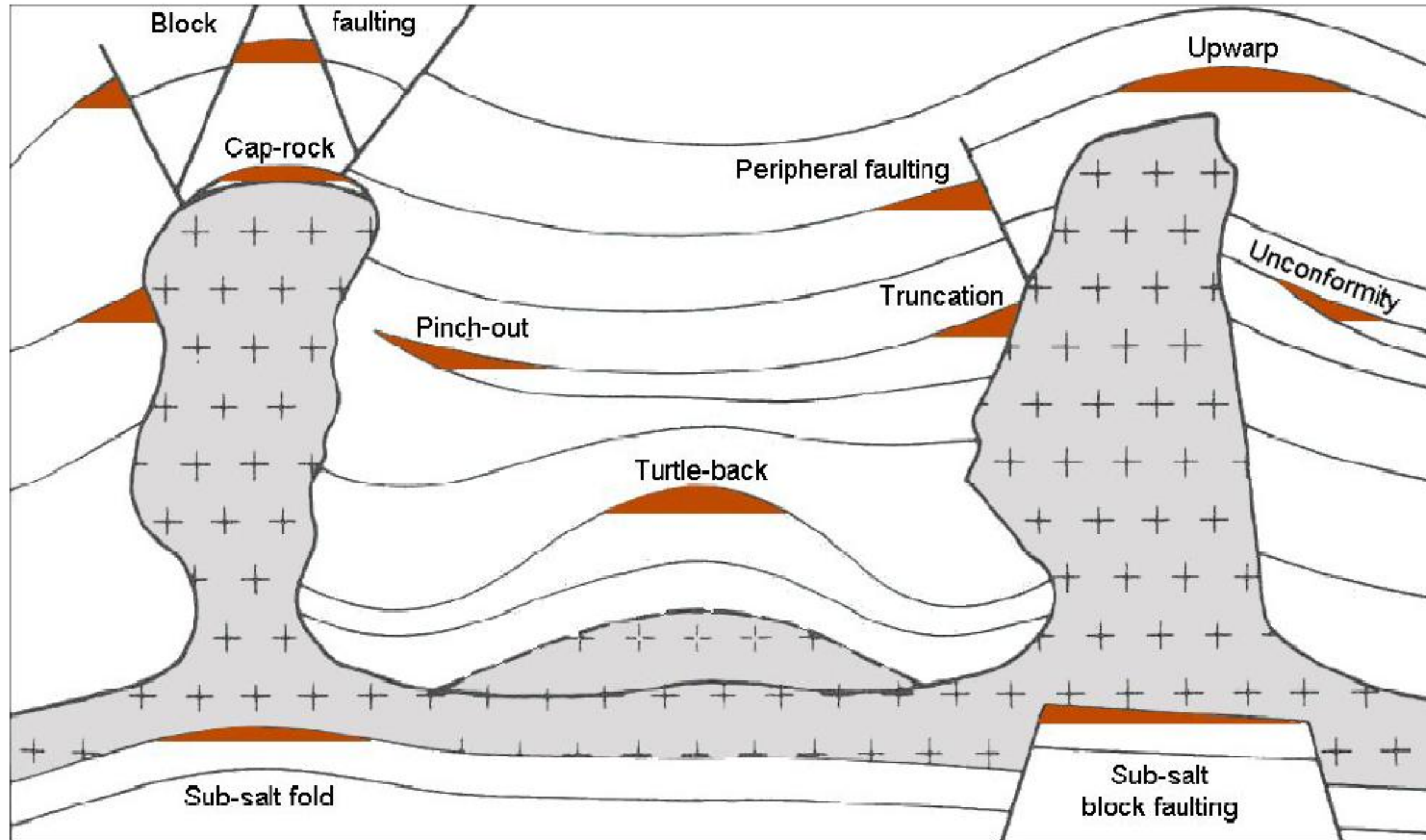
- ✓ Select the correct mud system
- ✓ Maintain sufficient MW
- ✓ Plan frequent wiper trips
- ✓ Consider eccentric bits
- ✓ Identify salt dome
- ✓ Monitor mud chlorides and mud resistivity
- ✓ Trip in slowly and ream

Potential Salt Hazards



Potential Salt Geomechanical Hazards (after Wilson SPE 95621)

Salt Dome Structural Traps



Squeezing Risks

Squeezing is not uniform throughout the section.

Effected by

- Salt type
- Pressure fluctuations
- Position in the structure

High Risk Areas

- First 10 meters into the salt
- Leaving the salt
- Between salt types interfaces
- E.g. Halite / Anhydrite / Carnallite / Tachydrite

Buckling Risks

- Pipe parting
- “Cork screwing” the Pipe
- Essentially caused by excessive down weights. Often accepted as normal in gauge holes.
- Early indications require to be acted upon.
 - E.g. Excessive salt re-crystallization.
 - Excessive bottoms time. (carbide detection)

Additional Problems

- Large wash outs can lead to casing hanging up in them.
- Cement jobs can be poor.
- Pipe sticking can be frequent.

”

Freeing Techniques

Determining Sticking Mechanism

- Movement at time of sticking – making a connection

- After sticking
 - No downward movement
 - No rotation
 - Full circulation

- Use the worksheet and establish the sticking mechanism

Stuck Pipe ID Worksheet

Pipe Motion Prior To Sticking?	Pack-Off/ Bridge	Differential Sticking	Wellbore Geometry
Moving Up	2	0	2
Rotating Up	0	0	2
Moving Down	1	0	2
Rotating Down	0	0	2
Static	2	2	0
Pipe Motion After Sticking?			
Down Free	0	0	2
Down Restricted	1	0	2
Down Impossible	0	0	0
Pipe Rotation After Sticking?			
Rotate Freely	0	0	2
Rotation Restricted	2	0	2
Rotation Impossible	0	0	0
Circulating Pressure After Sticking?			
Circulate Freely	0	2	2
Circulation Restricted	2	0	0
Circulation Impossible	2	0	0
Totals			

E.G. When pulling out of the hole on the fifth stand. 50K more than up drag was pulled. Unable to go up, unable to go down, unable to rotate, full circulation is established.

To Establish the Mechanism

Step 1

Identify your answer to the first question then highlight the numbers on that line.

Step 2

Repeat the above for each of the remaining questions.

Step 3

Add up the highlighted numbers in each mechanism column and enter it opposite totals.

Step 4

The sticking mechanism is the one with the highest score!!

Stuck Pipe ID Worksheet

Pipe Motion Prior To Sticking?	Pack-Off/ Bridge	Differential Sticking	Wellbore Geometry
Moving Up	2	0	2
Rotating Up	0	0	2
Moving Down	1	0	2
Rotating Down	0	0	2
Static	2	2	0
Pipe Motion After Sticking?			
Down Free	0	0	2
Down Restricted	1	0	2
Down Impossible	0	0	0
Pipe Rotation After Sticking?			
Rotate Freely	0	0	2
Rotation Restricted	2	0	2
Rotation Impossible	0	0	0
Circulating Pressure After Sticking?			
Circulate Freely	0	2	2
Circulation Restricted	2	0	0
Circulation Impossible	2	0	0
Totals			

Questions:

- 1. What was pipe motion prior to sticking?
- 2. What is pipe motion after sticking?
- 3. Can we rotate?
- 4. Can we circulate?

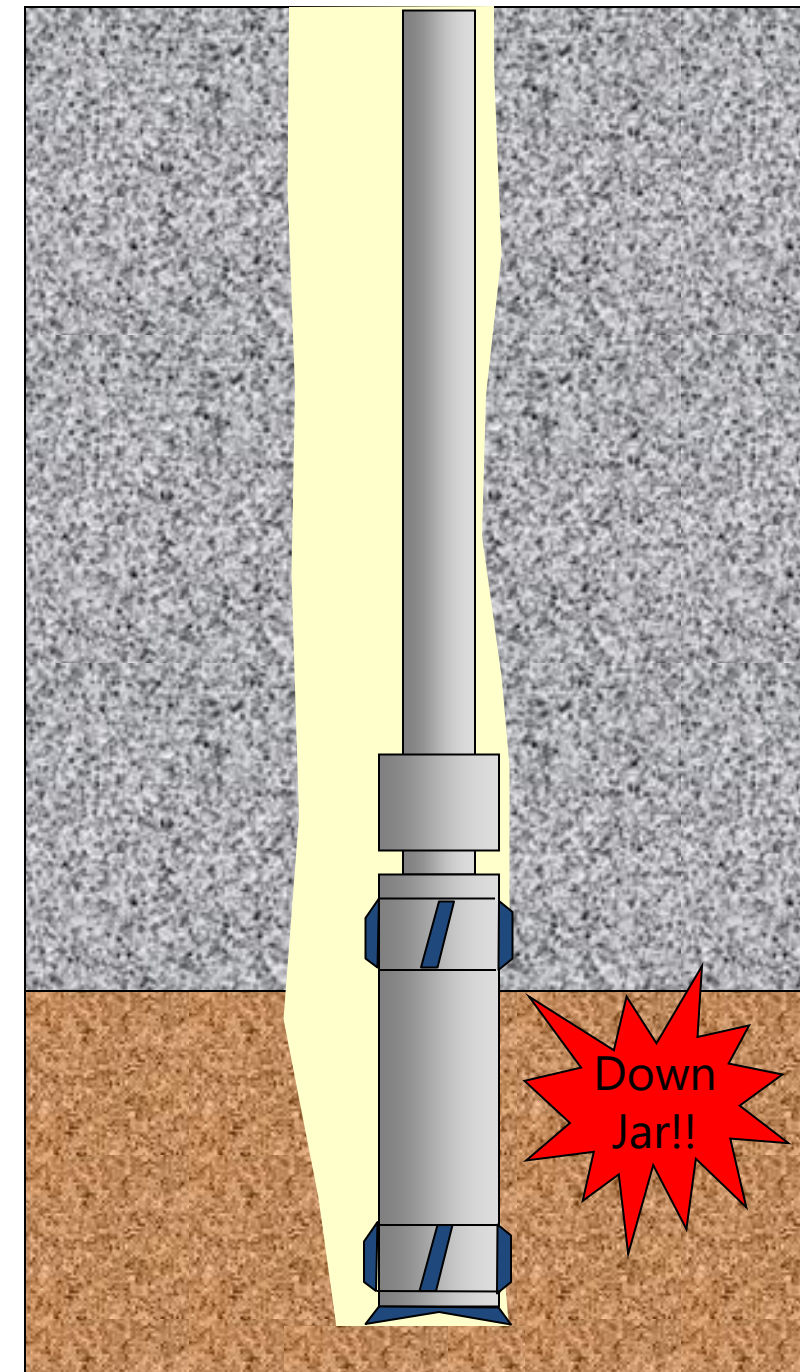
Freeing Methods – Differential Pressure

Initial action

- Circulate at maximum rate to clean cuttings from BHA.
- Work maximum torque down to the stuck point and hold the torque in the string.
- Stop or reduce pump speed.
- Slack off to maximum set-down and allow time for down jar blow.
- Continue working torque and jarring down.

Secondary action

- Spot pipe releasing pill and continue jarring down. Allow time for pill to work (18 - 20 hours).
- When pipe is free; rotate and work the string and circulate at maximum rate.



Freeing Methods - Differential Pressure

Torque and Slump

This can also be tried at the time the pipe becomes differentially stuck.

- STEP 1** Have the driller find neutral weight on the weight indicator. This is the point where the pipe is in tension under its own weight – no pick-up weight. Record this neutral weight.
- STEP 2** Make one rotation of the rotary for every 10,000# of weight but not to exceed the torque limits of the drill pipe tubes or tool joints.
- STEP 3** Drop the block. The block should be lowered at a rapid pace, not slowly and then suddenly stopped. Make sure the brakes are applied before the block reaches the rig floor. The object is to suddenly “shower down” a generous weight on the torqued drill string.

This can be done in conjunction with using jars applying downward blows. It may not be a bad idea to start preparing a spotting pill while this is taking place.

Freeing Methods - Differential Pressure

Spotting A Diesel Pill

- Quickest pill - diesel
- Penetrates the thick filter cake, wetting it and softening the cake
- Spot diesel around the entire BHA (unless the free point is known)
- Diesel lighter in density than the mud being used – make calculation with respect to BHP reduction

Freeing Methods - Differential Pressure

If the BHA is in a vertical hole the following can be performed:

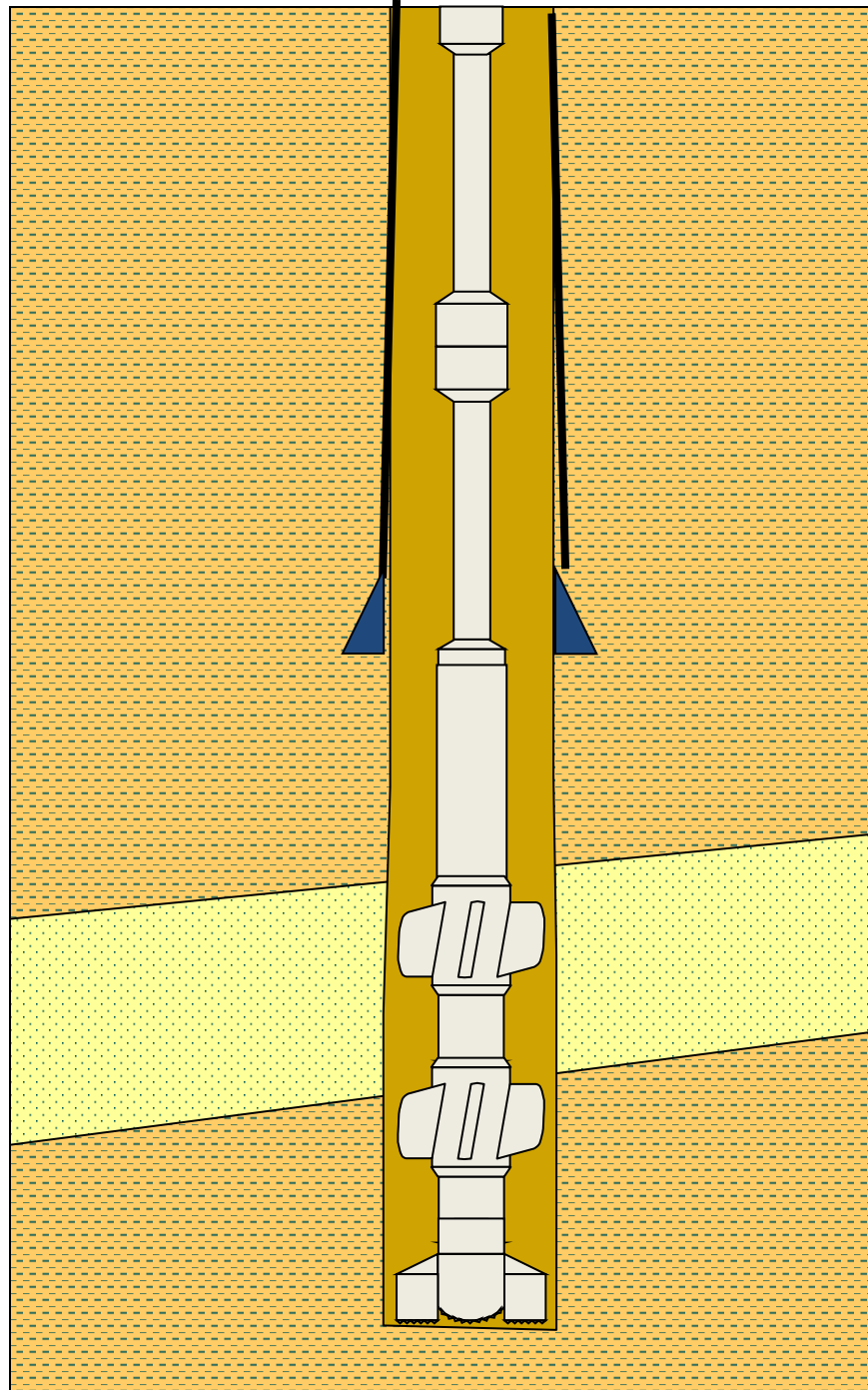
$$\left(MW_{PPG} - Diesel_{PPG} \right) \times .052 \times Length_{BHA}$$

If the BHA is in a deviated hole the following can be performed to estimate the BHP reduction:

$$\left(MW_{PPG} - Diesel_{PPG} \right) \times .052 \times \left(Length_{BHA} \times \cos \angle \right)$$

After the diesel or spotting agent is in place, torque up the pipe and hold the torque. Occasionally jar down. When using a spotting agent, follow the manufacturer's directions for use.

Freeing Methods - Differential Pressure

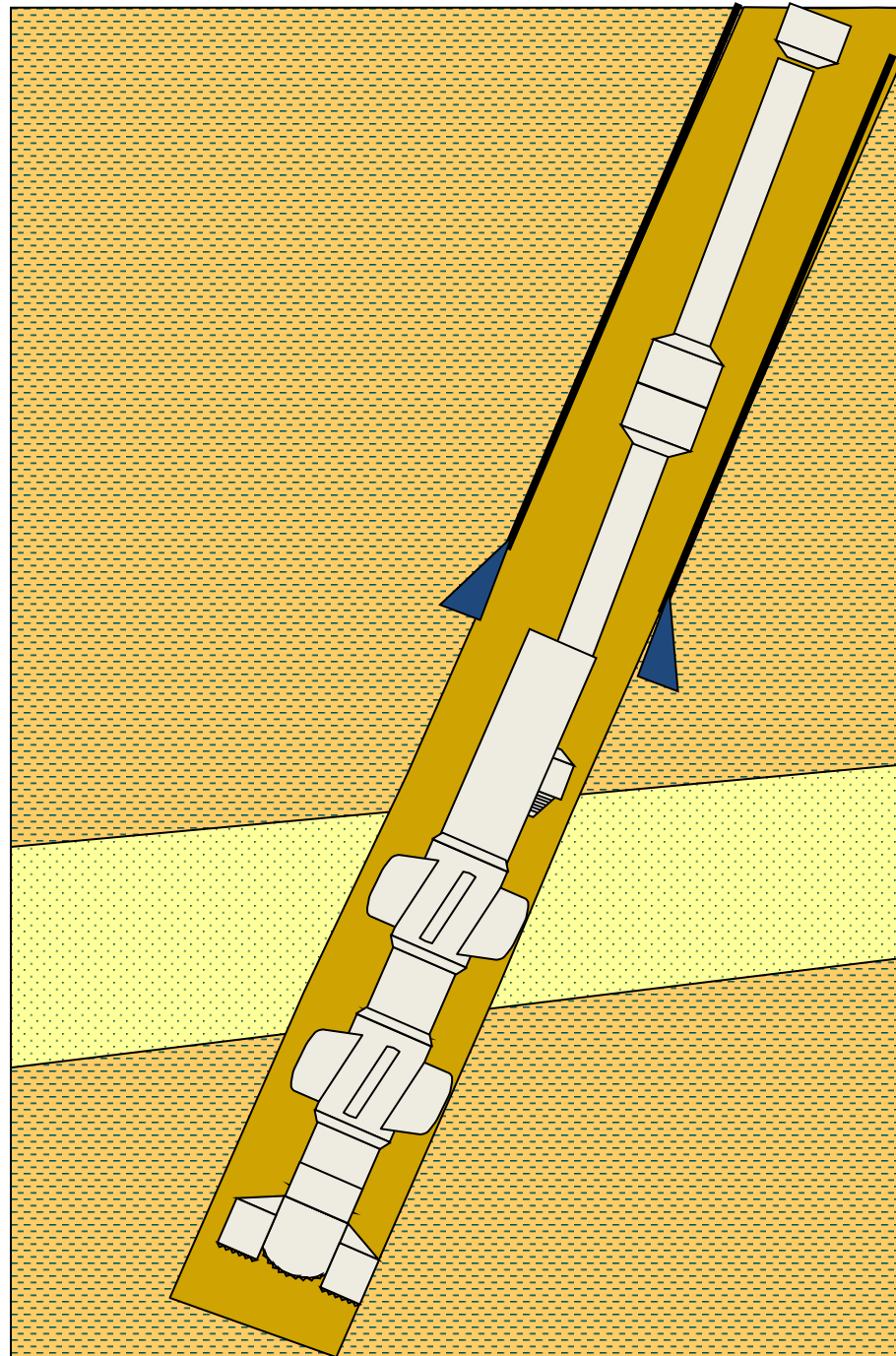


Vertical Hole Example

The mud weight in use is 11.5 ppg. The diesel weighs approximately 6.6 ppg. The BHA length is 762'. Based on this information, what is the reduction in BHP?

BHP Reduction = _____ psi

Freeing Methods - Differential Pressure



Deviated Hole Example

The mud weight in use is 11.5 ppg. The diesel weighs approximately 6.6 . The BHA length is 762' and the BHA is at a 32.5 degree angle. Based on this information, what is the reduction in BHP?

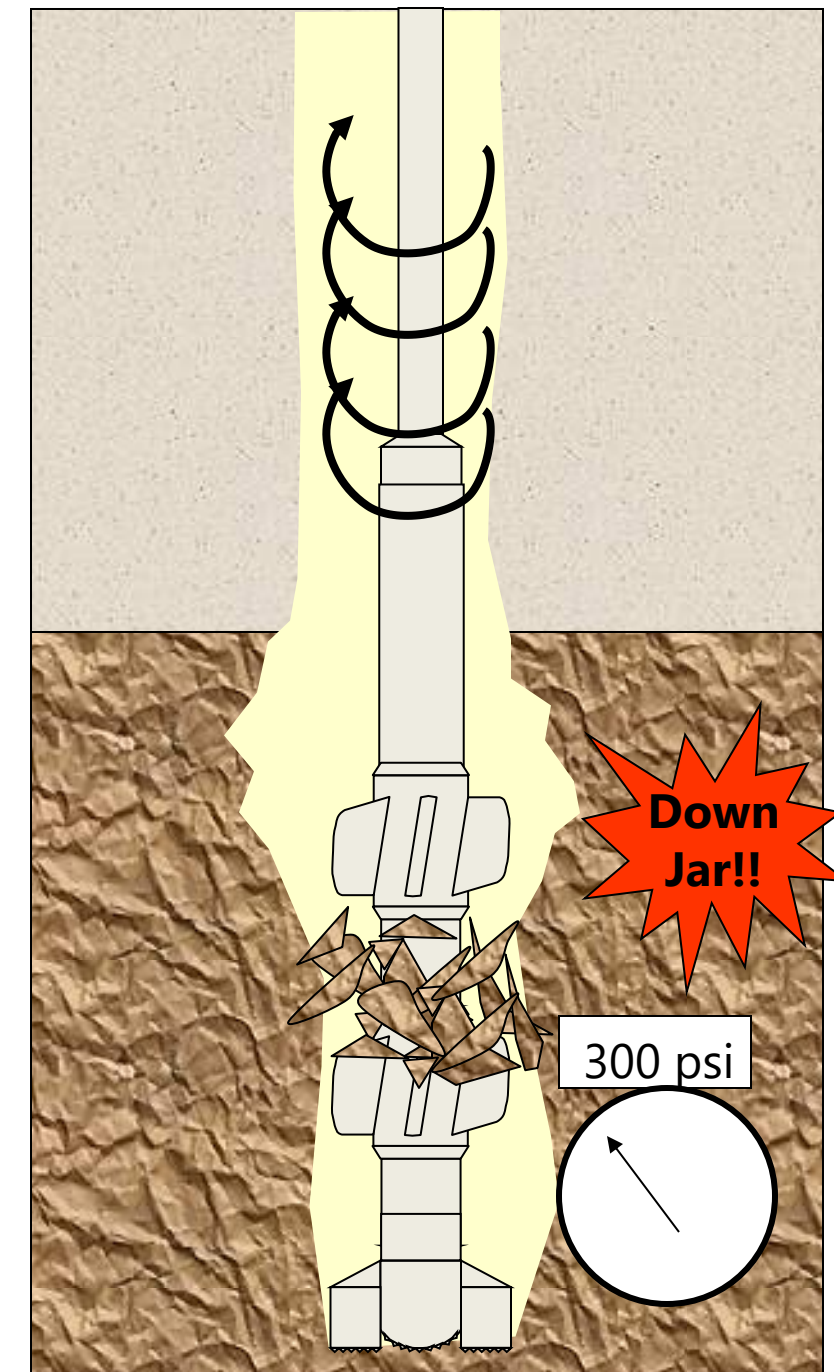
BHP Reduction = _____ psi

Make sure that the BHP reduction will not induce a kick.

Freeing Methods – Hole Pack-Off/Bridging

Initial Action

- Attempt to regain full circulation - if totally packed-off leave 200 - 400 psi pressure on standpipe and monitor for bleed off.
- Have string at free hanging weight and work maximum torque to stuck point and then release. Repeat torque and release until circulation is regained or pipe is free.



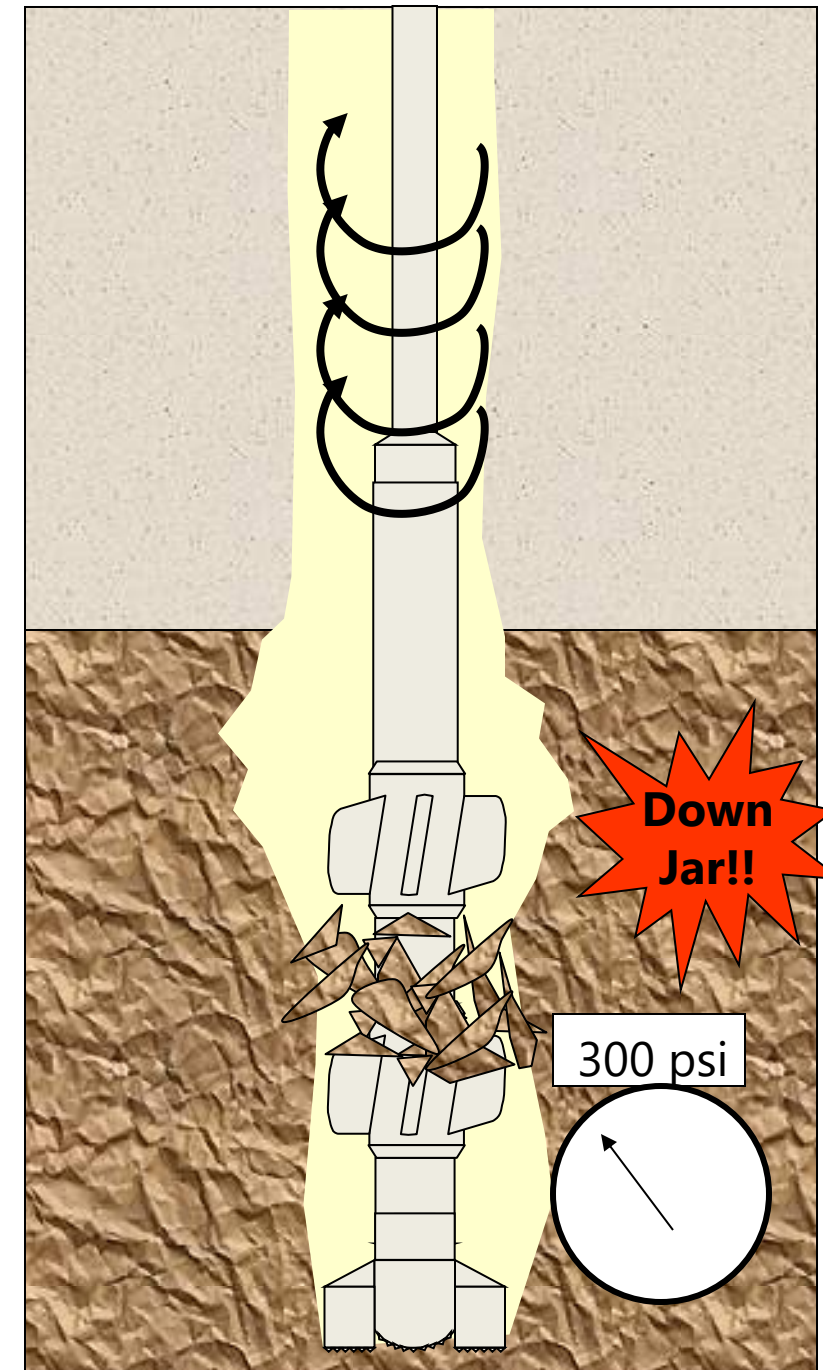
Freeing Methods - Hole Pack-Off/Bridging

Secondary Action

- Work torque into string and jar down with maximum loads
- Continue initial and secondary actions until pipe is free

When Pipe is Free

- Circulate hole clean, ream the section, and continue to clean the hole. Keep the pipe moving.



Freeing Methods - Wellbore Geometry

Initial action

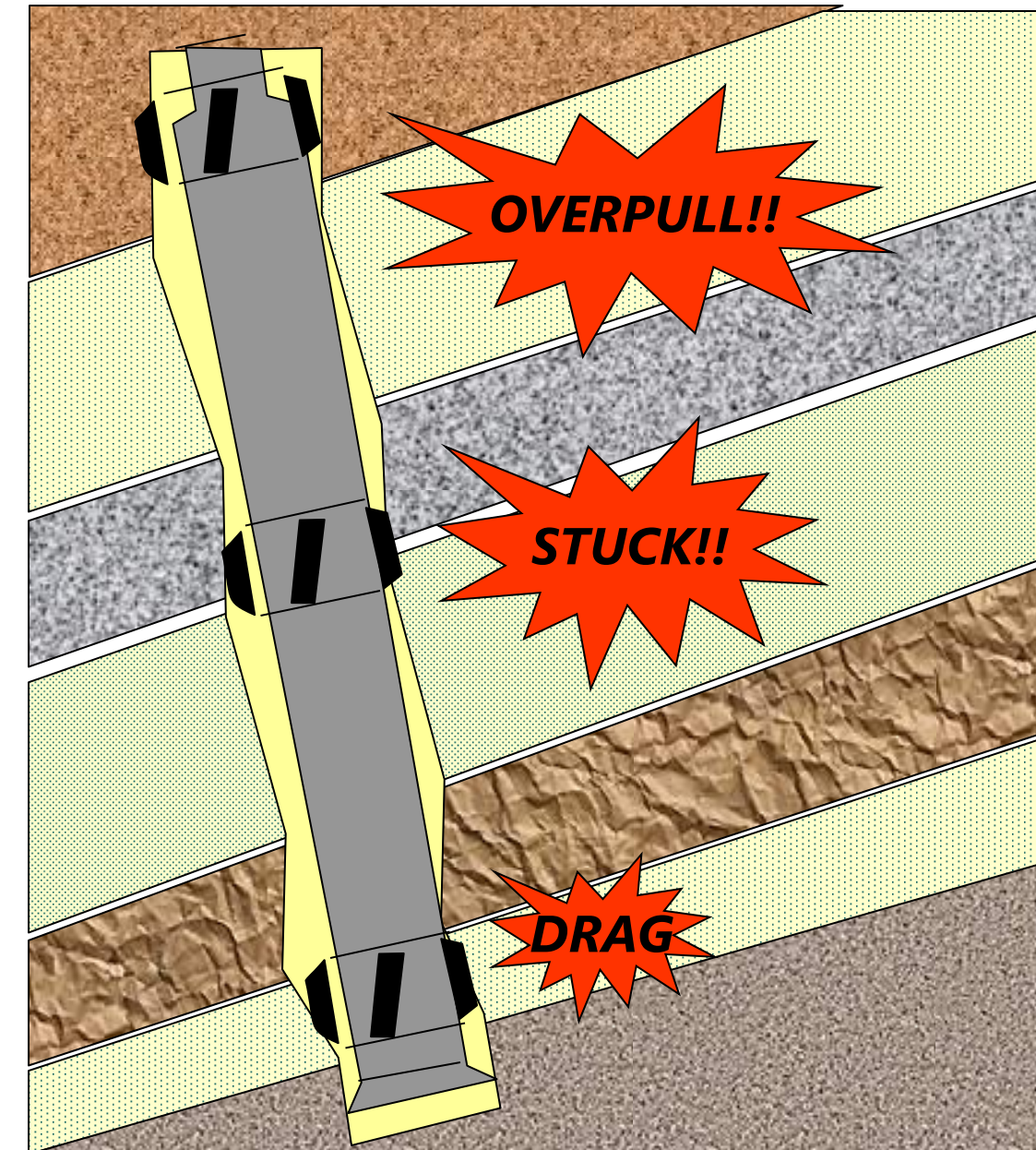
- If moving UP when sticking occurred - apply torque and JAR DOWN with maximum load.
- If moving DOWN when sticking occurred - ***do not apply torque***, JAR UP with maximum load.
- Continue jarring until pipe comes free (may require several hours).

Secondary action

- If stuck in limestone or chalk - spot acid
- If stuck in salt - spot fresh water pill

When pipe is free

- Circulate at maximum rate.



Freeing Methods - Mobile Formation/Bridge

Initial Action

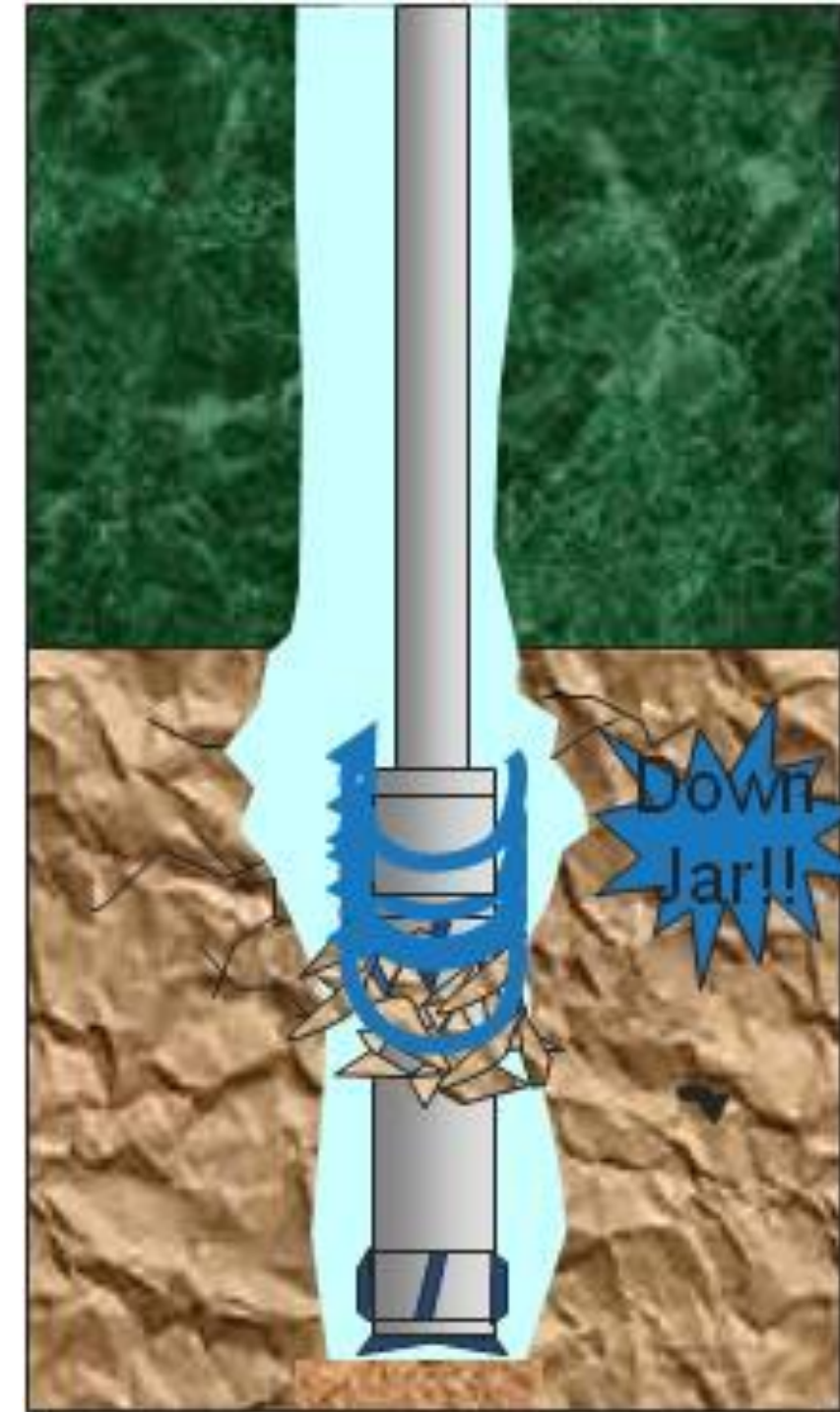
- ✓ Have string at free hanging weight and work maximum torque to stuck point and then release.
- ✓ Repeat torque and release until the pipe is free.

If Still Stuck

- ✓ Work torque into string and jar down with maximum loads.
- ✓ Continue until free or it is decided to stop.

Spot a Fresh water pill

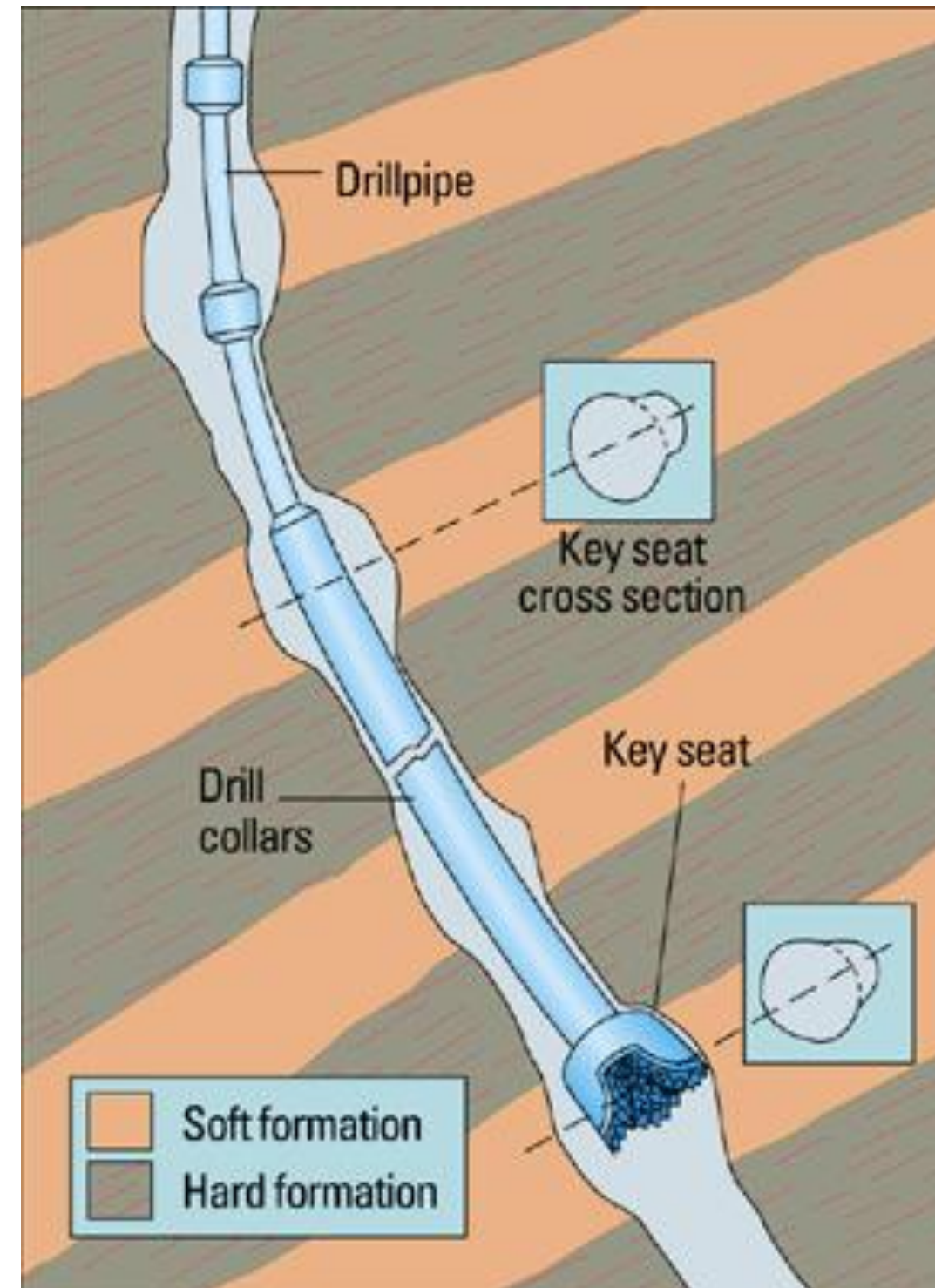
- ✓ Pull a little over up drag and wait for the salt to dissolve.



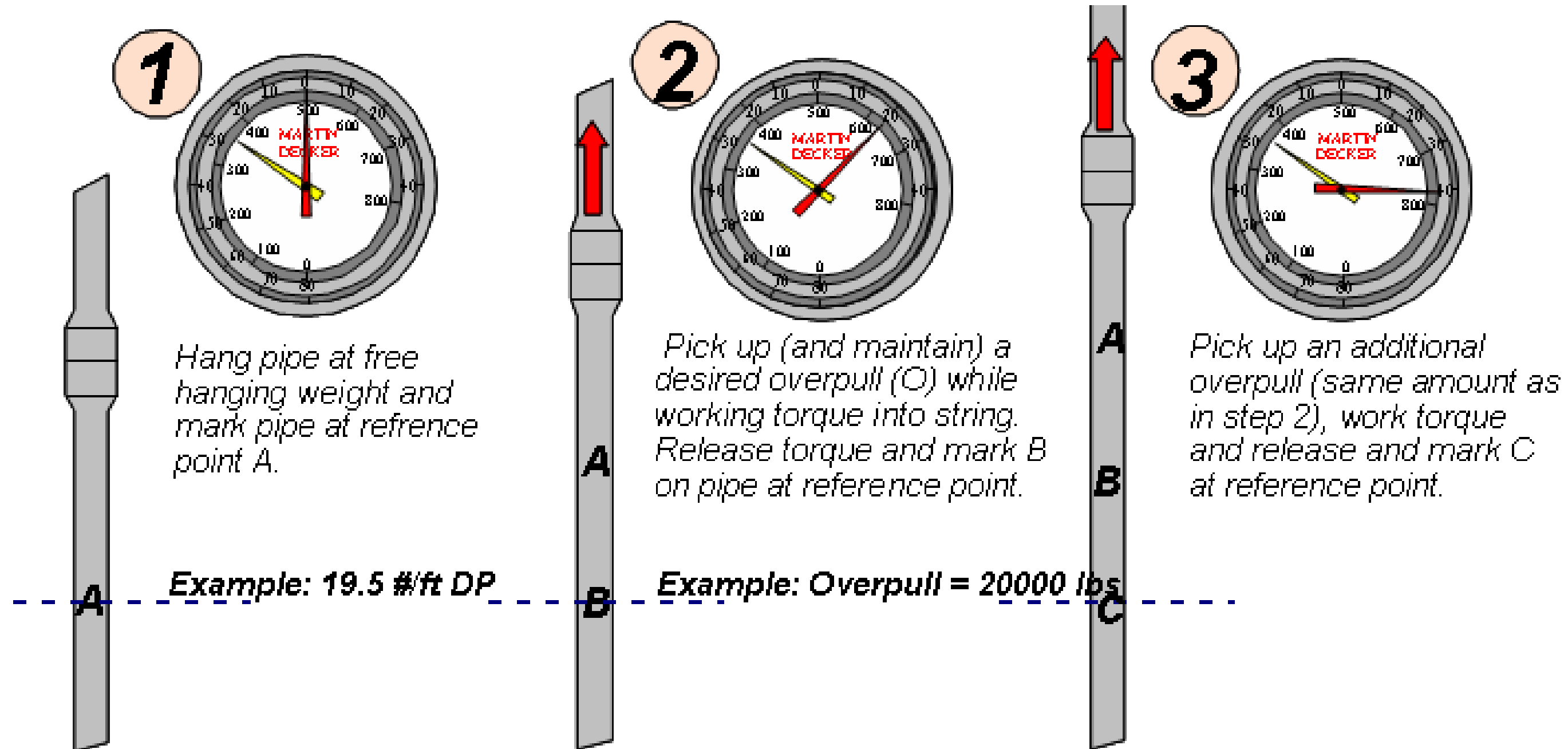
Stuck Pipe Prevention

Bottom Hole Assemblies:

- Operate BHA and string with limits
- Keep the BHA as simple as possible
- **Run a drilling jar** whenever possible
- Whenever possible **use spiral drill collars**
- Stabilize the BHA to minimize wall contact
- Always gauge bits and **stabilizers** accurately



Free Point Estimation



Q & A

Session 3



Thank You.



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